

Year	Gross Profit Margin (GPM)	Net Profit Margin (NPM)	Operating Profit Margin (OPM)
2016	69.56	20.22	33.60
2017	74.56	26.62	40.11
2018	74.43	16.38	41.69
2019	75.66	31.66	44.30
2020	76.75	32.29	46.64

Note: Numbers are in percent.

We can dissect Microsoft's profitability even further. The ROTC measures the profitability of the company's overall capital resources, whereas return-on-assets (ROA) and ROCE focus solely on the profitability of those types of resources. (These ratios also should be compared with the ratios of other firms within the industry to measure the capability of management to generate returns.) Microsoft has an ROA of 18.65%, an ROCE of 38.49%, and an ROTC of 27.18%. If these returns were below those of other rates available in the marketplace at a similar level of risk, then it would be better for the company to be broken up and have the assets reinvested where they would be more productive. For Microsoft, not surprisingly, these ratios only confirm that the company enjoys relatively high profitability.

4.2.5 Solvency Factors

In [Table 4.7](#) we list the values of popular solvency factors for selected stocks. The cash-flow-from-operations ratio (CFOR), the cash ratio (CR), the current ratio (CUR), and the quick ratio (QR) are all measures of a company's ability to cover liabilities with varying amounts of liquid cash. Companies with high solvency factor values are more liquid or solvent and less likely to be forced into bankruptcy by payments owed on debt and other liabilities. Many portfolio managers use one or more of these variables as a screen for financially sound stocks.

TABLE 4.7

Solvency Factors for the Top 10 S&P 500 Companies

Ticker	Cash-Flow-from-Operations Ratio (CFOR)	Cash Ratio (CR)	Current Ratio (CUR)	Quick Ratio (QR)
AAPL	0.77	0.86	1.36	1.22
MSFT	0.94	1.97	2.53	2.30
AMZN	0.54	0.67	1.11	0.84
GOOGL	1.18	2.75	3.41	3.28
TSLA	0.33	1.11	1.63	1.24
FB	2.83	4.68	5.51	5.35
BRK.B	N/A	N/A	N/A	N/A
JNJ	0.56	0.79	1.48	1.17
WMT	0.40	0.21	0.79	0.27
JPM	N/A	N/A	N/A	N/A

Note: Data are for end of 2020. For the company names, see the note to Table 4.1.

CUR and QR are very industry sensitive, but a general rule of thumb in assessing these ratios is that the more liquid the current assets are, the less crucial it is that those assets exceed the current liabilities by a wide margin. CUR, which divides current assets by current liabilities, gives a general sense of the company's liquidity versus its liabilities; QR refines the assessment by narrowing current assets down to cash, marketable securities, and accounts receivable. A high CUR generally indicates adequate short-term liquidity, but it should be examined alongside other ratios such as the inventory turnover (IT) ratio and return on assets (ROA) to ensure that there is not an inefficient use of cash or other marketable equivalents. Large inventories inflate CUR but may be a sign that the company's products are not selling and are even becoming obsolete. QR is a better measure of liquidity in relation to liabilities for companies that rely on large inventories.

Microsoft's CUR of 2.53 and QR of 2.30 contain several important facts about the company's liquidity situation. Microsoft is adequately funded on a short-term basis. The QR is only slightly less than the CUR as a result of excluding inventories from current assets, which indicates that the bulk of Microsoft's current assets are in the form of cash and marketable securities rather than shelf upon shelf of inventory. By contrast, Walmart has a CUR of 0.79 and a QR of 0.27. The significant difference, percentagewise, between CUR and QR reveals that most of Walmart's current assets are tied

up in inventory, and the company may be suffering from inadequate liquidity. Although QR by itself raises a red flag, we should analyze other factors in conjunction with this, such as Walmart's debt-to-equity and profitability ratios, which we also discuss in this chapter.

4.4.6 Financial Risk Factors

Table 4.8 lists the values of popular financial risk factors for selected stocks.⁹ By controlling for signs of financial risk, portfolio managers can identify which companies are not well equipped to weather the storm if sales temporarily slow down or if the economy hits a slump. A company with a high level debt-to-equity (D/E) ratio relative to other companies is more susceptible to bankruptcy even if its core business is solid. A high D/E ratio in conjunction with a low interest coverage ratio (ICR), which equals earnings before interest and taxes over interest expense, is cause for further concern because it means that the firm's current earnings will not cover many interest payments.

TABLE 4.8

Financial Risk Factors for the Top 10 S&P 500 Companies

Ticker	Cash-Flow Coverage Ratio (CFCR)	Debt-to-Equity Ratio (D/E)	Interest Coverage Ratio (ICR)	Total Debt Ratio (TDR)
AAPL	0.66	3.96	23.35	0.38
MSFT	0.80	1.44	22.21	0.28
AMZN	0.58	2.41	12.12	0.34
GOOGL	2.06	0.41	127.29	0.09
TSLA	0.29	1.67	1.33	0.33
FB	3.03	0.24	N/A	0.08
BRK.B	0.38	0.98	11.72	0.13
JNJ	0.57	1.65	95.82	0.22
WMT	0.48	1.92	9.50	0.29
JPM	0.06	10.97	N/A	0.17

Note: Data are for end of 2020. For the company names, see the note to Table 4.1.

A high D/E ratio alone does not automatically spell financial danger. Higher levels of debt generally lead to greater earnings volatility and a

higher risk of financial distress or possibly bankruptcy. However, businesses that generate large, stable cash flows can handle relatively large amounts of debt because they can finance the interest payments on the debt. They are therefore more likely to have higher D/E ratio values than companies that are short on cash. In 2020, Apple's D/E ratio equaled 3.96, compared with an industry average D/E ratio of 3.34. Apple's capital structure included a greater-than-average portion of debt, but its interest coverage of 23.35 times was also greater than the industry average. Its earning power covered its debt burden.

The best possible position to be in is to have little debt and plenty of cash on hand to cover the interest payments on it. Google's D/E ratio is a mere 0.41, and its interest coverage ratio is 127 times. The fact that the company is capable of taking on much more debt gives it room to start new projects. Low financial risk does not just avert disaster; it is a big plus for expansion and new investment.

4.2.7 Corporate Activity Factors

The final category of fundamental factors is a category that concerns corporate activity like stock buybacks, insider purchases, and decisions on research and development (R&D) expenditure, as well as other factors that do not neatly fit in the other factor categories. Although stock buybacks are public information, quantitative portfolio managers sift through the data better than the average investor, and this ability is an informational advantage that creates near-arbitrage opportunities. Stock buybacks could reveal that management is upbeat about the future of the company. Some people refer to the shareholder yield as dividends plus net buybacks of stocks. Buybacks have the additional advantage that they are not taxed as much as dividends. Insider purchases could also represent good news about a stock that is not yet public or could simply reflect management's confidence in the company. Either way, studies have shown positive excess returns from stocks experiencing insider purchases and lower returns for stocks experiencing insider selling.¹⁰ R&D expenditures reveal that management is actively betting on the future, which could be an indication of beneficial corporate activity.

4.3 TECHNICAL FACTORS

Most technical factors are constructed from past price and volume data (usually, open prices, high prices, low prices, closing prices, volume, open interest, and bid and ask prices) and other readily available financial information. One of the great advantages of technical factors is that they constantly update themselves. New fundamental data are available only quarterly at most, when a company files its financial statements with the Securities and Exchange Commission (SEC). Up-to-date technical indicators, on the other hand, are available as often as every few seconds—although most portfolio managers are happy with daily or monthly technical data. Portfolio managers typically use technical factors to capture very short-term changes in the relative value of stocks.

We group the technical factors into four subcategories: liquidity risk factors, price-based factors, volume-based factors, and overall market movement factors. Liquidity risk factors are used to understand the consequences and effects of trading a stock. Price-based factors are factors that are generated mainly from stock prices or stock returns and are used as indicators of potential future price movements in the stocks. Volume-based factors are generated principally from information in past trading or volume that might signal the intentions and changing behavior of market participants. Overall market movement factors are aggregate technical indicators that might help decipher overall stock market movements and their implications for the near future. Tables 4A.8 through 4A.11 in [Appendix 4A](#) provide complete lists of some of the most important technical factors for QEPM models.¹¹

4.3.1 Liquidity Risk Factors

[Table 4.9](#) shows the values of some important liquidity factors—trading turnover (TT), Amihud illiquidity, and invariance illiquidity—for selected stocks. Liquidity factors gauge whether a stock’s liquidity matches trading demand. One of the easiest and most common liquidity factors is TT. TT, which is calculated as the average daily dollar trading volume (ADDTV) divided by market capitalization, measures the percentage of outstanding shares traded in a day on average. It gives an indication of how easy or difficult it is to trade the stock. Other proxies for liquidity include market capitalization and float capitalization, the portion of the market capitalization available for public trading. Floating shares and total shares

traded are typically highly correlated, but some stocks have a very small float. The number of shareholders also indicates the ease with which an investor can enter and exit a position in a stock.¹² Amihud illiquidity (AILIQ) is a measure of illiquidity based on the ratio of the daily return of the stock to the daily volume. The idea is that the higher the daily return for a given volume, the less liquid is the stock due to price impact. A more recently discovered measure of liquidity, or lack thereof, is invariance illiquidity (INVIL), which measures illiquidity based on a theoretical model. This measure is related to the average dollar volume of the stock and its volatility. As we can see from the table, TT, AILIQ, and INVIL vary even among relatively large, well-known stocks. In 2020, Amazon’s measured trading turnover was 0.87% compared to Walmart’s 0.24%. This says that Amazon is more liquid than Walmart in that much more trading occurs given Amazon’s market capitalization. More precisely, on average, about 1% of Amazon’s shares are being traded daily compared to one quarter of 1% for Walmart’s shares. Similar conclusions are drawn by using Amihud illiquidity and invariance illiquidity. For these measures, higher values mean that the stock is less liquid. Amazon’s AILIQ value is 0.13 compared to Walmart’s 1, suggesting Walmart is even less liquid than was implied by TT. Finally, Amazon’s INVIL is 0.76 compared to Walmart’s 1.49. Thus, all of these measures suggest, to varying degrees, that Amazon stock is more liquid than Walmart’s stock. Generally, these liquidity factors are correlated with each other.

TABLE 4.9
Selected Technical Liquidity Measures for the Top 10 S&P 500 Companies

Ticker	Amihud Illiquidity (AILIQ)	Invariance Illiquidity (INVIL)	Trading Turnover (TT)
AAPL	0.14	0.83	0.61
MSFT	0.24	1.02	0.36
AMZN	0.13	0.76	0.87
GOOGL	0.57	1.27	0.48
TSLA	0.25	1.16	3.52
FB	0.38	1.16	0.77
BRK.B	1.00	1.55	0.35
JNJ	0.91	1.45	0.25
WMT	1.00	1.49	0.24
JPM	1.01	1.80	0.42

Note: Data are for end of 2020. For the company names, see the note to Table 4.1. Trading turnover is in percentage terms.

All these liquidity factors matter only in relation to the size of the manager's portfolio and the value of the trades he or she wants to place. Trading liquidity mainly becomes a concern when a manager wants to buy more than the available number of shares or sell more than the market will take. Some portfolio managers build liquidity factors into portfolio models as transactions costs, whereas others leave it to the trading desk to sort out the costs of the trades in light of liquidity constraints.

4.3.2 Price-Based Factors

Table 4.10 shows the values of some prevalent price-based factors for selected stocks.¹³ These factors attempt to capture short-term movements in stock prices or measures related to stock prices that might indicate something about future stock returns. Momentum is a very popular short-term indicator of future performance. Various academic studies have found positive auto-correlation in stock returns, meaning that positive returns in one period tend to lead to positive returns in the next period.¹⁴ There is also evidence of negative autocorrelation in individual stock returns over periods of one week and one month.¹⁵ Some portfolio managers measure momentum as the stock return over the last year, some measure it as the stock return over the last month, and others use more complicated variations. The same idea prevails: stocks that have performed well recently

tend to continue performing well in the near future. You might argue that momentum cannot last because no stock can continue to rise forever. The technician would tell you that a positive stock return in the last period contains useful information that may not be fully distributed throughout the investing population. It could represent insider buying by company executives who anticipate profit growth, or it could represent buying by some knowledgeable investors who have deduced some as yet unreleased piece of good news that will help the stock in the near future. Tesla's momentum (i.e., stock return for 2020) was 743%, while J.P. Morgan's was -5.5%. To the extent that momentum works, it would tell you to buy stocks like Tesla, Apple, Amazon, and Google and avoid the smaller-momentum stocks.

TABLE 4.10

Selected Technical Price-Based Factors for the Top 10 S&P 500 Companies

Ticker	Beta (BETA)	Bollinger Band (BB)	Twelve-Month Momentum (M12M)	Relative Strength Index (RSI)
AAPL	1.28	1.00	82.31	66.88
MSFT	0.82	1.00	42.53	56.43
AMZN	1.19	1.00	76.26	60.18
GOOGL	0.98	0.00	31.03	48.96
TSLA	2.18	1.00	743.44	47.44
FB	1.18	0.00	33.09	45.65
BRK.B	0.84	1.00	2.37	46.49
JNJ	0.70	1.00	10.85	50.56
WMT	0.42	-1.00	23.34	50.74
JPM	1.23	1.00	-5.52	66.29

Note: Data are for end of 2020. For the company names, see the note to Table 4.1.

Portfolio managers use various moving averages to capture information traveling through a stock's price. The relative strength index is a popular indicator that attempts to pinpoint when a stock is a buy and when it is a sell. Bollinger bands, which delineate limits around the mean price for buying and selling, improve on simple moving-average models. When a stock hits the top band above the mean, it is a signal to sell; when it hits the bottom band below the mean, it is a signal to buy. Both moving averages

and Bollinger bands may be useful for identifying short-term changes in the relative attractiveness of stocks. At the end of 2020, the Bollinger band signals for Apple and Microsoft are 1.00, meaning that there is positive movement in these stocks and they are a buy, while the signal is neutral on stocks like Google and Facebook and bearish on Walmart.¹⁶

Beta is not conventionally thought of as a technical factor, since it's based on the capital asset pricing model (CAPM) of finance. Beta is also different from other factors in that each stock's beta exposure is estimated by a regression. Nevertheless, research dating back to the early 1970s shows that beta might be useful in predicting future stock returns. In fact, in recent years, this strategy has become so popular, it's part of many QEPM portfolio models and tailored investment strategies for retail investors.¹⁷ Contrary to theory, people have found that low-beta stocks have higher risk-adjusted returns than high-beta stocks. As of December 2020, the beta of Apple was 1.28, while the beta of Johnson & Johnson was 0.70. Theory would predict that Johnson & Johnson would be less volatile in relation to the stock market than Apple. However, recent empirical evidence might indicate that a portfolio of low-beta stocks will outperform a portfolio of high-beta stocks on a risk-adjusted basis.

4.3.3 Volume-Based Factors

Table 4.11 shows the values of a very important factor—the short interest ratio—for selected stocks.¹⁸ Some people use a high degree of short interest as a contrarian signal. That is, when many investors are shorting a stock, if the stock continues to rise due to buying pressure, it may force the very investors who shorted the stock to buy it back—causing the stock to rally even further. This factor was much in the news in 2021 as many retail traders targeted stocks with large amounts of short interest.¹⁹ Consider data on Walmart, with a relatively small short interest ratio of 2.86%, and Tesla, which has a ratio of 1.97% but whose value declined significantly in 2020. In December 2015, Tesla's short interest was 64 and in December 2019 it was 23 as people strongly believed the company was overvalued. Since that time, the stock is up 18-fold. Amongst companies in the S&P 500 not in the table, insurance broker AON and specialty chemical company IFF both have short interest ratios around 20%.

TABLE 4.11

Selected Technical Volume-Based Factors for the Top 10 S&P 500 Companies

Ticker	Short Interest Ratio (SIR)	SRI Minus One Year Average (SIRX)
AAPL	1.32	−3.02
MSFT	2.15	0.12
AMZN	1.26	0.13
GOOGL	1.73	−0.73
TSLA	1.97	−5.11
FB	2.08	0.17
BRK.B	2.94	−0.76
JNJ	2.76	−0.24
WMT	2.86	−0.07
JPM	2.08	−0.10

Note: Data are for end of 2020. For the company names, see the note to Table 4.1.

4.3.4 Overall Market Movement Factors

Table 4A.11 lists the technical factors that are not computed for individual stocks but are rather computed for the overall market and are usually indicators of where the overall market is headed. These are appropriately called *overall market movement factors*. In order to compute the exposures of individual stocks, you must estimate regressions of individual stock returns on these market-wide factors. In Chapters 16 and 17 we actually compute these factors and their returns.

4.4 ECONOMIC FACTORS

The first arbitrage pricing models were based on economic indicators, and the popularity of economic factors endures in QEPM models.²⁰ Table 4A.12 gives a list of the most common economic factors used in QEPM models.

For macroeconomic models of stock returns, the portfolio manager should choose factors that affect all stock returns to some degree. GDP growth, the yield-curve slope, unemployment, and inflation are popular factors because they influence almost every corner of the market. Although each of these variables can be defined in different ways, there are some standard definitions. A common way of calculating inflation, for instance, is

to take the latest monthly inflation number and annualize it. Quantitative managers must be careful to use the right data when calculating macroeconomic figures. There are significant delays between the measurement and release of macroeconomic data. The most current monthly inflation data are usually data on the conditions one month prior. Other macro variables are reported at different delays. The portfolio manager should create a table of data release months and adjust models to account for the lag in information.²¹

4.5 ALTERNATIVE FACTORS

We group the remaining factors into a category called *alternative factors* for lack of a better name. There are three subcategories of alternative factors: analyst factors, captivus factors, and social responsibility factors. Tables 4A.13 and 4A.14 in [Appendix 4A](#) list the most common alternative factors used in QEPM models.

4.5.1 Analyst Factors

Analysts on Wall Street and elsewhere spend a lot of time researching individual companies. Wall Street analysts' earnings forecasts, buy–sell recommendations, and related information can be valuable in forecasting stock returns.²² There have been hundreds of studies documenting the success of analyst recommendations in predicting stock returns.²³ The beauty of an analyst recommendation is that it condenses hours and hours of research into a single directive, buy or sell. A portfolio manager can transform recommendations into factors that work in a quantitative model.

Some studies show that a portfolio of analyst-rated “strong buys” outperforms a portfolio of analyst-rated “strong sells.” Since the analyst recommendations are public information, one might wonder how portfolio managers could use them for statistical arbitrage. One argument is that quantitative managers have an advantage in filtering the vast amount of analyst data for useful recommendations. In mere seconds they can access newly released analyst ratings and use software to create or update a portfolio of the top-rated stocks. The average investor may not hear about ratings changes until the end of the trading day, and even then he or she

probably does not own the kind of software that can quickly sort through hundreds of ratings.²⁴

A strategy of purchasing stocks that have been upgraded by analysts in the last month has been shown to outperform on a risk-adjusted basis. Similarly, selling downgraded stocks outperforms on a risk-adjusted basis. One would expect even better results from a strategy of shorting the firms that were downgraded by analysts of their own underwriter. Other analyst-related strategies seem to produce strong returns as well. It also has been documented that purchasing a group of stocks with recent upward earnings revisions by analysts outperforms on a risk-adjusted basis. The degree of dispersion in analyst ratings also may create opportunities to outperform.

Table 4.12 shows the values of several analyst factors for selected stocks, including the median recommendation, the percentage of buy and sell recommendations for each stock, and the standard deviation of the analyst ratings. Analyst recommendations are given as numbers from 1 to 5, such that 1 is for “strong buy,” 2 is for “buy,” 3 is for “hold” or “neutral,” 4 is for “sell,” and 5 is for “strong sell.” Tesla has the lowest median analyst rating at 3 (i.e., neither a buy nor a sell, but a hold). The highest median rating of a buy (i.e., 2) is shared by many companies. The most bullish sentiment from analysts is for Amazon, with 94% of analysts suggesting a buy or strong buy. The most bearish sentiment is for Tesla, with 37% of analysts recommending a buy or strong buy and 31% recommending a sell or strong sell. Finally, the greatest disagreement amongst analysts concerns Tesla and Berkshire Hathaway, judging by the standard deviation of analyst ratings.

TABLE 4.12
Selected Analyst Factors for the Top 10 S&P 500 Companies

Ticker	Median Rating (ARMR)	Percent Buys (ARPB)	Percent Sells (ARPS)	SD of Rating (ARSD)	Number of Analysts (EPSFN)
AAPL	2.00	75.00	5.00	0.90	29.00
MSFT	2.00	91.43	0.00	0.64	27.00
AMZN	2.00	94.00	0.00	0.58	40.00
GOOGL	2.00	92.31	0.00	0.65	29.00
TSLA	3.00	37.14	31.43	1.32	15.00
FB	2.00	88.24	3.92	0.76	40.00
BRK.B	2.50	50.00	0.00	0.96	3.00
JNJ	2.00	72.22	0.00	0.80	14.00
WMT	2.00	79.41	2.94	0.81	28.00
JPM	2.00	62.96	7.41	0.74	22.00

Note: Data are for end of 2020. For the company names, see the note to Table 4.1. Each analyst rating is ranked 1–5. 1 = strong buy, 2 = buy, 3 = hold, 4 = sell, and 5 = strong sell.

If analyst coverage provides useful information for models of stock return, what should be done about stocks that receive little or no analyst coverage? For example, Amazon has about 40 analysts researching the company, Tesla has about 15, but at the bottom of the S&P 500 there are companies with no analyst coverage or just a few, such as one analyst for Under Armour and three for NRG Energy. Lack of analyst coverage may in fact be an advantage unto itself. There is a neglected-firm effect in which stocks followed by few analysts or owned by few institutions tend to have relatively higher long-run risk-adjusted returns than heavily covered stocks. There could be an informational story for this effect. If it takes time for good information to seep through the marketplace, then information about a stock that is not covered will spread more slowly than information that is publicized by analyst comments and recommendations. When there is good news about a neglected stock, the stock's price adjusts to the news more slowly than a well-known stock's price would, which gives investors more time to buy before the gain. The neglected-firm effect suggests a strategy of purchasing neglected firms on good news and selling them on bad news. In reality, many quants look for good neglected stocks using a multifactor model that examines fundamental characteristics and assigns a bonus point to neglected status. Given two stocks with identical fundamentals, the model would rate the neglected one more highly than the well-known one.

4.5.2 Captivus Factors

With the advent of the internet and the explosion of technology into everyday life in the form of personal computers, smart phones, satellites, cameras, and social networks, everyone from police to investors has realized that capturing this information can be very valuable. As of 2020, about 45% of asset managers use some form of *captivus* data.²⁵ This type of data is often called *alternative data* or *big data*, but we prefer to call it *captivus* data, which is the Latin word for “captured.” Whether it’s GPS or satellite data, social media data, or news feeds, the data are essentially captured by sophisticated programs. Some of the data is collected by specialized companies for resale and some is captured by portfolio managers for proprietary usage. One of the benefits of *captivus* data over other data used in QEPM is their usually higher frequency: *captivus* data are obtained earlier than typical fundamental data. Another advantage is that *captivus* data usually supplement or are complementary to more traditional data sources. The most popular method of obtaining *captivus* data is through *web scraping*. Web scraping is the process of locating web pages and extracting data from them, whether they are company sites, consumer sites, or social media sites. For example, a program that captures the posts of a group of Twitter and Instagram users and stores the data is web scraping. This data set is already large and growing. As of 2019, there were 5.5 billion mobile users worldwide, 4 billion internet users, and 3 billion users on social networks.

Since there is such a vast array of *captivus* data and these data are typically either proprietary to an institution or very expensive to purchase, we do not use them in this book directly. However, any serious investor could purchase the data from existing providers or hire a team to scrape the data themselves.²⁶ We can, however, provide some examples of how the data might be used. When attempting to predict the sales of a business in advance of earnings releases, you might use geolocation, foot traffic, and app usage data to predict the number of customers frequenting a particular store or stores. For example, as the 2020 Covid-19 pandemic unfolded, data on foot traffic from mobile phones and geolocation could be used to understand which cities were rebounding and which were not. For those interested in workout companies, you might use Instagram and Twitter posts to model trends in buying home workout equipment and other

personal trends that are related to public companies. For earnings calls, rather than read and/or listen to every transcript, you could use machine learning and textual analysis to determine if there are any key language patterns that might be undetectable to the average person that hint at potential future company health. You might use data on air traffic and other transportation resources to gauge the behavior of economies long before the actual GDP numbers are released.²⁷ And, of course, consumer trends can be analyzed for a host of retail companies through web traffic, Google Trends, and social media applications.

Captivus data are more varied than traditional data and are oftentimes expensive and more difficult to obtain. However, captivus data can offer many advantages to the quantitative manager, including higher frequency of information, earlier understanding of important metrics for a company, and significantly more accurate information for understanding a specific issue. Captivus data will continue to be an important part of the QEPM data tool kit in the future.

4.5.3 Social Responsibility Factors

Many investors believe that in order to maximize the return on their portfolios, they must separate financial decisions from social concerns. Some socially responsible business practices may in fact do nothing for a company's bottom line over the short term. Yet, contrary to the view that socially responsible investing necessarily sacrifices profits in the name of doing good, many studies show that factoring social concerns into investment decisions does not reduce the return of a portfolio.²⁸ Regardless of one's philosophical stance on socially responsible investing, data on what we might call "social responsibility" factors may offer useful information for portfolio managers.²⁹ Probably the most useful factor in this arena is the employee relations factor. Quantifying the state of management's relations with employees is admittedly not as straightforward as calculating the P/E ratio, and the quality of the measurement will affect the model.³⁰

This factor adds depth to a strictly financial model, though, because it contains information that may not be obvious from the most recent financials and yet may indicate a burgeoning operational problem (or, conversely, the seed of a future competitive advantage).

Table 4.13 shows the values of some prevalent socially responsible, or ESG, factors for selected stocks. For each category, we measure each company's percentage of strengths in that category minus their percentage of weaknesses.³¹ Many of the companies on the list are weak on corporate governance, which includes items such as exposure to bribery and fraud, controversial investment practices, and governance structure issues. Diversity includes elements such as the diversity of the board and discrimination practices. On the extreme positive side is Microsoft, with a 100% net positive reading, and on the negative side is Amazon with a -50% reading. Employee relations covers areas such as the health and safety of the workforce, as well as whether or not the firm engages in profit sharing with employees and human capital development and growth. By this score, Johnson & Johnson is on the positive side with a score of 44%, and Tesla is on the negative with a score of -22%. As for the environment and human rights, most firms are neutral to positive. QEPM's ultimate goal is portfolio outperformance. Thus, QEPM should focus on using these factors to find signals of future good and bad stock performance. Some readers might be skeptical about the validity of certain social responsibility variables, but they may help capture aspects of investor sentiment not captured by other variables. Therefore, we list them in Table 4A.14 for those interested in this area of analysis.³²

TABLE 4.13

Selected Socially Responsible Factors for the Top 10 S&P 500 Companies

Ticker	Community Contribution (CC)	Corporate Governance (CG)	Diversity (DIV)	Employee Relations (ER)	Environment (E)	Human Rights (HR)	Product (PRO)
AAPL	0.00	-25.00	50.00	0.00	14.29	0.00	-23.33
MSFT	0.00	25.00	100.00	5.56	35.71	50.00	10.00
AMZN	0.00	-25.00	-50.00	-16.67	0.00	0.00	-66.67
GOOGL	0.00	25.00	0.00	33.33	14.29	0.00	-23.33
TSLA	0.00	-50.00	0.00	-22.22	7.14	0.00	-16.67
FB	0.00	-25.00	50.00	0.00	0.00	50.00	-6.67
BRK.B	0.00	-25.00	0.00	0.00	7.14	0.00	0.00
JNJ	0.00	-25.00	0.00	44.44	14.29	0.00	-40.00
WMT	0.00	-25.00	-50.00	-16.67	14.29	0.00	-33.33
JPM	0.00	25.00	0.00	33.33	14.29	0.00	-16.67

Note: Data are for end of 2020. For the company names, see the note to Table 4.1. These factors are also known as environmental, social, and governance (ESG) factors. More details on these factors are supplied in Table 4A.14 of Appendix 4A.

4.6 FACTOR CHOICE

Earlier in this chapter we compared QEPM factors with cooking ingredients. Building an investment model, like creating a meal, is a mix of science and art, and many managers build models very intuitively, adding factors according to taste. When it comes to building most quantitative models, though, the cooking comparison only goes so far. The chef may not need to know chemistry, but the quantitative portfolio manager must understand financial and economic theory in order to choose and combine factors appropriately, always keeping in mind the seven tenets of QEPM.³³ The manager has to ask himself or herself why the market has not already identified the arbitrage opportunity that he or she sees. He or she needs to consider whether factors that worked in backtests using historical data will continue to work in the future. In this section we suggest several techniques of factor choice supported by theory. As we discussed in [Chapter 3](#), there are two basic types of quantitative stock return models: the fundamental factor model and the economic factor model. To choose factors for the fundamental factor model, we suggest using the univariate and multiple regression techniques. To choose factors for the economic factor model, we suggest using the unidimensional and multidimensional zero-investment portfolio techniques.³⁴ In addition to these techniques, we also suggest using a simple correlation statistic or Kendall's rank-correlation statistic to determine the correlation among factor choices. These techniques aid in combining and grouping factors.

4.6.1 Univariate Regression Tests

Many portfolio managers simplify the process of searching for relevant factors by performing a series of simple regressions of factors.³⁵ That is, they begin by first identifying a group of factors that they can theoretically justify explaining stock returns. They then run panel regressions on each factor versus the stock returns in the universe. Thus

$$r_{i,t} = \alpha + f\beta_{i,t} + \epsilon_{i,t} \quad (4.1)$$

where $\beta_{i,t}$ is the factor exposure of stock i at time t , and the estimate of f from this panel regression will show the relationship between the factor and stock returns. If a factor has a significant value of f , then the factor is useful in explaining stock returns.³⁶

The drawback of univariate regressions is that the portfolio manager may find many variables that are significant in explaining stock returns but are surrogates for each other. For instance, he or she may find that both the P/E and P/B ratios explain stock returns but that they represent the same idea. Thus, only one of them may be needed in the model. Ideally, one would like to find factors that help in explaining stock returns but are not highly correlated with each other. In terms of the fundamental law discussed in [Chapter 2](#), we know that the more factors that we find to explain stock returns, the higher will be our information ratio, *provided* that the average contribution of each factor *does not decrease*. Thus, if we add a factor that is highly correlated with an existing factor, we probably will decrease the contribution of each factor, which will not improve the model. In that case, why use simple regressions for factor searching at all? First, they are very simple to perform. Second, they are a way to get an early idea of what might be relevant and what might not be relevant. If there are too many potential factors, the simple regression can be used to make the first round of cuts.

4.6.2 Multiple Regression Tests

The multiple regression is more sophisticated for determining which factors to include because it considers the relationship among factors while estimating the impact on return. The greatest danger with this method is that if the portfolio manager includes too many factors in the regression, there is a chance of *misspecification bias*, which leads to misleading statistical inference.

If the portfolio manager has limited the number of explanatory factors to something reasonable, say, fewer than 10 factors, then a multivariate panel regression can be estimated, that is,

$$r_{i,t} = \alpha + f_1 \beta_{i,1,t} + \cdots + f_K \beta_{i,K,t} + \epsilon_{i,t} \quad (4.2)$$

where K is the number of factors to investigate, and $\beta_{i,k,t}$ is the exposure for stock i to factor k at time t . Once this regression is estimated, all insignificant factors are dropped. The significant factors go into the factor model.^{[37](#)}

One of the biggest dangers here is data mining, of course. Data mining is the process of taking historical data on stock returns and factors and testing a variety of models until the portfolio manager finds the model that best predicts or explains past stock returns. There are many ways to do data mining, and we discourage all of them for two reasons. First, by searching in a set of historical data for the factors that best explain stock returns, the portfolio manager is guaranteed to find a group of factors that would have worked historically, but this provides absolutely no guarantee that these factors will work in the future. The t -statistics or significance of these factors historically also is misleading because the manager did not take into account every factor he or she tested. If these were taken into account, the actual t -statistics would be much lower (see [Chapter 2](#) for more details on problems related to data mining). Second, by only finding the factors that explain stock returns statistically in the past, the portfolio manager has clearly failed to provide any theoretical explanation of why those factors were chosen. This is very bad practice for any serious quantitative portfolio manager. Data mining can be a serious problem when one uses the *stepwise regression* technique.³⁸ This method involves starting with a collection of factors and then running a regression on every possible combination of the factors and picking the set of factors with the largest $\bar{R}^2$. It's dangerous because, by searching for the best fit in a set of historical data, one might find a model that fits the past data very well but has no bearing on the underlying stock return generation and has a high likelihood of failing in predicting future stock returns.

A related technique is the sequential specification search. The portfolio manager begins with a list of factors that he believes explains stock returns, and, for this set of factors, he looks at the estimated coefficients. He may then decide to drop one factor because of insignificance and add another factor based on theory. The manager may do this a few times until he obtains a desired level of significance at explaining stock returns. This also leads to exaggerated statistical significance. Usually, the portfolio manager will not report the failed model attempts, which makes it very difficult for the investment committee to evaluate the truth or significance of the final stock return model.

The best advice for choosing factors is to begin with a strong explanation, grounded in theory, of why a model with certain factors will work. A manager might even start with a few potential models. She then

should test the models on the data but be wary of adding and dropping variables. She should test the model over a variety of time periods for model robustness.

4.6.3 Unidimensional Zero-Investment Portfolio

Portfolio managers may construct zero-investment portfolios based on a factor and study the characteristics of the portfolio. Typically, a manager will split the universe of stocks conditional on a particular factor exposure into thirds, quintiles, or deciles. Of course, any other division of the stocks is acceptable. Usually, a portfolio is created from the first division, and another portfolio is created from the last division. The returns of the top division are subtracted from those of the last division. These are the returns to a hypothetical zero-investment portfolio in which the top-division portfolio is bought and the last-division portfolio is shorted. It is called *zero investment* because, theoretically, no capital needs to be used to create the portfolio. The returns of this portfolio measure the benefits from using this factor to pick stocks.

We shall do an example with quintiles. Suppose that the factor of interest is the P/B ratio. The first task is to rank the universe of stocks by their P/B ratio exposure in each historical period. There are a variety of methods to do this. The stocks can be ranked monthly, quarterly, or yearly. The stocks should be ranked for every month for some historical period, say, 5 to 10 years of historical data to the present. The next step is to create an equal-weighted portfolio of stocks in the first quintile and an equal-weighted portfolio of stocks in the fifth quintile. The first quintile is just the 20% of stocks ranked highest according to the factor, which in this case is the P/B ratio. The fifth quintile is the bottom 20% of stocks ranked according to the P/B ratio. The next step is to compute the returns of the two portfolios for each monthly period. Of course, these portfolios will change over time on a monthly, quarterly, or yearly basis depending on the QEPM's choice of rebalancing period. The final step is to calculate statistics on the historical returns of the first-quintile portfolio minus the fifth-quintile portfolio. This procedure should be repeated for every factor that the manager is interested in using as a predictor of stock returns.³⁹

After the zero-investment portfolio returns have been calculated, one can do a statistical test of whether the average portfolio return is significantly

different from zero. Suppose that we calculated portfolio returns for T periods and obtained $r_1, \dots, r_T$. Then the t statistics can be calculated as follows:

$$\hat{t} = \frac{\bar{r}}{\hat{S}(\bar{r})} = \frac{\bar{r}\sqrt{T}}{\sqrt{\frac{1}{T-1} \sum_{s=1}^T (r_s - \bar{r})^2}} \quad (4.3)$$

where $\bar{r}$ is the sample average, i.e., $\bar{r} = \frac{1}{T} \sum_{s=1}^T r_s$, and $\hat{S}(\bar{r})$ is the standard error of the mean. The value of the t -statistic should be compared with the t distribution with $T - 1$ degrees of freedom. Although one should look at a t -statistics table for the actual critical values of the t -statistic to determine significance, a good rule of thumb is that if the absolute value of the t -statistic is greater than 2, the average portfolio return is significantly different from zero, and the factor is statistically significant.⁴⁰

4.6.4 Multidimensional Zero-Investment Portfolio

Zero-investment portfolios also can be created by considering many factors simultaneously. This approach is more rigorous than the unidimensional approach because we can examine the joint significance of factors. Construction of the zero-investment portfolio proceeds almost in the same way as before. The portfolio manager ranks all stocks by each factor of interest. If there are two factors, say, size and P/B ratio, then each stock will be assigned two rankings, one by size and the other by P/B ratio. Based on these rankings, the portfolio manager can group stocks into a number of joint quintiles or deciles. If the portfolio manager wants to create 10 groups out of each factor, then there will be 100 groups eventually. From the size factor, there will be 10 portfolios starting from the smallest to the largest. From the P/B ratio, there also will be 10 portfolios from the lowest to the highest. By taking intersections of these portfolios, one can obtain 100 portfolios.

Given 100 portfolios, the method to create the zero-investment portfolios depends on what the portfolio manager is interested in. If he or she is interested in whether small size and low P/B ratio together influence

stock returns, he or she could create the zero-investment portfolio by taking a long position on a small–low portfolio and a short position on a large–high portfolio. Once the zero-investment portfolio is constructed, t -statistics can be calculated to determine whether the joint effect of the factors is significant, as explained in the preceding subsection.

Alternatively, one may apply a multiple regression analysis. For each of the factors that we consider, we may define an indicator variable that takes the value of 1 if the stock belongs to the top division (e.g., quintile), a value of -1 if the stock belongs to the bottom division, and a value of 0 otherwise. If we run a regression of stock returns on the indicator variables, the estimated coefficients can be interpreted as the returns to suitably constructed zero-investment portfolios. Thus, the significance of the multiple regression can be taken as an indicator of the joint significance of selected factors.

4.6.5 Techniques to Reduce the Number of Factors

There are several methods to determine the correlation among factors. A quantitative portfolio manager could use these methods to determine whether certain factors are highly similar to one another either for the grouping of factors or to aid in reducing the number of factors in one's stock return model. We discuss two types of statistics here, one based on simple correlation and the other based on rank correlation.

In general, a perfect correlation (i.e., a correlation of 1) indicates that the two factors are almost identical, whereas a correlation of 0 would indicate that the factors are quite independent. In the extreme cases of perfect positive or negative correlation, the two factors are more likely to be redundant. In the in-between cases, the quantitative analyst would have to have a cutoff criterion.

The first statistic we discuss is called the *within-correlation coefficient*. It is the correlation coefficient adjusted for the panel structure of the data. We first subtract the mean from each variable for each period and then calculate the correlation coefficient. The adjustment is necessary to make sure that we are examining the contemporaneous correlation amongst variables. Specifically, given two variables X and Y , the within-correlation coefficient is calculated as

$$\hat{\rho}(X_{i,t}, Y_{i,t}) = \frac{\hat{C}(X_{i,t} - \bar{X}_t, Y_{i,t} - \bar{Y}_t)}{\hat{S}(X_{i,t} - \bar{X}_t) \hat{S}(Y_{i,t} - \bar{Y}_t)} \quad (4.4)$$

where $\hat{C}$ and $\hat{S}$ are sample covariance and standard deviation, and $\bar{X}_t$ and $\bar{Y}_t$ are the sample means of X and Y for period t , respectively.

The second statistic we discuss is based on Kendall's τ . Kendall's τ shows the rank correlation (i.e., whether the ranking by one variable is related to the ranking by another variable). Conceptually, Kendall's τ answers the following question: "Suppose that you choose a pair of observations. You rank these two observations based on the value of one variable. Then you rank the observations based on the value of another variable. Then what is the probability that the ranking by one variable is identical to the ranking by another variable?" Thus, if Kendall's τ is 100%, then for every pair of observations, the ranking by one variable is always identical to the ranking by another variable. If Kendall's τ is negative 100%, then for every pair of observations, the rankings by two variables are opposite. If Kendall's τ is 0, half the time the rankings by two variables are the same, and the other half of the time the rankings are opposite.

Kendall's τ needs to be adjusted for the panel structure of the data as well. We calculate the numerator (the number of pairs for which the rankings by two variables are the same minus the number of pairs for which the rankings by two variables are the opposite of each other) and the denominator (the total number of pairs) for each period and then aggregate over time. These techniques can be used to group factors or to reduce the number of significant factors in a model.⁴¹

4.7 CONCLUSION

In this chapter we went through a wide range of factors for QEPM models. Portfolio managers should have a good understanding of why each factor may or may not be helpful in predicting stock returns. As we have reiterated several times, what distinguishes a portfolio manager from the crowd is the combination of factors he or she chooses for stock return models. There is no automatic way to select the best factors. We presented a number of statistical methods to aid the selection process, but they do not substitute for

human judgment. Theoretical justification for including a factor in a model is as important as statistical proof. Finally, factor choice sometimes comes down to a matter of preference. Great managers, just like great chefs, often add things “to taste.” In [Chapter 5](#) we will examine the signature stock screens of some well-known managers who combined factors according to their individual investment styles.

¹ See [Table 2.2](#).

² In this chapter, when we characterize a ratio as “high,” we generally mean that it falls within the top quartile or top decile of the investment universe. When we describe a ratio as being “low,” we generally mean that it falls within the bottom quartile or decile of the universe.

³ For those readers who replicate the calculations, some notes about our calculations will help. First, we use quarterly data and lag the data to avoid look-ahead bias, as discussed in [Chapters 6](#) and [16](#). Thus, for December 2020, we use firm financial data from September 2020, with the exception of market data like prices. Second, when we discuss the S&P 500 averages, we remove companies that do not have a share code of 10, 11, or 12 in CRSP. This effectively removes REITs from our S&P 500 index. Thus, our averages might differ from the actual S&P 500. For industry and sector averages, we use the GICS code of the respective industry for only stocks in our S&P 500. Third, it must be remembered that many of the average statistics of the indices are weighted averages by the float-weighted capitalization, which is different than a straight average.

⁴ In the Practical Application section of this book (Part V; see [Chapters 16](#) and [17](#)), we give some idea of the correlation and relationships between many common factors.

⁵ Some portfolio managers also use the inverse of P/E for evaluating stocks since it allows for the consideration of negative-earnings companies. In footnote 2 of [Chapter 5](#) we expand on the reasons for choosing the inverse of certain ratios.

⁶ Another way of calculating this ratio is to divide the PEG ratio by the stock’s dividend yield.

⁷ The consumer staples sector of the S&P 500 has about 30 companies in it. If one compares Walmart to the smaller industry set of food and staple retailing, which includes companies like Walgreens and Costco, then Walmart’s inventory turnover ratio is slightly lower than the industry average of 11.78.

⁸ A strand of literature has noticed that while the market may treat all earnings the same, there is a difference between the accrual and cash-flow components of earnings, and it has an impact on future stock returns. In particular, the higher the accrual component of earnings, the lower are the future stock returns. For more information, see Sloan (1996).

⁹ The work of researchers such as Professor Edward Altman of New York University’s Stern School of Business, who used financial ratios to predict bankruptcies, has encouraged the QEPM community to look closely at financial risk factors. Financial risk factors are closely related to solvency factors. See Altman (2019).

¹⁰ See [Table 2.2](#).

- 11 There are thousands of potential technical factors. For good descriptions of many technical indicators and how to use them, we recommend Achelis (2001).
- 12 This type of data is not easily available.
- 13 For an expanded list, see [Table 4A.9](#) and [Chapter 16](#).
- 14 See [Table 2.2](#).
- 15 See Jegadeesh (1990).
- 16 The reader should remember that many technical signals, like the Bollinger band, can be computed on various time horizons but also that these signals are for the very short term and must be updated frequently. For more details on how we computed this indicator, see the formulas in [Chapter 16](#).
- 17 Also known as the *low-beta strategy* or *betting against beta*.
- 18 For an expanded list of volume factors, see [Table 4A.10](#) and [Chapter 16](#).
- 19 In 2021, a group of social network-connected amateur traders made millions targeting high-short-interest stocks to buy, including Gamestop (GME), AMC Entertainment (AMC), and others. As they bought the highly shorted stocks, these stocks soared in price.
- 20 Arbitrage pricing theory (APT) was invented by the late Stephen Ross of MIT, and his suggested factors include macroeconomic factors such as monthly growth in industrial production, inflation and expected inflation, real interest rates, risk premiums (measured as the spread in returns between Baa bonds and government bonds), and the term structure or difference in returns between long-maturity government bonds and short-term Treasury bills. See Chen, Roll, and Ross (1986).
- 21 The original arbitrage pricing models assumed that the factor premiums should be unanticipated variables because it is only unanticipated changes that should affect returns. For macroeconomic variables, this usually means constructing derivatives of the variables that are unanticipated or have an average of zero. Most economic variables do not have an average of zero. This does not necessarily create a problem, however. We can simply interpret the economic factors as something that is proportional to the true factor premiums. We also could attempt to construct economic factors that are more like innovations, such as constructing unexpected inflation variables.
- 22 IBES is the main provider of analyst data, although Bloomberg and other data providers provide similar information. IBES is owned by Refinitiv, which was purchased by the London Stock Exchange Group in 2019.
- 23 For a list of some of those studies, see [Table 2.2](#).
- 24 StarMine, which is now owned by Refinitiv, has made a name for itself by building models that weight analyst recommendations according to analysts' historical ability to predict stock returns, taking into account preannouncements and using changes in recommendations to filter the information efficiently. The company finds that its models improve on the basic earnings revision model and have much higher information coefficients than the basic model. For more information, see <https://www.refinitiv.com/en>.
- 25 See Thornton et al. (2020).
- 26 See Thornton et al. (2020) for a list of some companies that sell these types of data.
- 27 For example, using FlightRadar24 (<https://www.flightradar24.com/>) or something similar.

- ²⁸ See Hamilton (1993), Cohen (1995), Diltz (1995), Kurtz and DiBartolomeo (1996), Guerard (1997), Guerard et al. (2002), Derwall et al. (2005), and Whelan et al. (2020).
- ²⁹ Since the writing of the first edition of this book, new terminology has gained vogue, using the acronym ESG for environmental, social, and governance factors. Incorporating ESG factors is also called *sustainable investing* or *impact investing*. In addition, many more studies on this topic have been written, including Madhavan et al. (2020), Amel-Zadeh and Serafeim (2018), Briere et al. (2017), Pastor et al. (2021), Munoz et al. (2014), Berg et al. (2020), and Ezeokoli et al. (2017). Also, many of the leading investment banks are regularly publishing pieces on ESG, including J.P. Morgan, Sanford Bernstein, Bank of America Merrill Lynch, and others. One must also remember that investors can trade either side of these factors. For example, those who don't agree with diversity can reduce the number of diversity-proponent companies in their portfolios, while those who agree with diversity can increase the number of these companies in their portfolios.
- ³⁰ MSCI ESG KLD STATS is a database that classifies companies according to social criteria. The MSCI ESG database provides in-depth profiles of more than 3,000 U.S. corporations, including every company in the S&P 500, most of the Russell 3000, and all of KLD's Domini 400 Social Index. The profiles present analyses on a range of issues that reflect a company's overall social record and include social ratings that highlight a company's strengths and weaknesses. The following issues are covered: community relations, employee and union relations, environmental liabilities, corporate governance, human rights, environmental impact, diversity, product quality and safety, environmental policies and practices, and company involvement in businesses related to abortion, contraceptives, military weapons, adult entertainment, firearms, nuclear power, alcohol, gambling, and tobacco. This database is maintained by MSCI after it bought KLD and its competitor, the Investor Responsibility Research Center (IRRC). There are other providers of social responsibility data, including Bloomberg, Sustainalytics, Vigeo Eiris, RobecoSAM, and Eikon. More information on all of these data providers can be found at <https://www.msci.com/>, <https://www.bloomberg.com/>, <https://www.sustainalytics.com/>, <https://vigeo-eiris.com/>, <https://www.robeco.com/>, and <https://www.refinitiv.com/>, respectively.
- ³¹ For more details on the measures, the reader should see Table 4A.14 and Chapter 16.
- ³² In recent years, it has been popular to consider socially responsible, or ESG, factors regardless of the future returns of stocks. Even when these factors are used in this fashion, the investor must be extremely cautious in interpreting the data. First, many providers of socially responsible data differ on their analyses of the same companies [Berg et al. (2020)]. Second, when investors choose criteria for their particular social concern, the screen brings up companies or sectors that would not even have this issue as part of their business—which begs the question of whether one is really doing anything like impact investing. Third, it is not always clear if social screening criteria are really Pareto improving. That is, while investing based on that view may be positive from a particular investor's point of view, it might be detrimental to an entire other class of investors.
- ³³ See Chapter 2.
- ³⁴ For an additional discussion on testing the out-of-sample predictability of factors, see Chapter 9 of Campbell et al. (1997).
- ³⁵ Simple regression is a regression of one variable on another. Multiple regression is a regression of one variable on many other variables.

- ³⁶ When we typically estimate these factor models, we use information at time t to predict returns at time $t + 1$. Sometimes, in this book, for convenience, we write the equation with time subscript t everywhere, as we do here. The factor exposure and the returns are at time t . It should still be interpreted as using information at t to predict returns at time $t + 1$. In other words, the factor exposure at time t is really the factor exposure value as of the end of the previous period, $t - 1$, which we may refer to sometimes as the beginning of month exposure for time t .
- ³⁷ Appendix C describes some of the potential problems with including too few or too many factors in a factor model. Appendix C can be found at www.ludwigbc.com under QEPM Exclusive Content.
- ³⁸ Unfortunately, we have met many portfolio managers who do this without understanding its implications. Most econometric software packages actually have procedures that do this very easily. Even if one understands the pitfalls of stepwise regression, it may be difficult to avoid stepwise regression completely. The point that we want to make is that one should be cautious interpreting the reported statistical significance.
- ³⁹ We provide programs in R, MATLAB, and Stata that actually create these portfolios with real data. See https://ludwigbc.com/books/qepm/exclusive_qepm_content_2020/. We also discuss this topic more in Chapter 16.
- ⁴⁰ When considering the testing of many factors, researchers may wish to adjust the t -statistics as explained earlier. Some suggestions for how to do this are contained in Appendix 4B.
- ⁴¹ Other techniques for reducing factors are also discussed in Chapter 6 of Campbell et al. (1997), Bai and Ng (2003), and Barillas and Shanken (2018).

APPENDIX 4A

Factor Definition Tables

TABLE 4A.1

Fundamental Valuation Factors

Factor	Description
Dividend yield (DY)	The dividend per share of a company divided by its stock price. High dividend yield stocks are also known as <i>dogs</i> . High dividend-yielding stocks have been shown to outperform over the long run.
Enterprise-value-to-earnings-before-interest-taxes-depreciation-and-amortization ratio (EV/EBITDA)	Also known as the <i>enterprise multiple</i> . Looks at a firm from a buyer's perspective because it takes debt into account; it is used to find attractive takeover targets. Low ratios may indicate an undervalued company. Also good for comparisons of companies in different countries because it ignores tax effects.
Price-to-book ratio (P/B)	The price of the stock divided by book value of common equity per share. Low P/B ratio stocks are commonly referred to as <i>value stocks</i> and high P/B ratio stocks as <i>growth stocks</i> . Low P/B ratio stocks have been found to outperform high P/B ratio stocks over the long run.
Price-to-cash-flow ratio (P/CF)	The price of a stock divided by its cash flow per share (various measures of cash flow). Lower P/CF ratio companies have been found to outperform high P/CF ratio firms. Although P/CF ratio is correlated with P/E and P/B ratios, some portfolio managers use it because cash flow is less vulnerable to accounting manipulation.
Price-to-earnings ratio (P/E)	Price of a stock divided by the earnings per share. Low P/E ratio stocks are known as value stocks. High P/E ratio stocks are known as growth stocks. It has been found that low P/E ratio stocks outperform stocks with higher P/E ratios over the long run.

Price-to-earnings-before-interest-taxes-depreciation-and-amortization ratio (P/EBITDA)	The price of a stock divided by its earnings before interest, taxes, depreciation, and amortization. It is computed by adding depreciation and amortization to the earnings before interest and taxes. Lower P/EBITDA ratio companies have been found to outperform high P/EBITDA ratio firms. It is used because it is a proxy for cash flow; however, it suffers some problems, including variations in accounting methods, cash required for valuation purposes, debt service and other fixed-charge requirements, and the need to maintain productive capacity.
Price-to-earnings-to-growth ratio (PEG)	Price-to-earnings ratio divided by earnings growth rate. Historical earnings growth rates or forecasted consensus analyst growth rates are usually used. It incorporates anticipated growth rates of a company to adjust P/E ratios for different potential earnings' growth rates. EV/EBITDA, P/EBITDA, and P/E neglect a company's growth rate, which can be vastly different across companies.
Price-to-earnings-to-growth-to-dividend-yield ratio (PEGY)	The price-to-earnings-growth ratio divided by the dividend yield of the stock. Some analysts use P/E ratio divided by earnings growth plus dividend yield.
Price-to-sales ratio (P/S)	The price of a stock divided by its sales per share. This ratio is very industry sensitive. Low P/S ratio stocks have been found to outperform high P/S ratio stocks.

Note: Oftentimes, in quantitative research, in order to have a richer dataset, the inverse of many of these ratios is used (e.g., EBITDA/EV, B/P, CF/P, E/P, EBITDA/P, 1/PEG, 1/PEGY). For example, a company with zero earnings or negative earnings would make the relative comparison of P/E ratios confusing.

TABLE 4A.2

Fundamental Size Factors

Factor	Description
Common equity (CE)	Common equity, or book value of equity, is a measure of the size of a company.
Market capitalization (SIZE)	The market capitalization of the shares outstanding, which is the price of a share multiplied by the number of shares outstanding. Commonly referred to as <i>size</i> . Smaller size stocks have been found to outperform larger size stocks over the long run.
Total assets (TA)	Total assets represent the short- and long-term assets of a company. This includes cash, marketable securities, inventory, fixed assets like buildings, and more.

Note: Oftentimes, in quantitative research, due to the skewness of these variables, $\log_e(CE)$, $\log_e(SIZE)$, or $\log_e(TA)$ is used.

TABLE 4A.3

Fundamental Operating Efficiency Factors

Factor	Description
Cash conversion cycle (CCC)	Accounts receivable turnover in days plus inventory turnover in days minus accounts payable turnover in days. It is the time between outlay of cash and cash recovery. A company may have a negative cash conversion cycle because it is able to get money from customers quicker than it has to pay its creditors. This can be a sign of strong power in the industry.
Cost management index (CMI)*	Cost of goods sold plus selling and administrative fees divided by operating revenue.
Equity turnover (ET)	This is defined as net sales divided by average shareholders' equity.
Fixed asset turnover (FAT)	This is defined as net sales divided by average fixed assets. This reflects the level of sales from investments in productive capacity. For a variety of reasons, it can be rather erratic, including the discreteness of fixed-asset investment, the particular stage in a company's development, depreciation, and the timing of fixed-investment purchases.
Inventory turnover (IT)	This is equal to the cost of goods sold divided by average inventory. This is a measure of the efficiency of inventory management. A higher ratio indicates that inventory turns over rapidly from the time of acquisition to sale. This ratio is affected by the choice of accounting method, and, thus, when comparing companies, the analyst should make sure that the accounting methods are similar.
Receivables turnover (RT)	This is defined as net sales divided by average receivables. The calculation method may differ depending on the analyst, but an analyst may use the last 2 years (or more or less data) of receivables and average them. This measures the effectiveness of credit policies of the firm. It also is an indication of the level of investment necessary to maintain the firm's sales level.
Total asset turnover (TAT)	This is defined as net sales divided by average total assets.

*When dealing with banking or financial firms, an alternative measure of cost management might be used. The cost management index for banks is defined as noninterest expense divided by net interest income (interest income minus interest expense) plus noninterest income.

TABLE 4A.4

Fundamental Operating Profitability Factors

Factor	Description
Accruals (ACC)	Accruals are revenues earned or expenses incurred that impact a company's net income on the income statement, although cash related to the transaction has not yet changed hands. It has been documented that buying stocks with low accruals and selling stocks with high accruals produce excess returns.
Cash flow to total assets (CFTA)	This is defined as operating cash flow divided by total assets.
Cash flow to sales (CFTS)	This is defined as operating cash flow divided by total sales.
Gross profit margin (GPM)	This is defined as net sales minus the cost of goods sold divided by net sales. This measures how profitable a firm is. This is a historical measure. Portfolio managers implicitly assume that some form of this measure will continue into the future.
Gross profits to total assets (GPTA)	This is defined as gross profits divided by total assets.
Net profit margin (NPM)	This is defined as net income divided by net sales. It is known as the "bottom line." This represents the true income of the firm. A very important measure of profitability for companies.
Net profit to total assets (NPTA)	This is defined as net income divided by total assets.
Operating profit margin (OPM)	This is defined as operating profit (net sales minus the cost of goods sold and the sales and general administrative expenses) divided by net sales.
Operating profits to total assets (OPTA)	This is defined as operating profit (net sales minus the cost of goods sold and the sales and general administrative expense) divided by total assets.
Return on assets (ROA)	This is defined as net income plus after-tax interest cost divided by average total assets. Or it is defined as earnings before interest and taxes divided by average total assets. This measures management's ability and efficiency in using the firm's assets to generate profits.
Return on common equity (ROCE)	This is defined as net income minus preferred dividends divided by average common equity. This is a measure of return directly related to common equity holders.
Return on owner's equity (ROE)	This is defined as net income divided by average total equity. It measures the return to common and preferred shareholders.
Return on total capital (ROTC)	This is defined as net income plus gross interest expense divided by average total capital. This measures the total return to investors on their investment with the firm. It includes both debt and equity capital. Thus it's a return to all investors in the proportion of their contribution.

Note: Although we do not directly analyze them in this book, there are two additional measures that can help with company assessment. One is the return on net financial assets (RONA). This is computed as net financial income divided by average net financial assets. Net financial income (after tax) is interest income minus interest expense multiplied by one minus the tax rate, and average net financial assets is financial assets minus financial liabilities. This is the complement to the return on net operating assets. The other is the return on net operating assets (RNOA). This is computed as net operating income divided by average net operating assets. Net operating income (after tax) is net income minus net financial income, and average net operating assets is net assets minus net financial assets. Financial assets are cash and marketable securities, while financial liabilities are notes payable, short-term debt, current portion of long-term debt, long-term debt, and preferred stock. This measure is useful for nonfinancial companies in understanding that the returns to the business of the firm are independent of its financial activities. The actual way of computing these variables can vary by analyst.

TABLE 4A.5

Fundamental Solvency Factors

Factor	Description
Cash-flow-from-operations ratio (CFOR)	This equals cash flow from operations divided by current liabilities. Measures liquidity by looking at actual cash flows instead of current and potential cash resources.
Cash ratio (CR)	This is equal to the sum of cash and marketable securities divided by current liabilities. If inventories and/or accounts receivable are not liquid, then this is a better measure of solvency.
Current ratio (CUR)	Current assets of the firm divided by current liabilities. If accounts receivables and inventories are relatively liquid, then this ratio is a good measure of solvency. Firms that are more solvent have less likelihood of going into bankruptcy and thus can withstand paying their current liabilities.
Quick ratio (QR)	This is equal to the sum of cash, marketable securities, and accounts receivable divided by current liabilities. This measure is also known as the "acid test." It's a better measure of liquidity or solvency, especially when inventories are not that liquid. It is essentially the current ratio minus the ratio of inventories to current liabilities.

TABLE 4A.6

Fundamental Financial Risk Factors

Factor	Description
Cash-flow coverage ratio (CFCR)	This is defined as the cash flow from operations divided by the total debt of the firm. It measures the ability of cash flows to cover debt obligations.
Debt-to-equity ratio (D/E)	The long-term debt of a company divided by its common equity. This is a very common measure of the financial risk of a firm. Many portfolio managers view a high D/E ratio relative to the universe as a bad thing, all else equal.
Financial leverage ratio (FLR)	This is measured as the total assets of the firm divided by the total common equity.
Interest coverage ratio (ICR)	This is defined as the earnings before interest and taxes divided by the interest expense. Also known as the <i>times interest earned</i> . It gives an investor an idea of financial risk because it indicates how well interest payments are covered by the current earnings of the firm before taxes and, obviously, interest. It gives an indication of how capable the firm will be in making interest payments. A high ratio is comforting because even if the company struggles but is ultimately a good company, it will unlikely default on an interest payment, causing even more problems.
Total debt ratio (TDR)	This is defined as the total debt (short- and long-term debt) of the firm divided by the total capital (total liabilities plus total equity).

TABLE 4A.7

Fundamental Corporate Activity Factors

Factor	Description
Insider purchases (IP)	This factor is the number of insiders making purchases on stocks. The more insiders purchasing the stock is a positive signal for the stock return. The more insiders selling the stock is a negative signal for purchasing the stock. It has been documented that if a stock is bought when there is insider buying, this leads to abnormal returns. Similarly, selling a stock when there is insider selling leads to abnormal returns.
Research-and-development-to-sales ratio (RNDS)	Measured as the dollar amount spent by the company on research and development divided by the dollar value of sales revenue for the company. A high value relative to the industry might indicate a company that is poised for strong future growth owing to the investment in research and development.
Stock buybacks (SB)	This factor looks at companies that are buying back their stock. One proxy for this is the change in Treasury securities for a company. It has been documented that companies that buy back their shares tend to have subsequent abnormal returns.

TABLE 4A.8

Technical Liquidity Risk Factors

Factor	Description
Amihud illiquidity (AIIQ)	A measure of illiquidity based on the ratio of the daily return of the stock to the daily volume. The idea is that the higher the daily return for a given volume, the less liquid is the stock due to price impact.
Crowding (CROWD)	Crowding is a recently identified risk in investing that might occur when multiple market participants begin to follow the same trade in such a concentration that liquidity becomes fragile and it alters the risk and return dynamics of the trade. There is no one definitive measure of crowding, but rather many proxies.
Invariance illiquidity (INVIL)	Invariance is a measure of illiquidity based on a theoretical model. This measure relates the illiquidity of a stock to a constant multiplied by the ratio of the volatility of a stock to its average dollar volume.
Modified Amihud illiquidity (MAILIQ)	A measure of illiquidity that is exactly the same as the Amihud measure, but rather than using close-to-close prices to measure daily returns, uses open-to-close prices.
Pastor and Stambaugh liquidity (PSL)*	A measure of liquidity through an empirical regression of volume on daily stock returns. It has been documented that this may be a risk factor in the financial markets.

Trading turnover (TT)	The percentage of outstanding shares of the security traded during a given period. A common measure is the average daily trading volume figure divided by the market capitalization of the company. The average daily trading volume is usually measured as an average of the last 90 days' trading volume. This measure allows a gauge of liquidity. For larger funds, it also places limits on how much of a certain stock can be traded in the portfolio.
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Note: Although we do not directly analyze them in this book, floating shares is another measure of liquidity. This measure of liquidity considers only the shares available to the public for trading. It is the number of floated shares multiplied by the price per share. It is highly correlated with market capitalization, but not always, and this is the more relevant measure since it represents the amount of shares that can be traded by the public. In fact, many indices, including the S&P 500, switched from market capitalization-weighted to float-weighted during the 2000s. Another factor that can be useful is the number of security owners. This represents the number of investors who own the stock. This can be important for liquidity. For most securities, it is not a concern since securities are so widely held. However, for some securities it can be very important since if a few institutional shareholders own the majority of the security, then buying/selling the security may prove difficult and make the security less attractive for portfolio management strategies. Compustat does provide a variable from 10K reports of companies, CSHR, which is supposed to be the number of common shareholders. CSHR can be very misleading as it usually but not always fails to include the number of shareholders who hold the shares in "street name" at their brokers.

*This factor could also be considered an "economic" factor because we must estimate each stock's exposure from the factor via regression.

TABLE 4A.9

Technical Price-Based Factors

Factor	Description
Beta (BETA)/Market (MKT)*	The historical beta of a stock. There are numerous ways to compute the beta of a stock. This factor is known amongst practitioners as <i>low beta</i> , reflecting the belief that low beta is associated with a higher risk-adjusted return. Because the beta of a stock is derived from regressing returns against the market factor, we also share this space with the market factor (i.e., the return of the market minus the risk-free rate).
Bollinger bands (BB)	A moving average of the stock price with two bands around it, one below and one above, representing some number of standard deviations away from the mean. The mean over k days is given by $\bar{P} = \frac{\sum_{i=1}^k P_i}{k}$ The upper band is given by: $\bar{P} + l \sqrt{\frac{\sum_{i=1}^k (P_i - \bar{P})^2}{k}}$, where l is the number of standard deviations from the mean, and k is the number of historical days used in the Bollinger band. The lower band has the same formula, but the second term is subtracted. This concept is named after its inventor, John Bollinger. The idea is that one should not just buy or sell a stock when its price crosses its moving average but rather when its price moves away from its moving average by a sufficient amount (e.g., 1 standard deviation).
Channel breakouts (CB)	Channel breakouts are defined as conditions where a channel is breached. The channel is formed when the high price over the last n days is within $x\%$ of the low price for the same period. When a channel is followed by a new high level, $y\%$ above the old high, it signals a buy. If the opposite occurs, it signals a sell.
Momentum (MOM)	A positive return of a stock over a specified time horizon. There are numerous ways to compute momentum. One of the simplest ways is just the k -month total return. Studies have documented that stocks with high relative momentum have subsequent high relative returns.
Moving average (MA)	An average of the stock's price over a specific window in the past. For example, if you want a 60-day moving average, simply average the stock price over the last 60 days. Doing this every day creates a history of 60-day moving averages for the stock. The moving average can be weighted in various ways. Investors usually use the moving average as follows: When the actual stock price today rises above the moving average, this is a buy signal. When it falls below the moving average, this is a sell signal.
Moving-average convergence–divergence indicator (MACD)	Originally defined as a 26-day exponential moving average minus a 12-day exponential moving average. This trend following the momentum indicator shows the relationship between two moving averages. It was developed originally by Gerald Appel. There are many ways to interpret this indicator. One method is to buy when it goes above the zero line and sell when it goes below the zero line. It also can be used as an overbought/oversold indicator.

	line. It also can be used as an overbought/oversold indicator when it rises or falls very rapidly.
Price (P)	Some evidence has been found that stocks with low prices have high returns. One argument is that small investors feel that these stocks get them more. It's a psychological affordability. If stock price increments are limited to a certain movement, one might think that this could partially be responsible for these results. This factor is known amongst practitioners as <i>low price</i> .
Relative strength index (RSI)	The RSI is a measure of the strength of a security. This is an indicator that takes on values between 0 and 100. Generally, when the RSI is above 70, this is an overbought market, and one should sell the stock. An RSI reading of less than 30 indicates an oversold market, and one should buy the stock. This index was first introduced by Welles Wilder in an issue of <i>Commodities</i> magazine in June 1978. There are various ways to measure the RSI.
Support resistance (SR)	This measure looks for prices of stocks that are establishing new highs or lows. Generally, the investor constructs a look-back window, say, 30 days. If the price of the stock exceeds the previous high for the last 30 days, this is a buy signal. If the price of the stock falls below a previous low, this is a sell signal.
Volatility (VOL)	The historical volatility of a stock. There are numerous ways to compute the historical volatility of a stock. One of the simplest ways is to choose a look-back period and compute the standard deviation of returns. This factor is known amongst practitioners as <i>low vol</i> , reflecting the belief that low volatility is associated with a high rate of return.

*The beta factor is really like an "economic" factor in the sense that we must estimate regression equations of the returns of stocks on the market returns in order to obtain each stock's exposure, or beta. We left it in the technical category, since to compute it involves technical data and also so as not to confuse the reader by placing it in another category of data.

TABLE 4A.10

Technical Volume-Based Factors

Factor	Description
Odd-lot balance (OLBI)	This is a measure of the ratio of odd-lot sales to odd-lot purchases. Odd lots are transactions in a stock that are fewer than 100 shares. The idea is that small-investor money is “dumb” money. Thus, when the OLBI is high, this means that small investors are selling more than buying, and it’s a signal to buy. When the OLBI is very low, it’s a signal to sell. Some practitioners take a moving average of the OLBI.
On-balance volume (OBV)	If the security moves down, all the volume is considered down volume. The formula is $OBV_t = OBV_{t-1} + V_p$ if $P_t > P_{t-1}$, and $OBV_t = OBV_{t-1} - V_p$ if $P_t < P_{t-1}$. For zero price movements, the OBV remains the same. The investor should take an action when the OBV breaks out or changes trend. Thus, if the OBV were falling and it suddenly begins rising, this would be a buy signal.
Short interest ratio (SIR)	This measures the amount of short selling on a particular stock. Some people use a high degree of short interest as a contrarian signal. That is, when short interest is high, then a slight increase in the stock price might force the short sellers to buy the stock back, causing the price to increase even further. This is called a <i>short squeeze</i> . A general guide is to look at a stock’s short interest versus its historical average.
Volume lead (VL)	Volume is used in many technical indicators. One simple rule is that if the stock has a return on larger than average volume, it is overreactive and may reverse itself. If the stock has a return on lower than average volume, it is informative and may lead to subsequent returns in the same direction.

TABLE 4A.11

Technical Overall Market Movement Factors

Factor	Description
Advance–decline ratio (ADR)	This technical indicator measures the participation in market rallies. It is the ratio of the number of advancing issues (i.e., stocks whose stock price has increased) to the number of declining issues (i.e., stocks whose price has declined). Some investors smooth this ratio by taking a moving average of it. If the indicator is very high, this indicates an overbought situation and a sell signal. If the indicator is very low, this indicates an oversold condition and a buy signal.
Arms index (ARMS)	This indicator measure is a combination of two other well-known indicators and attempts to measure the flow of volume into stocks that are going either up in price or down in price. That is, it is computed by dividing the advance–decline ratio by the upside–downside ratio. It shows whether volume is flowing into advancing or declining stocks. A ratio below 1 indicates a buying opportunity, whereas a ratio above 1 indicates a selling opportunity. Many practitioners take a moving average of this variable and use that as their indicator. It is named after its developer, Richard W. Arms.
Upside–downside ratio (UDR)	This measure compares the volume of stocks that have upward price movements with the volume of stocks that have downward price movements. It is computed by dividing the daily volume of advancing stocks by the daily volume of declining stocks.
Upside–downside volume (UDV)	This measure compares the volume of stocks that have upward price movements with the volume of stocks that have downward price movements. It is computed by subtracting the total volume of stocks that increase from the total volume of stocks that decrease. Typically, the investor can look at the historical average of this variable. When the UDV is very positive, this is a buy signal. When it is very negative, this is a sell signal.

TABLE 4A.12

Economic Factors

Factor	Description
Consumption growth (PCG)	This measures the growth in real personal consumption in the economy based on the breakdown of GDP numbers.
Change in commodity prices (CIG)	This measures the change in oil or commodity prices in the economy. Typically, the percentage change in oil prices is used or the return on some commodity index is used.
Corporate bond spread [e.g., BBB spread (BBBS)]	This attempts to measure the risk premium in the economy as derived from corporate bond spreads. It can be defined as the spread between lower-grade corporate bonds (BBB or lower) and U.S. Treasury bonds. The higher the spread, the greater is the risk premium being paid in the marketplace for riskier securities. It can also be measured as the return of lower-grade bonds minus the return of U.S. Treasury bonds of a similar duration. Sometimes swap spreads, the lending rate of banks on collateralized loans versus the U.S. government interest rates, are used as well.
Industrial production (IPG)	This attempts to measure the growth rate of the economy. Usually, the change in industrial production or the growth in industrial production is used as the economic factor. The industrial production number is released monthly and therefore perhaps is more useful than GDP growth, which is released quarterly. Presumably, different stocks will have different exposures to different stages in the business cycle as measured by the growth in industrial production.
Inflation (CPIG)	This measures the monthly inflation rate in the economy. Typically, either the consumer price index (CPI) or the producer price index (PPI) is used to measure inflation. Some use the actual inflation rate as the economic factor, and some attempt to build a series of unexpected inflation rates and use this as the economic factor. Presumably, different stocks behave differently during different inflation cycles.

Real GDP growth (GDPG)	This measures the growth in real GDP from quarter to quarter. Some people also use the survey of economic forecasts to compute the innovation or surprise growth in real GDP.
Slope of yield curve (e.g., 10 year rate minus 2 year rate (TP2Y))	This measures the slope of the yield curve. Usually it is measured as the 10-year yield minus the 2-year yield or the 10-year yield minus the 3-month Treasury bill yield, although other variations are possible. The yield-curve slope is a proxy for measuring expected changes in the future state of the economy according to bond market participants.
Unemployment rate (UR)	This measures the unemployment rate in the economy. Some practitioners also may use the change in the unemployment rate.

TABLE 4A.13

Alternative Analyst Factors

Factor	Description
Analyst rating— median rating (ARMR)	A median rating of all analysts on a particular stock. Each analyst rating is ranked 1–5. 1 = strong buy, 2 = buy, 3 = hold, 4 = sell, and 5 = strong sell.
Analyst rating— percent buys (ARPB)	This represent the percentage of analysts that recommend a buy or better on the stock.
Analyst rating— percent sells (ARPS)	This represents the percentage of analysts who recommend a sell on the stock.
Analyst rating—SD of ratings (ARSD)	This is the statistical measure of standard deviation of all analyst recommendations in a cross section. It indicates the degree of heterogeneity in the analysts' forecasts.
Analyst rating— percent downgrades (ARPD)	This represents the percentage of analysts whose ratings on the stock have decreased in the last month.
Analyst rating— percent upgrades (ARPU)	This represents the percentage of analysts whose ratings on the stock have increased in the last month.
EPS forecast— normalized SD (EPSSD)	This factor represents the median EPS estimate by all analysts covering the stock divided by the statistical standard deviation of all EPS estimates for the stock. Since EPS estimates will vary by type of company, this measure normalizes the measure for interpretation.
EPS revision— percent downward (EPSFPD)	This factor represents the percentage of analysts who have decreased their EPS estimates on the stock over the last month.

EPS revision— percent upward (EPSFPU)	This factor represents the percentage of analysts who have increased their EPS estimates on the stock over the last month.
Number of analysts (EPSFN)	Also called the <i>indicator of neglected firms</i> . The factor is the number of analysts covering a firm. The idea is to choose firms with less analyst coverage. It has been documented that firms with low analyst coverage or low institutional ownership tend to have higher risk-adjusted returns. There is some controversy over whether this is a small-cap effect or something else.
Standardized unanticipated earnings (SUE)	$SUE = \frac{EPS^a - \overline{EPS}}{S(EPS)}$, where $\overline{EPS}$ is the mean earnings-per-share estimates by analysts prior to the earnings announcements, $S(EPS)$ represents the standard deviation of analyst earnings-per-share estimates, and EPS^a is the actual earnings per share reported. It has been documented that companies that exhibit positive earnings surprises continue to perform well over the next 13 to 26 weeks; similarly, companies that exhibit negative earnings surprises continue to underperform into the future.

Note: Although we do not directly analyze them in this book, when detailed analyst information is available, it can also provide additional valuable information. High EPS and low EPS forecasts represent the highest and lowest analyst estimates for the EPS of a company. High and low recommendations represent the highest and lowest analyst recommendations for the company. In addition to these simple statistics, other statistics related to individual analyst data can also be valuable.

TABLE 4A.14

Alternative Social Responsibility Factors

Factor	Description
Community contribution (CC)	This factor measures how much the company contributes to charitable giving and other such causes.
Corporate governance (CG)	This factor measures how much the company compensates executives. Higher compensation is seen as a bad indicator.
Diversity (DIV)	This factor measures how much the company promotes women and minorities.
Employee relations (ER)	This factor measures whether the company shares profits with employees, promotes safety and the health of employees, and has a strong retirement program and strong union relations.
Environment (E)	This factor measures how much the company is involved in activities that attempt to reduce pollution or benefit the environment.
Human rights (HR)	This factor measures how well the company considers human rights.
Product (PRO)	This factor measures how much the company promotes the quality of its products, whether it's a leader in R&D for innovative products in the market, and whether the products have notable social benefits.

On Data Mining and Techniques to Adjust the Significance of Factors

One way to describe the danger of data mining is as follows: If we determine a factor to be statistically significant whenever the absolute value of the t -statistic associated with the factor is greater than 1.96, we are most likely overstating the statistical evidence. In many real-life situations, we do not test a factor in isolation but consider many factors jointly. A particular factor that we like may have been selected because of its large t -statistic in comparison with the conventional critical value of 1.96. This may not be the appropriate critical value. Such a comparison will lead us frequently to declare a factor to be significant even when the factor is in reality not significant. The probability of false discovery will be much higher than our intended target of 5% (i.e., our selected significance level).

One straightforward solution is to choose a higher critical value so that the probability of false discovery declines to 5% even when we focus on one factor out of many due to its sample t -statistic. While this sounds straightforward in principle, applying this idea in practice is not so easy. The biggest challenge is that, quite often, we are not able to determine the total number of factors under consideration. If another portfolio manager conducted the analysis and only reported a factor with a large t -value, we cannot know the total number of factors that he or she tested. Even if we conducted the analysis ourselves, our choice of candidate factors may have been influenced by our reading of others' research reports, and it is not easy

to decide how many factors were considered in the numerous research reports that we may have read over the years.

Despite these difficulties, we briefly describe two approaches to data mining that have been discussed in the academic literature. These approaches will not shield us from the danger of data mining completely, but may help us have a better sense of the magnitude of potential problems. The first approach, which we refer to as the *Bonferroni-type adjustment*, is based on the traditional, frequentist statistical theory in which the calculated t -statistic is compared to the distribution of all the t -statistics that we could have obtained assuming the truth about the null hypothesis. The second approach is based on Bayesian statistical theory in which our subjective probability of the null hypothesis being true is derived from the data and our personal beliefs.

4B.1 THE BONFERRONI-TYPE ADJUSTMENT

In the Bonferroni adjustment, the critical value for the t -test is determined by the total number of factors tested. The critical values are shown in [Table 4B.1](#). These values are computed in the following way. Let M be the number of factors tested. Let α be the significance level that we choose (e.g., 0.05). Then the critical p -value in the Bonferroni adjustment is calculated as α / M . From this critical p -value, we can recover the critical t -value as $t^* = \Phi^{-1}(1 - p^* / 2)$, where p^* is the critical p -value, Φ is the standard normal distribution function.

TABLE 4B.1

Critical t -Values for Bonferroni and Holm Adjustments

Number of Factors Tested	Bonferroni Critical t	Holm Critical t
10	2.81	2.81
20	3.02	3.02
30	3.14	3.13
40	3.23	3.22
50	3.29	3.28
60	3.34	3.33
70	3.38	3.37
80	3.42	3.41
90	3.45	3.44
100	3.48	3.47
200	3.66	3.65
300	3.76	3.75
400	3.84	3.82
500	3.89	3.88
600	3.93	3.92
700	3.97	3.96
800	4.00	3.99
900	4.03	4.02
1,000	4.06	4.04

Note: For the Bonferroni adjustment, you may simply use the reported critical values for each factor under consideration. For the Holm adjustment, you need to order all the factors from the one with the highest absolute t -value to the one with the lowest absolute t -value and then choose the j th factor such that $p_1 \leq \dots \leq p_j \leq \alpha/(M-j) < p_{j+1} \leq \dots \leq p_M$, where $p_1, \dots, p_M$ are the p -values for the M factors. The critical t -values reported correspond to the p -value of $\alpha/(M-j)$ assuming $j = 0.05M$.

The Holm adjustment, a variation of the Bonferroni adjustment, proceeds in a slightly different way. When M factors are considered, we may order these factors from the one with the highest absolute t -value to the one with the lowest absolute t -value. Thus, we may denote these t -values by $t_1 \geq \dots \geq t_M$. If we denote the p -values of these M factors, $p_1 \leq \dots \leq p_M$. To proceed with the Holm adjustment, we first need to determine the index j such that $p_1 \leq \dots \leq p_j \leq \alpha/(M-j) < p_{j+1} \leq p_M$. Once we determine j , we may conclude that the first j factors are significant. Note that the critical p -value, $\alpha/(M-j)$, is a function of the entire set of p -values and that it is not possible to convert this into a generic critical t -value. What we show in [Table 4B.1](#) instead is the t -value corresponding to $\alpha/(M-j)$ assuming $j = 0.05M$.

As mentioned earlier, the most difficult part of implementing the Bonferroni or Holm adjustment is the choice of the number of factors tested. Some researchers have suggested using the total number of published and unpublished research articles since 1967 for this purpose. By some counts, this is about 313 articles (or 316 factors) through 2012. The Bonferroni critical t -value corresponding to the previously discovered factors is 3.78. If the rate of research articles produced continues at this same rate, the Bonferroni critical t -value will be 4.00 by 2032.¹

4B.2 A BAYESIAN ADJUSTMENT

Consistently applying the Bayes rule to assess the subjective probability of a null hypothesis may alleviate the problems associated with data mining. In the Bayesian approach, the posterior probability of a null hypothesis being true is determined as the product of the prior probability of the null hypothesis being true and its likelihood (i.e., the probability of the observed data being generated when the null hypothesis is true). The same can be said about the posterior probability of an alternative hypothesis. Combining these two ideas leads to this statement: the posterior odds ratio (the posterior probability of the null hypothesis being true divided by the posterior probability of the alternative hypothesis being true) is the product of the prior odds ratio and the likelihood ratio. Among these three concepts—the posterior odds ratio, the prior odds ratio, and the likelihood ratio—the likelihood ratio can be calculated without reference to a particular individual's beliefs. Thus, this is a good starting point to construct a test statistic.

One potential choice to guide the portfolio manager is the minimum possible value of the likelihood ratio, determined by considering all the possible alternative hypotheses. This ratio might be called the *minimum Bayes factor* (MBF).² Once the value of MBF is determined, a new critical value of the t -test can be determined. At the significance level of 5%, a critical value of t is the one corresponding to the 95% posterior probability of the null hypothesis being true. Table 4B.2 presents the critical t -values for different prior odds ratios. The prior odds ratio of 0.01 indicates that the null hypothesis is highly likely with 99:1 odds. The prior odds ratio of 0.50 indicates that the null hypothesis is as likely as the alternative hypothesis. In

other words, the higher the odds ratio, the more the analyst or portfolio manager is convinced that the factor is important in predicting stock returns. A value of 0.99 or 99% would mean that he or she is almost certain of it.

TABLE 4B.2

Critical t -Values for a Type of Bayesian Adjustment

Prior Odds Ratio	MBF Critical t	SD-MBF Critical t
0.01	3.88	4.29
0.02	3.70	4.11
0.04	3.50	3.93
0.05	3.43	3.86
0.20	2.94	3.41
0.50	2.43	2.93

Note: The prior odds ratio is the prior probability of the null hypothesis being correct divided by the prior probability of the alternative hypothesis being correct. The minimum Bayes factor (MBF) critical t is relevant when there is no information about the appropriate alternative hypothesis; the symmetric, descending minimum Bayes factor (SD-MBF) critical t is relevant when one believes that the prior probability density for alternatives should be symmetric and descending around the null hypothesis.

QUESTIONS

- 4.1. What is a factor, and why is it important in QEPM?
- 4.2. What are the four basic categories of factors in QEPM?
- 4.3. What are some common fundamental valuation factors?
- 4.4. Suppose that there is a group of stocks, group A, with an average P/B ratio that is lower than the industry average, and another group of stocks, group B, that has a P/B ratio that is higher than the industry average. Which group would a growth portfolio manager choose? A value manager?
- 4.5. What would be more appropriate for a P/E ratio factor, a trailing P/E ratio or a forward P/E ratio? Explain.
- 4.6. Using concepts from Appendix B, at www.ludwigbc.com under QEPM Exclusive Content, about the dividend discount model, combined with the discussion in this chapter, answer the following questions:

- (a) Derive an expression for the P/E ratio in terms of the payout ratio k , the required rate of return on equity r_{ce} , and the growth rate of earnings g .
 - (b) Rearrange this expression so as to obtain the implicit growth rate of earnings implied by the actual P/E ratio of the stock and other parameters.
- 4.7. Why would fundamental solvency factors be included in a stock return model?
- 4.8. What might you expect to be more effective at explaining stock returns, gross profit margin or the change in gross profit margin? Explain.
- 4.9. All else equal, would you prefer a low D/E ratio or a high D/E ratio company in your portfolio? What are some other measures of fundamental financial risk?
- 4.10. (a) Why might fundamental liquidity factors be important for QEPM?
- (b) What do you think is more relevant in this regard, average daily trading volume, market capitalization, or float-weighted capitalization? Explain.
- 4.11. Technical factors are being used increasingly in QEPM.
 - (a) What is the main reason for using technical factors?
 - (b) Name four types of technical factors.
 - (c) Which type of quantitative portfolio manager is likely to use a technical indicator as a factor, one who rebalances monthly or one who rebalances quarterly?
- 4.12. Given the data in the following table on the stock AALC, please do the following:
 - (a) Compute the mean, upper, and lower bands of a Bollinger band with two standard deviations (i.e., $l = 2$) and a 10-day rolling window (i.e., $k = 10$) for the dates July 26, 2020–July 29, 2020.
 - (b) Suppose that the quantitative portfolio manager is a contrarian. Thus she believes that markets overreact to information. Should she buy or sell or do nothing on each of those days according to the Bollinger band?
 - (c) Using the same data, compute the relative strength index (RSI) for each of those days, assuming a nine-day Wilder period.

- (d) Some technicians would argue that the Bollinger band should not be used in isolation but in conjunction with other indicators, such as the RSI. Given this philosophy, would you buy or sell or do nothing on this stock on July 26, 2020?

Date	Price
7/13/2020	11.31
7/14/2020	11.41
7/15/2020	11.28
7/16/2020	11.57
7/19/2020	11.43
7/20/2020	11.41
7/21/2020	10.92
7/22/2020	10.51
7/23/2020	10.35
7/26/2020	9.35
7/27/2020	9.65
7/28/2020	10.01
7/29/2020	9.67

- 4.13. (a) What are three of the biggest drawbacks of using macroeconomic factors?
(b) What is one of the biggest benefits of using economic factors in stock return models?
- 4.14. A quantitative manager is collecting data for use with a stock return model. He finds that there is a positive relationship between this month's inflation and stock returns. What might you be worried about in use of these data?
- 4.15. Many factors are lumped into an area called *alternative factors*. Please respond to each of the statements below.
(a) Analyst factors are very useful because they incorporate a lot of analysis of a company into one number.
(b) Analyst factors were more useful prior to 2000 than they are today due to the SEC rule known as Reg FD (fair disclosure) that became effective in October 2000.

- (c) A change in analyst recommendation describes stock returns better than the recommendation itself (strong buy, buy, sell, etc.) does.
 - (d) Downgrades of a stock by some analysts probably serve as more useful factors than downgrades of a stock by all analysts.
 - (e) What are standardized unanticipated earnings (SUE)? Why might this be useful as a factor? (*Hint*: Use the idea of information diffusion in your explanation.)
- 4.16.** Does investing in socially responsible companies hurt portfolio returns?
- 4.17.** What kinds of tests or techniques can an analyst use to choose factors for
- (a) a fundamental factor model?
 - (b) an economic factor model?
- 4.18.** Why might univariate factor tests not be sufficient to determine the factors in a fundamental model?
- 4.19.** True or false? The stepwise regression technique is very powerful in identifying the optimal combination of factors for use in a stock return model. Portfolio managers always should use it.
- 4.20.** What might the Kendall statistic or other rank-correlation statistic be used for?
- 4.21.** From a cross-sectional regression of stock returns on the exposure to factor A , we obtained the slope estimate of 0.5 and the standard error of 0.2.
- (a) If you apply the conventional t -test, will you conclude that factor A is a statistically significant factor?
 - (b) Prior to running the regression, you have read through 100 research reports regarding 100 different factors including factor A and have learned that factor A has the highest t -statistic. In fact, this is why you became interested in factor A . Thus, your regression analysis is likely to be criticized as a data-mining exercise. In response to such criticism, you decided to apply the Bonferroni adjustment described in [Appendix 4B](#). Will you conclude that factor A is a statistically significant factor?
 - (c) Instead of applying the Bonferroni adjustment, if you adopt the Bayesian adjustment discussed in this chapter, will your answer be different?

¹ See Harvey et al. (2016).

² See Harvey (2017).

CHAPTER 5

Stock Screening and Ranking

Our life is frittered in detail ... simplify, simplify.

—Henry David Thoreau

5.1 INTRODUCTION

The aim of quantitative equity portfolio management (QEPM) is to refine investing to a reliable, repeatable, quantitative method that avoids the biases and errors of purely qualitative stock picking. *Stock screening* is the ranking of stocks within the investment universe such that they are separated easily into those that are more favorable and those that are less favorable according to the views of the portfolio manager. Screening also can be used to narrow the investment universe. There are two kinds of stock screening, sequential and simultaneous. In *sequential* screening, the portfolio manager prioritizes his or her stock-picking criteria. He or she eliminates stocks from the investment universe first based on the most important criterion, then based on the second most important criterion, and so on and so forth until he or she has reduced the investment universe to a list of stocks that meet all his or her criteria. The more advanced version of stock screening is multifactor simultaneous screening. In *simultaneous* screening, the manager applies all the criteria to the investment universe at once. As a result, all the stocks in the universe receive rankings according to their overall scores on the entire set of criteria.

One of the most popular simultaneous screening methods is what we call the *aggregate Z-score approach*. It involves choosing factors that explain stock returns and aggregating the factors into one score with which to rank the stocks in the investment universe. A Z-score ranking either can serve as the basis for a whole model of stock returns or can add information to an existing model. The stand-alone Z-score model relates the return of a stock to its Z-score and is sometimes equivalent to the fundamental factor model that we introduced in [Chapter 3](#). If a manager is already using a fundamental or economic factor model, the Z-score can be added to the model as the constant term α when certain conditions are satisfied.

The stock screening and ranking methods in this chapter are useful starting points for people just beginning to build quantitative portfolios. They are not the most sophisticated tools of portfolio construction, but they are simple to apply because they do not require mastering a great deal of mathematics or statistics. Many practitioners also will find these approaches to be intuitive.

5.2 SEQUENTIAL STOCK SCREENING

Imagine for a moment that a portfolio manager has only one criterion for investing: He or she believes that low P/B ratio stocks provide the highest excess returns. To create his or her portfolio, the manager would compute the P/B ratios of all the stocks in his or her investment universe.¹ He or she then would rank all the stocks from lowest to highest P/B ratio. If the portfolio were to include 60 stocks, then the manager would invest in the 60 lowest P/B ratio stocks on the ranked list. Alternatively, the portfolio manager could have used the inverse of the P/B ratio, that is, the book-to-market ratio (a.k.a., the B/P ratio). Only the criterion for selection would change: the portfolio manager would pick the 60 highest B/P ratio stocks.² This rudimentary process is stock screening in its most basic form.

What are the attributes of a good stock screen? A good stock screen should be easy to automate. In other words, it should be easy to run the screen regularly as stock data change. The screen should be easy to replicate as well. Any analyst who has the screening procedure and the raw stock-specific data should be able to run the screen and come up with the

same stock list that another analyst would come up with using the same data.

A good stock screen, of course, should accurately reflect the portfolio manager's preferences. If the goal is to create a *value* portfolio, then the stock screen should use value factors, such as a high B/P ratio.

Finally, in a sequential screen, the factor screens should be run one at a time in order of importance. This means that the first screens should be for the factors that speak most directly to the manager's investment goals. If market cap is the most important factor in the manager's mind, then that should be the very first screen.³

Suppose that we would like to invest in the companies that have the highest positive profit margins (which we suppose we can capture by selecting the top 30% of the S&P 500 by net profit margin) and a third of those companies with the highest B/P ratios. We would first screen our investment universe, the S&P 500, by net profit margin and choose the top 150 companies (30% of 500). We would eliminate the other 350 companies from our investment universe. The next step would be to sort the top 150 stocks by the B/P ratio. We would keep the 50 with the highest B/P ratios and eliminate the remaining 100.

We implemented this screen of S&P 500 stocks using data as of December 2020. Table 5.1 shows some of the stocks selected by the screen. First, we chose the 150 stocks with the highest profit margins. It turned out that all 150 had positive net profit margins. The highest (147.9%) belonged to Bio-Rad Laboratories and the lowest (14.7%) to Keysight Technologies. Then we chose the 50 stocks in this group with the highest B/P ratios. Neither Bio-Rad nor Keysight made it through the B/P screen. Among the final 50 stocks, the lowest ratio was 0.23 for American Water Works (which is not included in the table), and the highest was 1.38 for People's United Financial. With this final list of 50 stocks, our hypothetical portfolio would have been complete.⁴

TABLE 5.1

Selected Stocks from a Sequential Stock Screen

Ticker	Book-to-Price Ratio (B/P)	Net Profit Margin (NPM)
PBCT	1.38	21.64
BK	1.07	24.10
FITB	1.06	18.21
AFL	1.04	20.87
KEY	0.99	16.61
CMA	0.96	16.17
TFC	0.96	17.01
BAC	0.94	19.50
MTB	0.91	21.49
STT	0.90	19.63

Note: These represent the preferred stocks according to the sequential stock screen. The ticker symbols are for the following companies: People's United Financial (PBCT), Bank of New York Mellon (BK), Fifth Third Bancorp (FITB), Aflac (AFL), KeyCorp (KEY), Comerica (CMA), Truist Financial (TFC), Bank of America (BAC), M&T Bank (MTB), and State Street (STT). The data are from December 2020.

5.3 SEQUENTIAL SCREENS BASED ON FAMOUS STRATEGIES

The portfolio manager's success lies in his or her ability to identify what it is that leads to superior stock returns. Many managers follow an investment style or philosophy that guides their choices, and they may think that this style or philosophy cannot really be quantified. Yet it is possible to devise quantitative stock screens from a wide range of investment approaches. In this section we examine some of the most successful equity managers' investment strategies—many of them qualitative or fundamental—and translate them into stock screens.⁵ The philosophies of each manager and the screens that we built from them are also presented in Tables 5A.1 through 5A.6 in [Appendix 5A](#).

Peter Lynch has come up several times already in this book, and as we have noted, he found success with a fairly qualitative portfolio strategy. However, in his books *One Up on Wall Street* (1989) and *Beating the Street* (1993), Lynch also expounds on certain quantitative methods that can be collected in a valuable screening methodology. The Lynch screen is a combination growth-and-value strategy that seeks a P/E ratio that is less

than the industry average but also desires a price-to-earnings-to-growth (PEG) ratio that is less than 1.0. This indicates that the P/E ratio multiple is less than the growth rate in earnings and that the underlying company is generating strong bottom-line growth given its current market valuation. The P/E ratio to dividend yield is limited to stocks with a ratio of less than 4.0, which helps to provide adequate income and downside protection against further market deterioration. By restricting the growth rate in earnings to greater than 0% but less than 50%, we are including stocks with positive earnings momentum and excluding stocks that Lynch would believe to be at risk for failing to meet expectations owing to overly optimistic forecasts. Stocks with higher growth also may be candidates for accounting manipulation. An insider buying-to-selling ratio of greater than 1.5 also may shed light on management's view of the future of the business. Insider selling can occur for a variety of reasons, including a diversification of holdings. However, insider buying generally occurs only when management is optimistic about the future of the company and believes that the share price is undervalued. Further screening criteria include a long-term debt ratio of less than the industry median, a market capitalization of less than \$5 billion, and institutional ownership of less than 50%. These criteria increase the possibility that the stocks are undervalued and improperly priced because they lack a large or sophisticated following of investors.

Warren Buffett is perhaps the most widely emulated investor in history. Numerous books have been written about his buy-and-hold strategy. By applying Robert Hagstrom's *The Warren Buffett Way* (1994) and perusing Berkshire Hathaway's annual reports, we were able to construct a simple screen based on Buffett's investment principles. We first decided to limit the screen to the top 30% market capitalization of listed equities on the New York Stock Exchange (NYSE), American Stock Exchange (AMEX), and NASDAQ, which would aid us in excluding small-cap securities. Buffett generally believes in purchasing stocks that have a strong competitive advantage and are capable of outperforming other companies and demanding a premium for their services. Most small-cap firms fail to possess such strong competitive advantages or the economies of scale necessary to compete against large industry titans. They therefore fail to generate the necessary free cash flow to fund expansion and wield pricing power. Buffett is extremely concerned with management's ability to operate

efficiently and produce superior returns on equity. High free cash flow indicates management's ability to generate sufficient cash to fund its operations and reward its shareholders for taking on risk. Companies that do not generate sufficient free cash flow may be suffering from liquidity crunches or failing to collect payments on their sales. Hence two screening categories include return on equity (ROE) that is greater than or equal to 15% and free cash flow that ranks in the top 30% of listed securities in the investment database. Net profit margins should exceed industry averages, and the current D/E ratio should be in line with or lower than the industry median. Finally, forecasted free cash flow for the next five years should be greater than the current market price of the stock when discounted back to the present. This may indicate that the stock is undervalued and provides a "margin of safety" for uncertainty over future returns.

The Lakonishok screen is a value screen based on the work of Josef Lakonishok, the William G. Karnes Professor of Finance at the University of Illinois at Urbana-Champaign. Professor Lakonishok is also the CEO and founder of LSV Asset Management and has published widely in many prestigious academic journals. The crux of the Lakonishok screen is that some companies go out of favor with the market owing to overly pessimistic forecasts that depress their share prices. There is a great potential for profit for the patient investor who can identify which companies eventually will rebound from such depressed prices. The screen aims to avoid small companies that may not possess the necessary financial strength or customer base to weather the downturn. Therefore, part of the screen restricts our investment list to companies with market capitalizations in the top 30% of the NYSE, the AMEX, and the NASDAQ. To screen for undervalued stocks, the price-to-earnings (P/E) ratio and the price-to-book (P/B) ratio are set below their respective industry median values. The consensus earnings estimate for the next fiscal year must be greater than the forecasted earnings estimate for the current year. This helps to eliminate companies that are expected to continue on a dismal trajectory. The final step of the Lakonishok screen is to find companies whose stock returns have performed better than the S&P 500 over the last few months. This upward momentum may indicate the beginning of the turnaround, and it will help to filter out companies whose stock prices will continue to languish.

Another value screen similar to the Lakonishok screen but with an increased emphasis on the underlying financial condition of the firm is a screen based on the work of Stanford Professor Joseph Piotroski. Professor Piotroski is a world-renowned expert in financial statement analysis, and the Piotroski screen seeks to uncover undervalued firms with the necessary financial strength to succeed. As in the Lakonishok screen, market capitalization is restricted to the top 30% of the NYSE, AMEX, and NASDAQ. However, this criterion now takes a backseat to other financial ratios. The P/B ratio is restricted to the lowest 30% of listed securities in the database (the investment universe). Return on assets (ROA) must be positive, and both the debt-to-equity (D/E) ratio and the long-term debt-to-equity ratio are evaluated so as to exclude companies with overly burdensome debt obligations and inadequate financing. The current ratio and the cash ratio must be improving from the previous fiscal year to offer adequate liquidity to fund current operations. Finally, the asset turnover must show an improvement on a year-over-year basis, which helps to avoid companies that have failed to manage their assets efficiently and realize adequate top-line growth.

A seasoned money manager, David Dreman was the founder and chairman of Dreman Value Management LLC (the value strategies and funds were purchased by Foundry Partners in 2016). He is the best-selling author of several prominent books on the low P/E ratio contrarian value investment approach, including *Contrarian Investment Strategies: The Next Generation* (1998) and *Psychology and the Stock Market* (1977). The Dreman screen attempts to adopt some of the crucial tenets of Dreman's contrarian philosophy. He favors mid- to large-sized companies because he feels that they have greater visibility in a market rebound, and by being highly scrutinized by a variety of investors, they are less susceptible to accounting manipulation. We therefore have restricted our market capitalization to the top 25% of the S&P 1500. The contrarian strategy seeks out-of-favor stocks, and thus we have limited our investment options to stocks whose P/E ratios are in the lowest 40% of the S&P 1500. This helps to identify undervalued securities. A requirement that the D/E ratio be less than 1 and that the total-liabilities-to-total-assets ratio be less than the industry average ensures that the companies have adequate financial strength. A dividend yield above the S&P 500 dividend yield offers a cushion and provides downside protection to companies in a prolonged

downturn. The final major criterion is that earnings growth must continue to outpace the S&P 500 and industry median levels on a current and forecasted basis. All these screening steps attempt to identify firms that may be trading below their historical price levels but still show potential for future development and bottom-line growth that has not yet been priced into the stock.

Another famous investor and one whom Warren Buffett studied extensively in developing his own investment philosophy is the late Philip Fisher. Fisher authored several books on his investment strategy, including *Common Stocks and Uncommon Profits* (1958) and *Conservative Investors Sleep Well* (1975). Fisher believed that the business cycle made annual sales growth inherently unpredictable and failed to convey the proper underlying ability of the business to succeed. Year-over-year metrics could be distorted easily by slight alterations in the business environment and therefore should be judged in light of other, longer-term trends. We therefore have decided to screen for companies whose sales have increased on a year-to-year basis for the last five years and whose three-year compound growth rates in sales are greater than their industries' median sales growth rates over the same period. This helps to smooth out the effects of the business cycle and include only companies that have consistently performed above average in a variety of economic conditions. Fisher also was interested in identifying growth that was a result of a low-cost advantage, strong marketing organization, a strong research and development enterprise, and excellent management depth with the utmost integrity. Growth was important, but only in the context of value, so we confined the PEG ratio to be greater than 0.1 but less than 0.5. This helps to ensure that the stock multiple neither gets ahead of the growth rate in earnings nor falls too far below it. Fisher believed that research and development (R&D) was an important component for maintaining a competitive advantage, but only if it could generate a high level of sales per R&D expense. Hence we have decided to add the criteria that R&D expenses as a percentage of sales be greater than the industry median ratio and that sales growth keep up with or exceed the growth in R&D expenses. Finally, we screen for companies that are not expected to pay a dividend and therefore are likely to reinvest any retained earnings for the future growth of the business.

Bill Miller was the CEO of Legg Mason Capital Management, Inc., a division of the Baltimore financial services firm Legg Mason (purchased by

Franklin Templeton Investments in 2020). He managed the Legg Mason Value Trust from 1982 to 2016, beating the S&P 500 in 21 of those 35 years. In 2016, he left Legg Mason and created Miller Value Funds. Miller adheres to a value strategy that aims exclusively for securities that trade below their intrinsic value, as indicated by his multifactor valuation analysis. His strategy differs from that of many value managers in that it focuses on cash earnings rather than on accounting earnings. Many of his holdings, including positions in several “new economy” growth stocks such as internet high fliers Google, Amazon, and Facebook (as of the end of 2020), continue to be controversial among “pure value investors” owing to their speculative and/or growth qualities.

The Miller screen attempts to capture the core principles of his strategy by combining various financial metrics aimed at discovering undervalued companies that possess the necessary competitive advantage to sustain a high level of growth. The first major criterion in the Miller screen is that the current market capitalization of the company cannot exceed three times the total estimated free cash flow for the next five years. This is a value ratio that attempts to eliminate companies whose current market valuation appears overly optimistic and exceeds the amount of discretionary cash the business is expected to generate over the next several years. Free cash flow also should be increasing on a year-to-year basis. This helps to reaffirm Miller’s emphasis on cash flows and cash earnings as the impetus behind the dynamics of valuation. The PEG ratio then is restricted to a level of 1.5 or below. A maximum of 1.0 may be too restrictive in this screen because we do not want to eliminate growth companies that may have P/E ratio multiples slightly higher than their growth rates but nevertheless still may possess vast potential that has failed to be discounted properly into the stock price. A skilled management team is highly crucial to success, and therefore, the Miller screen takes into account that gross margins should be increasing and that ROE should exceed the industry median. Finally, the long-term debt ratio should be below industry average levels to prevent future liquidity difficulties or cash-flow shortages.

The late Richard Driehaus was the founder of Chicago-based Driehaus Capital Management. It was one of the top small- to mid-cap money managers. His superior investment performance resulted in his acceptance to the elite *Barron’s* All-Century Team of fund managers, whose members include such luminaries as Peter Lynch and Sir John Templeton.

The Driehaus screen is structured around earnings and price momentum. The initial steps focus on year-over-year positive and increasing earnings growth. The next major aspect of the Driehaus screen incorporates earnings surprises. Earnings surprises to the upside are taken as a positive signal of a company's ability to generate income and capitalize on its position in the market. A failure to meet expectations is taken as an indication that the company is failing to execute its strategy and that it may be facing increasing costs or intensified competition. The screen looks for companies that have had positive earnings surprises over the last fiscal year, meaning that the actual rate of earnings growth has exceeded analyst estimates during the most recent time period. The strategy also values surprises in which the range or standard deviation of the estimates is tighter or more significant on subsequent returns. Price momentum then is evaluated to ensure that there is sufficient buyer pressure on the stock price to encourage further demand and maintain an upward trajectory. The screen requires that the current market price be greater than its 200-day moving average, which indicates a long-term uptrend in the stock. The 50-day moving average then is set at greater than the 200-day moving average so that we can eliminate stocks whose momentum has already peaked. The on-balance volume (OBV), a technical indicator that compares price to volume, then is used to determine if the momentum is a net accumulation of shares or a distribution. Finally, the screen requires that the company be followed by fewer than five analysts so that there is a good chance of improper valuation, giving the astute investor an opportunity to purchase neglected securities and profit immensely when the market recognizes their potential.

Several other screens listed in the tables in [Appendix 5A](#) include the Neff contrarian screen, the Muhlenkamp screen, the Templeton screen, and the Dogs of the Dow screen. The Neff screen draws on the fundamentals of legendary Vanguard Windsor Fund manager and former *Barron's* Roundtable member John Neff. Neff's value strategy, explained in his book *John Neff on Investing* (1999), targets low P/E ratio stocks with strong sales growth, healthy operating margins, increasing free cash flow per share, and underhyped estimated earnings growth between 7% and 20%. The Muhlenkamp screen is a growth-and-value screen based on Ronald Muhlenkamp's investment strategies and mutual fund performance. It seeks an above-average and stable ROE figure over the last five years. It looks for strong earnings growth, low P/E and P/B ratios, and adequate liquidity, as

indicated by the current ratio and others. The O'Shaughnessy value-and-growth screens are named after seasoned investor Jim O'Shaughnessy, who still manages the O'Shaughnessy Funds. Their strategy is to gather data on companies and turn those data into custom factors that can be used to evaluate each stock; these custom factors are then used to build a portfolio with the highest-ranking factor profiles while also managing risk exposures and implementation costs. The Templeton screen has an international focus rooted in the strategies of global pioneer investor Sir John Templeton.⁶ It seeks value—through low P/B and P/E ratios, as well as strong forecasted and actual earnings-per-share (EPS) growth—in an entire universe of securities such as the Compustat database.⁷ ROE and operating margins for the trailing 12 months should be above average, and financial strength should be strong enough to face a downturn in market conditions. Finally, the Dogs of the Dow screen is a simple strategy of choosing the top 10 highest-yielding stocks in the Dow Jones Industrial Average at the beginning of the year and holding them.

Further information on these screens and others, including background information, a description of the screen, and the actual screening steps, is included in Tables 5A.1 through 5A.6 of [Appendix 5A](#).

5.4 SIMULTANEOUS SCREENING AND THE AGGREGATE Z-SCORE

In the preceding section we discussed the simplest type of screening, sequential stock screening. A more sophisticated way to refine the investment universe is to use multifactor simultaneous screening (a.k.a. *simultaneous screening*). In simultaneous screening, the portfolio manager still picks stocks according to a list of factors, but he or she screens for all the factors at once rather than one at a time. He or she therefore does not need to prioritize the factors in advance of screening. The entire list of factors is taken as a single set of criteria with which to evaluate all stocks in the investment universe.

Since the stocks undergo just one screen rather than multiple rounds, there is no chance of eliminating an otherwise good stock simply because it does not measure up during an early round of screening. For example, suppose that the portfolio manager believes that both the B/P ratio of a

stock and its change in gross profit margin are equally important in selecting stocks. With sequential screening, he or she would have to make a choice to screen first for one factor or the other. If he or she screened first for high B/P ratio stocks, he or she might throw out a lot of companies with tremendous profit growth. Simultaneous screening sidesteps this problem by considering stocks for both their B/P ratios and their profit growth concurrently.

Another benefit of simultaneous screening is that the portfolio manager is in no danger of running out of stocks. In sequential screening, the stock list gets smaller and smaller with each successive factor screen. When screening with four or more variables, it is not unusual to end up with a list of fewer than 20 stocks—hardly sufficient for an entire portfolio and probably not diversified in any case. Simultaneous screening preserves all the stocks in the universe and assigns each of them a score based on the whole set of factors.

Once a portfolio manager has chosen the list of factors, how does he or she screen for all of them at once? Unfortunately, since different types of factors are expressed in different units, adding all the factor values together does not work. For a two-factor list of the B/P ratio and size, trying to add a B/P ratio of 0.08 (P/B of 12.5) and a size of \$36,026,000,000 would yield a meaningless number. It does not scale the factors, so the size, which is by far the larger factor in terms of absolute value, dominates the stock ranking.

The right way to aggregate factors is to *standardize* or *normalize* them. Standardization or normalization refers to transforming variables so that they are comparable with each other by converting them to a standard unit of measurement. The most commonly used standardization method is known as the *Z-score method*.⁸

5.4.1 The Z-Score

A population of data has a cross-sectional mean and variance. For example, as of December 2020, the average value of the B/P ratio for all stocks in the S&P 500 was 0.3577. The standard deviation of the B/P ratio of all firms in the S&P 500 was 0.3794.⁹ We can use this information to standardize any value of B/P ratio for any stock in December of 2020. We obtain the Z-score value of any stock's B/P ratio by taking the difference between the stock's B/P ratio and the mean value of the B/P ratio for the universe of

stocks and then dividing the difference by the standard deviation of the B/P ratios of all stocks in the universe. Standardizing the values of factors allows us to make clear statements about how far any particular observation is from the population mean. For example, if we find a company with a B/P ratio Z-score of 2, then we can say that the company's value is 2 standard deviations away from the population mean. The Z-score therefore shows how far a stock is from the norm.

The Z-score also allows us to compare the values of two different factors. Suppose that, as a portfolio manager, you think that companies with high B/P ratios and high profit growth are buys. There is one company whose normalized B/P ratio value is 2 (i.e., the company's B/P ratio is 2 standard deviations from the mean of all the companies' B/P ratios) and whose normalized value of profit growth is also 2. Since the values are normalized (or standardized), you can say that this particular company is as desirable from the perspective of the B/P ratio factor as from the perspective of the profit-growth factor. Also, since both factors are measured in the same units, you will be able to combine these factors into an aggregate Z-score.

In certain situations, standardization is truly normalization, and this is in part why this terminology is used. If the original cross-sectional distribution of the stocks' factor exposures is a normal distribution—and many things in society follow a normal distribution—then we can make clear statements about how probable a particular observation is or how rare it is. For example, a normalized factor score of 2 would mean that the probability of obtaining a stock with a higher value would be less than 2.27%.¹⁰

Let's look at the mechanics of assigning Z-scores. First of all, a portfolio manager should make sure that the Z-scores match the underlying factor values. That is, if high Z-scores are going to represent good stocks and low Z-scores bad stocks, then the factors themselves must be expressed such that high values are good and low values are bad. For instance, if a high book-to-price (B/P) ratio is considered a good quality in a stock, then the B/P ratio must be used to create the Z-score, not the P/B ratio.¹¹ Once this issue is resolved, to create a Z-score for a factor, a portfolio manager needs to compute the mean (or average) value of the factor for all stocks in the investment universe.¹² Call this value $\bar{\beta}_k$ for factor k .¹³ The next step is to compute the standard deviation of the factor across all stocks in the

investment universe. Call this value $S(\beta_k)$ for factor k .¹⁴ Putting these pieces together, the Z-score computation for stock i for factor k is

$$z_{i,k} = \frac{\beta_{i,k} - \bar{\beta}_k}{S(\beta_k)} \quad (5.1)$$

where $z_{i,k}$ represents the Z-score of factor k for stock i , which is the normalized value of the factor value.

5.4.2 The Aggregate Z-Score

Multifactor simultaneous screening means screening for more than one factor (hence the name *multifactor*), but since all factors must be screened for simultaneously, they need to be combined into a single screening value for each stock. Once we have calculated a Z-score for every factor of every stock in the investment universe, calculating each stock's aggregate Z-score is fairly straightforward. Since Z-scores are *scale independent*, we can simply add them together. The sum of a stock's Z-scores is its *aggregate Z-score*.

There may be some individual factor or stock aggregate Z-scores with extreme values. Different portfolio managers will deal with these in different ways. If it seems that a Z-score is an *outlier*, then one control mechanism is to round all Z-scores above 3 down to 3 and similarly to round up all Z-scores below -3 to -3 .¹⁵ It is also possible to throw out stocks that have extreme Z-score values, but this may result in the loss of good information about potentially very high-return or very low-return stocks. In [Appendix 5B](#), we discuss several methods to deal with outliers.

We can choose how to weight each of the factors within a stock's aggregate Z-score. One option is to weight all the factors equally. The aggregate Z-score with K factors then would equal

$$\bar{z}_i = \left(\frac{1}{K} \right) (z_{i,1} + z_{i,2} + z_{i,3} + \cdots + z_{i,K}) \quad (5.2)$$

where $\bar{z}_i$ is the aggregate Z-score for stock i , and $z_{i,k}$ represents the Z-score for factor k and stock i . The equal weighting of Z-score factors is a common method portfolio managers use to compute aggregate Z-scores. Some portfolio managers prefer equal weighting to more sophisticated weighting schemes because it produces relatively more stable results.

5.4.3 Ad Hoc Aggregate Z-Score

There are also a number of ad hoc ways to weight the individual factor Z-scores within the aggregate Z-score. The portfolio manager can weight the factors according to his or her ex-ante beliefs or investing style. If he or she prefers the value factor above all others, he or she can amplify its importance by assigning it a weight of 80% and dividing the remaining 20% weight among the rest of the factors. Factor Z-scores also can be weighted according to some interpretation of their relative levels of importance in influencing stock returns. It is important not to let this sort of weighting scheme be unduly influenced by past research or past reading, which could amount to “data snooping.”¹⁶ In any case, since weighting according to general ideas of factor importance and weighting by preference are not methods of quantitative portfolio management, we do not recommend these methods.

Some portfolio managers weight the factor Z-scores according to the factors’ information ratios, obtained by creating decile or quintile portfolios of each factor and computing the factor’s historical information ratio.¹⁷ The higher the information ratio, the more important the factor is in predicting stock returns. Factors with high information ratios are weighted more heavily. This ad hoc procedure has more merits than weighting according to general notions of factor importance, although it still neglects the effect of the correlation between factors.

5.4.4 Optimal Aggregate Z-Score

Equally weighting the Z-score factors is a simple procedure. However, it ignores certain information contained in the data. It ignores the importance of each factor Z-score in predicting the returns of stocks. The ad hoc methods of weighting attempt to assign weights that vary with the perceived importance of the factors, but these methods are highly subjective. Both the

equal weighting and ad hoc methods also fail to account for the correlation between factors. We can find better ways to weight the factor Z-scores. There are several *optimal weighting* methods.

One very common approach to optimal weighting is to use econometrics to estimate the optimal exposures using a historical sample data set. The portfolio manager takes a series of monthly returns on all the stocks in the universe, combined with the factor Z-score values for each stock at the beginning of the month, and runs a set of cross-sectional regressions over the sample period to find the optimal Z-score weights.¹⁸

Suppose that we have K factor Z-scores for each of the N stocks in the universe and that we have T periods of return information (e.g., 120 months of sample data). We can use the following econometric techniques to estimate the parameter exposures of each factor. This will tell us the best way to combine Z-scores so as to use the information contained in the variance-covariance matrix of Z-scores and returns. One can estimate the following regression with a sample of historical data:

$$r_{i,t} = \gamma_i + \delta_1 z_{i,1,t-1} + \delta_2 z_{i,2,t-1} + \cdots + \delta_K z_{i,K,t-1} + \epsilon_{i,t} \quad (5.3)$$

where γ_i represents a constant term, δ_k is the coefficient estimate for the contribution of the factor Z-score k to the stock returns, and ϵ_i is a typical error term.¹⁹ The regression can be estimated as a panel regression. The δ estimates are the optimal combination of the factor Z-scores.²⁰ The portfolio manager might run this type of regression over various horizons and time periods to check the robustness of the results.

A variation on the optimal weighting scheme is to determine the optimal weights for various economic scenarios. The portfolio manager constructs three or more hypothetical economic environments and calculates a set of optimal Z-score weights for each of them. This method requires the exercise of more subjective judgment than the original method does, but it gives the manager a chance to consider which way the economic winds are blowing and to weight the factors accordingly.

It might be useful to provide a simple example of the aggregate Z-score procedure using real stock data. For simplicity, we took a snapshot of stocks from the S&P 500 as of December 2020. [Table 5.2](#) shows 10 selected stocks from the S&P 500 with their corresponding factor values for E/P, B/P, and

D/E ratios, the natural logarithm of market capitalization (SIZE), and the 12-month momentum (M12M), i.e., the average monthly stock return over the previous 12 months.²¹ These factors do not necessarily represent some phenomenal α model; they were chosen merely for illustration. One will notice that the table gives the mean and standard deviation of all stocks in the S&P 500 for the month of December 2020. We will need these values to compute the Z-scores.

TABLE 5.2

Selected Factor Exposures and Z-Scores of Selected Stocks

Ticker	Earnings-to-Price Ratio (E/P)	Earnings-to-Book Ratio (B/P)	Debt-to-Equity Ratio (D/E)	Log of Market Capitalization (SIZE)	Momentum (M12M)	Z-Score of E/P	Z-Score of B/P	Z-Score of D/E	Z-Score of SIZE	Z-Score of M12M	Aggregate Z-Score
AAPL	0.0254	0.0290	3.96	14.6291	82.31	0.117	-0.866	-0.056	3.962	1.431	-0.645
MSFT	0.0282	0.0734	1.44	14.3353	42.53	0.136	-0.749	-0.224	3.688	0.582	-0.699
AMZN	0.0106	0.0507	2.41	14.3066	76.26	0.016	-0.809	-0.159	3.662	1.302	-0.599
GOOGL	0.0323	0.1927	0.41	13.9152	31.03	0.164	-0.435	-0.292	3.297	0.337	-0.588
TSLA	0.0008	0.0240	1.67	13.4134	743.44	-0.051	-0.880	-0.208	2.830	15.535	2.397
FB	0.0385	0.1793	0.24	13.3949	33.09	0.206	-0.470	-0.303	2.813	0.381	-0.479
BRK.B	0.0659	0.7637	0.98	13.2060	2.37	0.393	1.070	-0.254	2.637	-0.274	-0.239
JNJ	0.0410	0.1556	1.65	12.9344	10.85	0.223	-0.533	-0.210	2.384	-0.094	-0.516
WMT	0.0439	0.1847	1.92	12.9186	23.34	0.242	-0.456	-0.191	2.369	0.173	-0.444
JPM	0.0659	0.6223	10.97	12.8670	-5.52	0.392	0.697	0.410	2.321	-0.443	-0.417
S&P 500											
Mean	0.0083	0.3577	4.80	10.3737	15.24	0	0	0	0	0	0
SD	0.1468	0.3794	15.04	1.0741	46.88	1	1	1	1	1	

Note: The Market capitalization is in millions of dollars. Momentum is defined as the monthly cumulative return (in percentage terms) over the past 12 months. The stocks were selected from the S&P 500. Ticker symbols are for the following companies: Apple Inc. (AAPL), Microsoft Corporation (MSFT), Amazon (AMZN) Alphabet Inc. or Google (GOOGL), Tesla Inc. (TSLA), Facebook Inc. (FB), Berkshire Hathaway B shares (BRK.B), Johnson & Johnson (JNJ), Walmart Inc. (WMT), and J.P. Morgan Chase & Co. (JPM). All values were computed as of December 2020. The data were obtained from Compustat.

To compute the Z-score for the B/P ratio factor of the tech company Apple, we need the value for $\beta_{AAPL,B/P} = 0.0290$ for December 2020. The mean for all the S&P 500 stocks is 0.3577 (i.e., $\bar{\beta}_{B/P} = 0.3577$), and the standard deviation is 0.3794 [i.e., $S(\beta_{B/P}) = 0.3794$]. Thus the Z-score for the B/P ratio factor for Apple is -0.866 [$z_{AAPL,B/P} = (0.0290 - 0.3577)/0.3794$]. Although the sign is negative for the Z-score, this does not necessarily imply that it will be a negative contribution to the aggregate Z-score. This will depend on the portfolio manager's ultimate belief regarding this factor. If the portfolio manager believes that a high B/P ratio is a negative for picking stocks based on his or her previous analysis, he or she should multiply this factor Z-score by -1 when computing the aggregate Z-score so that higher Z-scores imply a better stock.

One can continue computing the Z-scores for the other factors for Apple. They are all listed in the table. The aggregate Z-score then will be a combination of the individual Z-scores for each factor. In this example, we assume that the portfolio manager *equal-weights* the individual Z-scores. Thus the aggregate Z-score for Apple in December 2020 is equal to -0.645 . It is important that in aggregating the individual Z-scores, the portfolio manager uses the appropriate sign for each individual Z-score according to his or her beliefs. For example, to obtain the aggregate Z-score, we did the following:

$$\begin{aligned} z_{AAPL} &= \frac{1}{5} [z_{AAPL,E/P} + z_{AAPL,B/P} - z_{AAPL,D/E} - z_{AAPL,SIZE} + z_{AAPL,M12M}] \\ &= \frac{1}{5} [(0.117) + (-0.866) - (-0.056) - (3.962) + (1.431)] \\ &= -0.645 \end{aligned} \tag{5.4}$$

A negative sign precedes several factors' Z-scores because these factors are considered detrimental to stock returns. All individual factor Z-scores must possess the correct sign. The portfolio manager in this example likes stocks with high E/P and B/P and low D/E ratios, so the negative of the D/E Z-score is added to the total Z-score, while the E/P and B/P Z-scores are added as is. The manager also likes smaller-cap stocks, hence the negative sign in front of the Z-score for the SIZE. He believes that momentum in stock returns is a positive technical indicator, so the momentum Z-score is added to the total Z-score. For a different portfolio manager, the signs might be different. The important thing is consistency. If high exposure to a factor is supposed to boost stock returns, add the factor's Z-score to the total; if high exposure to a factor is supposed to hurt stock returns, subtract its Z-score.

[Table 5.2](#) goes through the computations of Z-scores for nine other stocks from the S&P 500 in December 2020. A portfolio manager would do this for all stocks in his or her investment universe and then rank the stocks by their aggregate Z-scores to determine which stocks are the most and least attractive.

5.4.5 Factor Groups and the Aggregate Z-Score

Some quantitative portfolio managers choose to separate the K factors that they use into M factor groups. For example, suppose that we have a B/P

ratio factor, an E/P ratio factor, and a price-to-sales (S/P) ratio factor. A manager might create a composite group known as the *valuation group* and place all three of these factors into the valuation group. He or she may do this for other factors as well.

Although dividing the factors into groups might seem at first like an arbitrary step, it offers several benefits. It organizes the screening process. Rather than having to keep track of a large group of K factors that all represent different attributes of stock returns, the manager will be able to look at the factors in sets of M groups that represent the essential forces that are believed to affect stock returns. This simplifies both modification of the model and presentation of performance results to an investment committee.

Second, a manager might be tempted to use only one factor to represent a category such as value. However, to the extent that there are idiosyncracies in the way individual factors represent the same concept, it might be better to collect more than one factor to represent the concept (e.g., using B/P, S/P, and E/P ratios to represent value rather than just the B/P ratio in isolation).

Third, by creating factor groups that represent fundamental forces on stock returns, a manager can easily change the weights (which represent the relative importance) of factor groups according to changing economic conditions or for other reasons.²¹ Although one could change the relative weights of each of the K factors individually, it is easier to change the weights of M factor groups, where $M < K$.

Formally creating composite Z-scores using factor groupings is relatively straightforward. The following steps are very similar to the procedure for aggregate Z-scores discussed in the last section.

1. Determine the number of factor groups M . For example, if $M = 4$, we might have a *valuation composite*, a *profitability composite*, a *financial-soundness composite*, and a *technical composite*. These four composites or groups represent the four main themes that we believe predict stock returns in our investment universe.
2. Decide which factors to include in each factor group. For example, the manager might believe that B/P, E/P, and S/P ratios are all factors that determine value and place those in the valuation composite. He or she then might decide that gross profit margin belongs in the profitability composite. The inverse interest

coverage ratio (IICR) and the firm's debt-to-equity ratio then might go in the financial soundness composite. Finally, the manager might put a factor such as momentum in the technical composite.

3. Compute all the factors for every stock in the investment universe. Thus, if there are N stocks in the investment universe and K factors, the investor will have to make $N \times K$ computations.
4. Compute the mean and standard deviation of each factor exposure in the cross section of all the stocks in the investment universe. These values should be stored. Label these as $\bar{\beta}_k$ and $S(\beta_k)$, respectively, for each factor k .
5. Compute the Z-scores for every factor and every stock. This is similar to the preceding section, where $z_{i,k} = (\beta_{i,k} - \bar{\beta}_k) / S(\beta_k)$ represents the Z-score of factor k for stock i , which is the normalized value of the factor value.
6. Compute the aggregate Z-score for *every factor group* for every stock. Thus, for each factor group, compute the group Z-score for stock i , that is,

$$\bar{z}_{i,m} = \sum_{k \in S_m} w_k^m z_{i,k} \quad (5.5)$$

where S_m is the set of factors in the m th factor group, w_k^m is the weight of factor k in the m th factor group (so that $\sum_{k \in S_m} w_k^m = 1$), and $z_{i,k}$ represents the Z-score of factor k for stock i . This weighted sum is computed for every factor group to obtain a factor group Z-score for every stock and every factor group.

7. Compute the aggregate Z-score for every stock in the universe. This is the final step for the ranking of the stocks. The final formula is to compute

$$\bar{z}_i = \sum_{m=1}^M w_m \bar{z}_{i,m} \quad (5.6)$$

where $\bar{z}_i$ is the aggregate Z-score for stock i , w_m is the weight given to the factor group m , and $\bar{z}_{i,m}$ is the factor group Z-score for group

m for stock i .

This procedure is almost identical to the procedure for computing aggregate Z-score. The only real difference between the two is that here we have classified the individual factors into specific factor groups. We have not discussed how to weight the relative factor groups or how to weight the factors within factor groups. The methods available are similar to the ones we described in the preceding section. If the weighting scheme of the factors within groups and across groups is similar to the weighting scheme in the normal Z-score aggregation without groups, this final step should produce results similar to the results of the procedure that does not divide factors into groups.

Let's walk through a simple example of grouping factors. We return to the example from an earlier section that uses data for S&P 500 stocks for December 2020. Following the seven steps of factor grouping, our first step is to choose the number of groups. We choose four factor groups ($M = 4$). The groups are a *valuation composite*, a *profitability composite*, a *financial-soundness composite*, and a *technical composite*.

Our second step is to decide which factors will go in each composite. Suppose we decide on the following factors for our model: E/P, B/P, and S/P ratios; gross profit margin (GPM); IICR and D/E; and M12M.²² It makes sense to place the first three factors in the valuation composite, the next one in the profitability composite, the next two in the financial-soundness composite, and the last one in the technical composite. See [Table 5.3](#).

TABLE 5.3

A Possible Categorization of Factors into Composite Groups

Group Number	Factor Group	Factors
1	Valuation composite	Earnings-to-price ratio (E/P) Book-to-price ratio (B/P) Sales-to-price ratio (S/P)
2	Profitability composite	Gross profit margin (GPM)
3	Financial-soundness composite	Inverse interest coverage ratio (IICR) Debt-to-equity ratio (D/E)
4	Technical composite	12-month momentum (M12M)

The third step is to compute the factor values for all stocks. We have listed these for selected stocks in Table 5.4. The fourth step is to compute the mean and standard deviation of all the factors for the investment universe. In this specific example, the S&P 500 is our investment universe. The mean and standard deviation of each factor are listed in the table as well. The fifth step is to compute the Z-scores for every factor, and the sixth step is to compute the Z-scores for each factor group. We equally weighted the Z-scores of each factor within a group to obtain the factor group Z-scores for each stock listed in the table.

TABLE 5.4

Selected Factor Exposures and Group Z-Scores of Selected Stocks

Ticker	Valuation Composite			Profitability Composite	Financial Soundness Composite		Technical Composite	Valuation Composite Z-Score	Profitability Composite Z-Score	Financial Composite Z-Score	Technical Composite Z-Score	Aggregate Z-Score
	E/P	B/P	S/P		GPM	IICR						
AAPL	0.0254	0.0290	0.1217	42.26	0.0428	3.96	82.31	−0.426	−0.038	0.148	1.431	0.279
MSFT	0.0282	0.0734	0.0875	76.75	0.0450	1.44	42.53	−0.393	1.273	0.231	0.582	0.423
AMZN	0.0106	0.0507	0.2129	47.06	0.0825	2.41	76.26	−0.408	0.144	0.188	1.302	0.307
GOOGL	0.0323	0.1927	0.1554	61.31	0.0079	0.41	31.03	−0.254	0.686	0.276	0.337	0.261
TSLA	0.0008	0.0240	0.0421	29.89	0.7534	1.67	743.44	−0.515	−0.508	0.016	15.535	3.632
FB	0.0385	0.1793	0.1203	89.23	N/A	0.24	33.09	−0.265	1.747	0.152	0.381	0.504
BRK.B	0.0659	0.7637	0.5136	25.73	0.0853	0.98	2.37	0.452	−0.666	0.234	−0.274	−0.064
JNJ	0.0410	0.1556	0.1952	74.65	0.0104	1.65	10.85	−0.253	1.193	0.234	−0.094	0.270
WMT	0.0439	0.1847	1.3290	26.71	0.1052	1.92	23.34	0.185	−0.629	0.197	0.173	−0.018
JPM	0.0659	0.6223	0.3437	73.54	N/A	10.97	−5.52	0.267	1.151	−0.205	−0.443	0.192
S&P 500												
Mean	0.0083	0.3577	0.6131	43.26	0.4523	4.80	15.24					
SD	0.1468	0.3794	0.9311	26.31	1.7088	15.04	46.88					

Note: For factor and ticker symbols, see Table 5.2. All values are as of December 2020. The data were obtained from Compustat. Some companies, such as J.P. Morgan and Facebook, had a missing value for interest expense for December 2020. We thus gave them a “neutral” Z-score of zero for this particular factor.

The next step is to calculate the factor group Z-scores. As in the aggregate Z-score procedure, the factor Z-scores should be added together (or subtracted from each other) according to the way they are expected to contribute to (or detract from) stock returns. For example, to obtain the valuation Z-score of Microsoft ($z_{MSFT,V}$), we take our three valuation factors of Microsoft and equal-weight them, making sure the signs are correct.²³ Since the portfolio manager believes that higher E/P, B/P, and S/P ratios signal “good” attributes of stocks, each individual Z-score can be added together without multiplying by -1 , that is,

$$\begin{aligned}
z_{MSFT,V} &= \frac{1}{3} [z_{MSFT,E/P} + z_{MSFT,B/P} + z_{MSFT,S/P}] \\
&= \frac{1}{3} \left[\left(\frac{0.0282 - 0.0083}{0.1468} \right) + \left(\frac{0.0734 - 0.3577}{0.3794} \right) + \left(\frac{0.0875 - 0.6131}{0.9311} \right) \right] \quad (5.7) \\
&= \frac{1}{3} [(0.36) + (-0.799) + (-0.564)] = -0.393
\end{aligned}$$

Applying the same process to each factor group, we can calculate other factor group Z-scores for Microsoft as well, including the profitability Z-score ($z_{MSFT,P} = 1.273$), the financial-soundness Z-score ($z_{MSFT,F} = 0.231$), and the technical Z-score ($z_{MSFT,T} = 0.582$). Microsoft receives a poor valuation rating owing in particular to its unusually low B/P and S/P ratios. It scores a very good rating for profitability given its gross profit margin of 76.25% compared to the S&P 500 average of 43.26% and also receives a good rating for financial soundness due to its lower-than-average debt-to-equity ratio and lower-than-average inverse interest coverage ratio. Microsoft also earns decent marks for its technical group, directly due to the factor M12M: relative to the investment universe, its stock price has grown over the last year by 42.5% compared to the S&P 500 average of 15.24%. With an aggregate Z-score of 0.423, Microsoft is overall a mildly attractive stock as of December 2020, according to the set of criteria used in this example (see [Table 5.4](#)).

5.5 THE AGGREGATE Z-SCORE AND EXPECTED RETURN

5.5.1 Expected Return Implied by the Z-Score

Expected stock returns can be found from aggregate Z-scores by performing a regression of actual stock returns against the aggregate Z-scores. This can be done using a panel series regression of stock returns on the Z-score of the prior periods (e.g., of the previous months). The regression would take the form

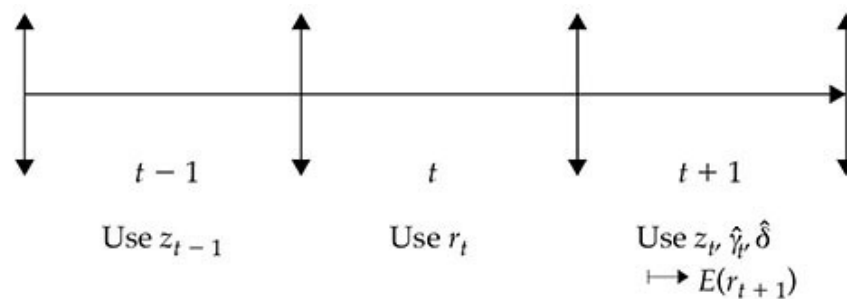
$$r_{i,t} = \gamma_i + \delta z_{i,t-1} + \epsilon_{i,t} \quad (5.8)$$

where γ_i represents a constant term,²⁴ δ is the coefficient that relates the aggregate Z-score to the stock return, and $\epsilon_{i,t}$ is a typical error term. Given the estimates of γ_i and δ , the expected return of stock i for time $T + 1$ can be written as (see Fig. 5.1)

$$E(r_{i,T+1}) = \gamma_i + \delta z_{i,T} \quad (5.9)$$

FIGURE 5.1

Timeline.



There are some problems with this methodology. First, Z-scores may not change much from one period to another, but factor premiums may change substantially. Thus one's estimated coefficients may not be stable or reliable. One way to reduce this problem is to run panel regressions on a larger period of historical data rather than just one single cross-sectional regression over one particular period.²⁵

The second problem is that there may be a low correlation between the Z-score and the subsequent returns. While we are using Z-scores at time t to predict stock returns at time $t+1$, the link might be rather weak because the equation is not based on a rigorous theory.

The third problem with this procedure is that it adds complexity to the process, which is a weakness given that the biggest strength of the aggregate Z-score model is its simplicity.

5.5.2 Forecasting Rule of Thumb

Many portfolio managers are familiar with the *forecasting rule of thumb* and love to talk about it.²⁶ We personally do not understand what makes this equation so exciting because it is essentially an algebraic manipulation

of a simple regression equation. We will make a few comments about it, though, because it may have some limited use in QEPM.

One can manipulate [Eq. \(5.8\)](#) and obtain the following:

$$\begin{aligned}
 E(r_{it}|z_{i,t-1}) - E(r_t) &= \delta [z_{i,t-1} - E(z_{t-1})] \\
 &= \frac{C(r_t, z_{t-1})}{V(z_{t-1})} z_{i,t-1} \\
 &= \rho(r_t, z_{t-1}) S(r_t) z_{i,t-1} \\
 &\equiv IC \cdot \text{volatility} \cdot \text{score}
 \end{aligned} \tag{5.10}$$

In this equation, IC is by definition the correlation between the aggregate Z-score or raw signal and the actual security returns, the volatility in this particular case represents the cross-sectional volatility of the returns of the securities, and score refers to the aggregate Z-score.²⁷

Although this is a neat formula, the portfolio manager should be sufficiently well versed in econometrics not to need this transformation of variables to make a connection between the raw scores or aggregate Z-score and the actual returns or residual returns of the security. This equation would be helpful if, for some reason, it were easier to compute IC and volatility than to run a regression. Otherwise, the equation does not add much to the screening process.²⁸

5.5.3 The Equivalence between the Z-Score Model and the Fundamental Factor Model

We can show that the Z-score model of [Eq. \(5.8\)](#) produces an expected return identical to that of the fundamental factor model under certain conditions. Specifically, if *the factor premium is inversely proportional to the standard deviation of the factor exposure*, then the aggregate Z-score model produces the same result as the fundamental factor model.

Let us assume that the factor premium is inversely proportional to the standard deviation of the factor exposure. Let β_k be the exposure to the k th factor and f_k be the premium on the k th factor.

Then

$$f_k = \frac{c}{\sqrt{V(\beta_k)}} \quad (5.11)$$

where c is a constant. Then the fundamental factor model can be rewritten as

$$\begin{aligned} r_i &= \alpha_i + \beta_{i,1}f_1 + \cdots + \beta_{i,K}f_K + \epsilon_i \\ &= \alpha_i + c \frac{\beta_{i1}}{\sqrt{V(\beta_1)}} + \cdots + c \frac{\beta_{iK}}{\sqrt{V(\beta_K)}} + \epsilon_i \\ &= \tilde{\alpha}_i + cZ_i + \epsilon_i \end{aligned} \quad (5.12)$$

where $\tilde{\alpha}_i$ is a constant defined as

$$\tilde{\alpha}_i = \left[\alpha_i + c \frac{E(\beta_1)}{\sqrt{V(\beta_1)}} + \cdots + c \frac{E(\beta_N)}{\sqrt{V(\beta_N)}} \right] \quad (5.13)$$

Equation (5.12) shows the relationship between the parameters of the aggregate Z-score model and the parameters of the fundamental factor model.

Certain conditions may give rise to an inverse proportionality between the factor premium and the standard deviation of the factor exposure. When the factor exposure variation is large, even a small factor premium can affect the excess return tremendously. Thus, to preserve the overall excess return of the universe, it might make sense to have the factor premium be proportional to the inverse of the standard deviation of the factor exposure. This would imply that a factor with extremely high variation has a low factor premium, thus in some sense preserving the balance of excess return.

5.6 THE AGGREGATE Z-SCORE AND THE MULTIFACTOR α

Ultimately, most portfolio managers will have to optimize their portfolios versus a benchmark and cannot apply ad hoc weightings to the stocks in the portfolio. Typically, many portfolio managers will use commercial software to perform the optimizations versus a benchmark.²⁹ Commercial software allows the user to supply the multifactor α (i.e., the part of the expected return that a multifactor model cannot explain). Many portfolio managers transform the aggregate Z-score into a multifactor α . In this section we examine three popular methods for doing this, discussing the strengths and weaknesses of each method.

Perhaps the simplest method is a one-to-one transformation that uses the actual Z-scores as α 's. Although the aggregate Z-scores clearly do not represent the actual excess returns of the stocks, they do represent the relative ranking of each stock in the manager's mind. Thus, we could just use the Z-score as a surrogate for the α of each stock. One immediate problem with doing this is that if the Z-scores do not really represent returns, they could distort the stock-selection process by favoring stocks with extremely high predicted Z-scores much more than is justified by real excess return potential. Fortunately, though, if the Z-scores are treated for outliers, such as using either windsorization, the IQR method, or the Rank method, the Z-scores will reside within a limited range (e.g., +3 to -3). In the extreme case of limiting Z-scores to a range of +3 to -3, the highest α a particular stock can have is 3% above its expected return, and the worst it can have is 3% below its expected return, with most values falling somewhere between +1 and -1.³⁰ Therefore, one could add the aggregate Z-score to the expected return of each stock predicted by a risk model. If one uses a standard commercial software package to build the portfolio, then the software will add these aggregate Z-scores to the expected returns of each stock according to the software's risk model. One then performs an optimization to form the portfolio.³¹ The basic point is that if the portfolio manager has constructed trimmed aggregate Z-scores, one method to convert aggregate Z-scores to the α of each stock is just to do a one-to-one mapping in which the α of each stock equals the aggregate Z-score of each stock.

The second method is to use the expected excess return from Eq. (5.9) as α . This is a more accurate method since the empirical relationship between the Z-score and actual stock returns is taken into account.

The third method is more complicated. This method attempts to link the Z-score to residual stock returns rather than actual stock returns. The residual stock returns can be computed with respect to a multifactor model or a benchmark. The idea is to obtain all stocks' factor exposures and factor premiums and use them to calculate historical α . Then one can estimate the relationship between α and the aggregate Z-score from the following regression:

$$r_{i,t} - \hat{\beta}_i' \mathbf{f}_t = \gamma_i + \delta z_{i,t-1} + \epsilon_{i,t} \quad (5.14)$$

where $r_{i,t}$ is the return of stock i at time t , $\hat{\beta}_i$ is the estimated factor exposures of stock i , $\mathbf{f}_t$ represents the realized factor exposures at time t , γ_i represents a constant term, δ is the coefficient that relates the aggregate Z-score to α , and $\epsilon_{i,t}$ is a typical error term.³² Then the portfolio manager can use the estimates from this equation to forecast the α 's of all the stocks for the next period ($t + 1$) given their aggregate Z-scores for time t . Thus, with the estimated coefficients, the portfolio manager can translate aggregate Z-scores into α 's. These values eventually will be used with a risk optimizer to create a portfolio of stocks.

There are drawbacks to all three of these methods. It may introduce serious distortion if the data behind the Z-score and the data behind the factor model are not sufficiently different. In such a case, the information criterion will be violated.³³ The second and third methods also require more computation, and the improvement in the optimization may not compensate for the added complication. Finally, the third method requires more estimation periods and more stability of estimates through time to be meaningful.

5.7 CONCLUSION

This chapter introduced the aggregate Z-score model of stock selection. We began by discussing the concept of sequential stock screening because it is

one of the simplest quantitative methods of picking stocks and easy for most investors to understand. We also discussed some of the investing styles of famous portfolio managers and translated their approaches into sequential stock screens.

While sequential stock screening is a very useful starting point, it is not as efficient or as commonplace in the portfolio management world as multifactor simultaneous stock screening. Using what we call the aggregate Z-score model, we showed how to standardize or normalize all stock return factors in order to perform multifactor simultaneous stock screening. We focused on Z-score standardization across stocks in the entire universe but also discussed the fact that some portfolio managers might choose to standardize the Z-scores by industry or sector and then select representative samples of stocks from each industry or sector. Finally, we introduced the concept of factor groups as a means of further structuring and clarifying the screening process.

The aggregate Z-score model is helpful in its simplicity, but it has the drawback of ranking stocks relatively instead of directly estimating the stocks' expected returns or α 's. The portfolio manager who does not manage against a benchmark does not need to worry about this because he or she simply can pick the stocks that are most attractive and weight them using any of a variety of methods (e.g., equal weighting the 100 top-ranked stocks). However, for a portfolio manager who manages against a specific benchmark (e.g., the S&P 500), it is necessary to know the expected returns in order to perform a risk optimization versus the benchmark. We have suggested some ways to convert the aggregate Z-scores into the stocks' expected returns or α 's. The last step in the portfolio process for portfolio managers who use the aggregate Z-score model is to combine the model with a risk optimizer and create the portfolio of stocks with stock weights.

Stock screening can serve as the basis for a model of stock returns, but in QEPM it is usually just a preliminary step in creating the portfolio. In the next chapters we delve into the core of QEPM, which is the factor model. We introduced factor models in [Chapter 3](#). Now that we are familiar with the various kinds of factors at our disposal, some methods for choosing them, and the option of screening the investment universe, we are ready to take a closer look at how to gather factors into a quantitative model of stock returns. We look first at the fundamental factor model in [Chapter 6](#) and then at the economic factor model in [Chapter 7](#).

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- ¹ The investment universe corresponds to the set of stocks that a portfolio manager is allowed to trade in his or her portfolio. For example, a portfolio manager who manages against the Standard & Poor's (S&P) 500 benchmark might only be allowed to purchase the 500 stocks in the S&P 500. His or her *investment universe* would be the 500 stocks that make up the S&P 500. Other managers' investment universes might follow other indices, such as the Russell 1000, or more complicated definitions.
 - ² Oftentimes, it is useful to use one version of a factor over another. For example, in [Chapter 4](#), we used the price-to-book (P/B) and price-to-earnings ratio (P/E) because they are familiar to most people. However, we find that using the E/P ratio is more effective. Many companies have negative earnings, which renders a P/E ratio meaningless: it makes no sense to argue that a low P/E ratio company is "cheaper" than a high P/E ratio company when companies with zero or negative earnings are included. However, if we calculate the E/P ratio of companies, we don't have to exclude companies with negative earnings. For example, a company with a very high P/E ratio will have a very small E/P ratio but will still be considered relatively "cheaper" than a company with a negative E/P ratio. The E/P ratio thus gives us a ratio that can be compared across many more companies. Another reason for using the E/P ratio rather than the P/E ratio is that a company's earnings for a particular period could be close to zero. This leads to an enormous value of P/E that isn't really representative of the company. Using the E/P ratio resolves this problem. As with P/E, it is more effective to use the inverse of the P/B ratio, the B/P ratio. In the rest of this book, we shall use both B/P and E/P rather than the more common ratios. In the Practical Application part of this book, Part V, we will use the inverse of many other common ratios for the exact same reason.
 - ³ Some people argue that the first screens should be the ones that eliminate the most data possible, but we do not agree.
 - ⁴ These correspond to P/B ratios of 0.72 and 1.11, respectively.
 - ⁵ Our descriptions of these famous portfolio managers and professors and our interpretations of their approaches in the form of quantitative stock screens are based partly on our readings of books written by the managers and on information supplied by the American Association of Individual Investors (AAII). The stock screens we present may not be the actual methods by which these managers select equities. The managers have in no way endorsed the stock screens that we present here.
 - ⁶ Templeton died in 2008. Prior to that, he requested that a large part of his portfolio be placed in a short strategy, since he was convinced that there was trouble in the markets. In fact, he turned out to be correct, although he died before seeing the entire realization of his prediction.
 - ⁷ A low price relative to expected earnings in three to five years is also used as a forward-looking measure of value.
 - ⁸ This term originates from the idea of converting any normal distribution to a standard normal distribution. The new variable created by adjusting the raw data to a standard normal distribution is traditionally called a *Z-score*. The concept of aggregating Z-scores of different variables is loosely linked to the Altman Z-score due to Edward Altman's original model to try to predict stocks that would go bankrupt. The use of the Z-score in QEPM, however, is very different from the use originally proposed by Altman (1968). In Altman's original work, he used a concept known as *multiple discriminant analysis* (MDA) to aggregate financial ratios that would help to predict whether a firm would go bankrupt or not. The weights of the financial ratios were chosen through the statistical optimization of past data. A final Z-score

would result for each stock. There was usually a cutoff value below which a stock would be predicted to go bankrupt. Our analysis is slightly different from Altman's, but the growth of these type of models in QEPM is most attributable to his original work.

- 9 The corresponding P/B ratios would be 2.795 for the average and 2.635 for the standard deviation of P/B (just the inverse of the numbers listed in the text). [Table 5.2](#) shows the values for other representative factors for the S&P 500. The numbers for both E/P and B/P may be different than expected for several reasons. First, we are using the inverse of the common ratios. Second, and most important, we are using equal-weighted averages of stocks in the S&P 500 for the Z-score, whereas most published metrics for the S&P 500 are market-capitalization weighted. Third, some published indices eliminate negative P/E or P/B stocks and just average the remaining stocks—or they average the values substituting a value of zero for negative P/E or P/B stocks. Fourth, our data avoid look-ahead bias, since we use the most recent market value of a company and the book or earnings of three months prior, rather than the coincidental values. Fifth, our S&P 500 universe does not include REIT stocks or any stocks that do not have a CRSP share code of 10, 11, or 12. The reason for excluding REITs is that most quant models would not apply to REIT securities. It should be noted that none of these issues alter the basic composition of our S&P 500 universe. Using this S&P 500 universe, we divide the S&P 500 total market value by total earnings or total book value, which results in a P/B of 4.20 and a P/E of 36.82—figures closer to what is more familiar.
- 10 Standardizing or normalizing the factor values is based on the idea of converting a normal distribution to the standard normal distribution for analysis purposes. The reader can refer to any basic statistics book such as DeGroot (1986). The Z-score concept still works even if the underlying cross-sectional factor distribution is not normal. The Z-score still gives a sense of how far a variable is from the mean of the distribution. If it turns out that the factor distribution is normal, it simply means that you can make statements about the probabilities of factor Z-scores.
- 11 The portfolio manager could use the P/B ratio to calculate the Z-scores, but he or she would need to remember to multiply all the Z-score values by -1 so that high Z-scores represented good stocks.
- 12 Some portfolio managers who wish to have industry neutrality or believe that their factors are very industry sensitive will create Z-scores for each stock relative to its industry average. They will rank each stock within its industry rather than within the entire investment universe and for the portfolio select the best stocks within each industry.
- 13 Most computer programs and software packages compute the mean very easily. The typical formula is $\bar{\beta}_k = (1/N) \sum_{i=1}^N \beta_{i,k}$, where N equals the number of stocks in the universe, and $\beta_{i,k}$ represents the factor exposure k for stock i .
- 14 Most computer programs and software packages compute the standard deviation very easily. The typical formula is $S(\beta_k) = 1/(N-1) \sum_{i=1}^N (\beta_{i,k} - \bar{\beta}_k)^2$, where N equals the number of stocks in the universe, and $\beta_{i,k}$ represents the factor exposure k for stock i .
- 15 An outlier is an observation in a distribution that is very far from the mean and has an extremely small probability of occurring. An outlier looks inconsistent with the rest of the data.
- 16 The phenomenon of data snooping, which we discussed in [Chapter 2](#), involves picking factors solely based on past reports of their usefulness without any statistical evidence that they continue to relate significantly to stock returns.
- 17 For those unfamiliar with this process, it is discussed in more detail in [Chapter 15](#).

- ¹⁸ The researcher should be very careful to avoid look-ahead bias, which is sometimes a problem when a portfolio manager uses historical factor data. The data set may contain a factor value for a certain month, but in that month, that piece of data would not have been available. By using data that the market did not have in that month, the quantitative portfolio manager has “looked ahead,” potentially causing the model to estimate parameters incorrectly. A typical solution to look-ahead bias is to know the delivery lag of all factors that one is using and place a lag on the factor. For example, if you’re using a factor that relates to March but is only reported in April, you may wish to use the $t - 1$ value of that factor in any historical estimations. This is discussed in more detail in [Chapter 16](#).
- ¹⁹ When we typically estimate these factor models, we use information at time t to predict returns at time $t - 1$ or similarly use information at time $t - 1$ to predict returns at time t . Sometimes, in this book, for convenience, we write the equation with time subscript t everywhere and sometimes we explicitly write the expressions with factor exposure $t - 1$ and returns of time t . In all cases, the reader should understand that we are using information at t to predict returns at time $t - 1$. In other words, the factor exposure at time $t - 1$ is the exposure value as of the end of period $t - 1$, which we may refer to sometimes as the beginning of month exposure for time t .
- ²⁰ This regression also tells us how Z-scores are related to stock returns. In some cases, like when dealing with returns accounting for transactions costs, it is beneficial to convert Z-scores to returns. In [Appendix 5C](#) we suggest a crude way to do this.
- ²¹ Using the natural log of market capitalization might also be uncomfortable to some. In order to get the more standard market capitalization, just take the exponential of the value. For example, for Apple, $\exp(14.6251) = \$2,255,991$ million or \$2.25 trillion.
- ²¹ For some procedures related to common tests for structural change, see Chow (1960), Toyoda (1974), Schmidt and Sickles (1977), Toyoda and Ohtani (1986), Ploberger et al. (1989), Banerjee et al. (1992), Stock (1994), Chu et al. (1995), and Hoyo and Llorente (1997). Some of these papers are also useful for parameter stability tests discussed earlier in the book.
- ²² These acronyms for factors can be found in Abbreviations of Factor Names at the beginning of the book or in [Appendix 16A](#).
- ²³ Of course, alternative weighting schemes could be used.
- ²⁴ If a single-period cross-sectional regression were used instead of a panel series regression, then there would not be a γ for each stock; thus γ_i would be replaced by γ .
- ²⁵ Panel regressions consist of using a time series of cross-sectional data to estimate the coefficients.
- ²⁶ This concept was popularized by Richard Grinold (1989). Significant research discussing and expanding our understanding of this law has been done since it was proposed [see Bolshakov and Chincarini (2020), Buckle (2004), Chincarini and Kim (2007), Clarke et al. (2002), Ding and Martin (2017), Hallerbach (2014), Van Loon (2018), Ye (2008), and Zhou (2008a, 2008b)].
- ²⁷ Because $E(z_{t-1})$ is zero and $S(z_{t-1}) = 1$, we can achieve the second to last part of the derivation. This is so because the Z-score is already normalized.
- ²⁸ The equation we present here differs from Grinold and Kahn’s original description because they use actual signals or raw forecasts prior to normalizing the variables. The actual regression of the returns on the raw signals or factors creates this equation where by definition the raw forecasts are normalized. This is why regressions are convenient: You don’t have to normalize prior to estimating them. They normalize for you. It is therefore difficult to see the magic in this formula.

- ²⁹ Common commercial software platforms include Factset, FTSE BIRR, MSCI Barra, Northfield, and Qontigo's Axioma. For more details, see Appendix D at www.ludwigbc.com under QEPM Exclusive Content. In [Chapter 9](#) we show how a quantitative portfolio manager can build his or her own portfolio optimization models.
- ³⁰ If the factor distribution were truly normal, then 68% of the stocks would fall into this area.
- ³¹ The process of portfolio optimization is discussed in significant detail in [Chapter 9](#).
- ³² There are several ways to estimate this equation. For example, one could use a set of data to estimate the factor exposures, $\hat{\beta}_i$, to factor premiums (or benchmark returns) of each stock. In the next sequential period of data, given the stock returns and realized factor premiums (or benchmark returns), the residual returns are regressed against Z-scores to obtain the relationship between Z-score and residual returns.
- ³³ We discussed this issue in [Chapter 3](#). Also see Chincarini and Kim (2007).

APPENDIX 5A

A List of Stock Screens Based on Well-Known Strategies

TABLE 5A.1

Well-Known Stock Strategies of Individuals

Screen Name	Background	Description
Buffett	Perhaps the most ubiquitous and famous investor ever, Warren Buffett is currently chairman of Berkshire Hathaway. Berkshire was once a stalwart textile company that Buffett acquired in 1965 and converted into an investment vehicle. Over the period 1965–2020, Buffett earned an average annual rate of return of 20%, pummeling the S&P 500 by 9.8% per year.¹ These extraordinary returns helped propel Berkshire Hathaway to a total annual compounded rate of return of 2,810,526% vis-à-vis the S&P 500's total return of 23,454% over the same period. His "Letter to Shareholders" in Berkshire Hathaway's Annual Reports is one of the most widely read market commentaries on the Street. Buffett's witty stories, common-sense investment approach, and continued success have made him a modern-day legend among investors of all classes.	Known to many as the "buy and hold" approach, this strategy concentrates its efforts on a relentless search for companies selling below their intrinsic values, or what Buffett describes as the "discounted value of the cash that can be taken out of a business during its remaining life."² This measure of value is dependent on a variety of conditions that the analyst must forecast, including interest rates, estimates of future cash flows, business prospects, and other valuable competitive advantages. Some measures of value may include a P/E ratio below those of comparable firms, a high book-to-market value, or a high amount of free cash flow to equity. Buffett believes that the more one knows about a particular business, sector, or industry, the better one will be at making assumptions to value it. Therefore, following the Buffett strategy involves purchasing only stocks whose underlying business one truly understands and avoiding all others, even if they possess huge growth potential. With the exception of Apple Computers (in 2016), Buffett generally avoids investing in technology stocks because he does not believe he understands the companies' business models well (even though one of his best friends is Microsoft Chairman Bill Gates!).
Dreman contrarian value	A seasoned money manager, David Dreman was founder and chairman of Dreman Value Management LLC (the value strategies and funds were purchased by Foundry Partners in 2016). He is the author of several prominent books on the low P/E, contrarian value investment approach, including <i>Contrarian Investment Strategies: The Next Generation</i> (1998) and <i>Psychology and the</i>	This contrarian strategy attempts to pick stocks that have gone out of favor with the public and avoid stocks that are "hot." This usually involves starting with stocks with low P/E ratios relative to the stock universe. Dreman advocated starting with stocks with P/E ratios in the bottom 40% of the universe. Dreman seeks companies with high dividend yields to contribute to total return. He favors large to midsized companies that have good visibility in a market rebound and, due to investor scrutiny, are less susceptible to

(1988), and *Technology and the Stock Market* (1977).

investor certainty, are less susceptible to accounting manipulation. Dreman favors companies of financial fortitude that can endure even when they fall out of favor with investors. Just because it's cheap doesn't mean it's good, so Dreman seeks companies with high earnings growth relative to the S&P 500. Finally, Dreman believes diversification is extremely important. Thus the investor should pick at least 20 to 30 stocks and spread the picks across a variety of industries.

Driehaus growth

Richard Driehaus was the founder of Chicago-based Driehaus Capital Management. It was one of the top small- to mid-cap money managers. He was part of *Barron's* All-Century Team, a group of 25 fund managers that includes such great investors as Peter Lynch and the late John Templeton. His strategies are discussed in *Investment Gurus* (1997) by Peter J. Tanous.

This approach favors companies with momentum in earnings and price. At the core of this strategy are earnings surprises. Companies with positive earnings surprises are buys, and companies with negative earnings surprises are sells. An earnings surprise can be defined many ways. One common definition is actual earnings versus the consensus of all analysts' forecasts. The strategy also values surprises in which the range or standard deviation of estimates is tighter, which has a more significant impact on subsequent returns. Investors also should focus on early earnings announcements that could signal upcoming earnings. In this strategy, one must determine what constitutes a sufficiently large surprise and what is the minimum number of analysts that should follow a stock to consider it valid. These are subjective variables. This strategy looks for companies with positive price momentum over the last four weeks. The strategy also considers how stocks do on a relative strength

basis versus the S&P 500 and how the stocks' industries do versus the S&P 500. Driehaus preferred small- to mid-cap stocks for this strategy and avoided international stocks, thereby excluding ADRs. The investor also may wish to monitor liquidity in terms of trading volume to make sure the chosen stocks are

Fisher
growth/value

Philip A. Fisher's investment philosophy is described in several books that he wrote, including *Common Stocks and Uncommon Profits* (1958), *Conservative Investors Sleep Well* (1975), and *Developing an Investment Philosophy* (1980). His works have subsequently been published in *Common Stocks and Uncommon Profits and Other Writings by Philip A. Fisher* (1996).

volume to make sure the chosen stocks are tradable for the size of the portfolio.

Fisher speaks about relying on recommendations of investment professionals for some screening, so analyst recommendations are a good source of data for a strategy based on his thinking. The Fisher-based strategy involves identifying growth stocks along several dimensions, including excellent management, long-term growth potential due to the company's product line, and a competitive advantage over others in the industry. Fisher sought companies that passed his 15-point system. This system consists of a product that could result in a significant increase in sales in the future, a low-cost advantage in the industry, a strong marketing organization, a strong R&D department, effective cost analysis and accounting, financial strength, management with superior skills, good working relationship among the management team, management with a depth of skills, management that works well together, management with integrity, good labor relations, good and honest investor relations, above-average profitability, and an ability to prevent competitors from entering the company's space in the market. Fisher focused primarily on the manufacturing industry, which he understood best. However, his approach can be applied to other industries. The strategy does not discriminate based on the size of the firm. Fisher believed in limits to diversification and thought that 20 stocks were sufficient in a portfolio. He recommended selling when you realize your analysis is wrong but not out of short-term considerations.

Lakonishok
value

Josef Lakonishok is currently the William G. Karnes Professor of Finance at the College of Business, University of Illinois at Urbana-Champaign. He is also the CEO and founder of LSV Asset

The Lakonishok strategy rests on the belief that certain companies that go out of favor will return to favor if the investor is patient. The idea is to purchase companies with P/B, P/E, P/S, and P/CF ratios lower than the industry averages. The strategy could lead the investor to purchase absolute disaster companies. In order to prevent

	Management. For more information, see Lakonishok et al. (1994).	this, Lakonishok advises picking companies that are starting to rebound, either by price return or by analyst revisions.
Lynch growth/value	Peter Lynch is known for his success as the portfolio manager of the Fidelity Magellan Fund from 1977 to his retirement in 1990. His record was impressive. Peter Lynch earned an annual average rate of return of 29.23% compared to the S&P 500 annual return of 15.81%. When Lynch took the helm at Fidelity Magellan, the fund had a mere \$20 million in assets. When he retired in 1990, Magellan had grown to a mammoth \$14 billion in assets. Lynch is a fundamental portfolio manager, not a quantitative one. He did a lot of research that is hard to model with data (e.g., on Hanes pantyhose). His strategies are detailed in his books <i>One Up on Wall Street</i> (1989) and <i>Beating the Street</i> (1993).	Lynch's strategy is to know the businesses he invests in very well. A screen based on this sort of strategy is slightly more involved than one based on a quantitative approach. One of the first steps involves computing a modified PEG ratio, which Lynch defines as the P/E ratio divided by the growth rate of earnings plus the dividend yield. A ratio below 1.0 is considered attractive and above 1.0 unattractive. Lynch also seeks stable and consistent earnings. The strategy avoids stocks with enormous, probably unsustainable earnings growth. The target firms are relatively neglected. Thus one might look for stocks with low institutional ownership or analyst coverage. One must observe the size of the firm or its geographic reach, since Lynch thinks that a firm with a small market share has more room to grow than one with a large market share. The Lynch strategy involves finding companies with strong balance sheets and low debt relative to their industries. The strategy considers net cash per share; a high value relative to the industry is best. Finally, Lynch likes to check whether insiders are buying the stock or if the company is buying back shares, either event a potentially positive sign for the company's future.
Miller value	Bill Miller was the CEO of Legg Mason Capital Management Inc., a division of the Baltimore financial services firm Legg Mason (purchased by Franklin Templeton Investments in 2020). He currently manages	Miller's strategy focuses on identifying securities that are trading below their intrinsic value, as indicated by his multifactor valuation analysis. Miller differs from many value managers in that his approach focuses on cash earnings, not accounting earnings. The strategy seeks to uncover firms that

2020). He currently manages money at his own firm, Miller Value Funds. During his tenure managing the Legg Mason Value Trust from 1982 to 2016, he beat the S&P 500 in 21 of those 35 years.

strategy seeks to uncover firms that may be undervalued based on the present value of future cash flows to the firm and its shareholders. Diversification represents a crucial element in Miller's strategy. However, the strategy aims for diversification among the stocks it incorporates rather than sheer quantity. In fact, Miller's strategy generally has been to hold very few stocks (at the end of 2020, the Miller Opportunity Trust fund had only 42 holdings) while placing a larger emphasis on the research that goes into valuing the companies he does include. Annual turnover generally has been around 20%, which is the equivalent of holding a stock for 5 years. By focusing on companies that are being shunned by the market, this strategy takes on higher risks in hope of higher returns. The value moniker also can be misleading because Miller screens for a wide variety of stocks and has purchased many internet "growth" stocks.³

Muhlenkamp
growth/value

Ronald Muhlenkamp founded Muhlenkamp & Company in 1977. His firm manages accounts for individuals and institutions. The Muhlenkamp Mutual Fund (MUHLX) was launched in 1988. The Muhlenkamp Fund has underperformed the S&P 500 for the last 15 years (through December 2020).⁴

This strategy involves searching for companies with ROEs above the historical average for all companies. One might also look for ROEs that have remained stable over the last five years. The company should be well priced according to the P/E ratio. The strategy looks for companies with higher earnings growth than that of their industry peers. One also should look at profit and cost control via a factor such as the operating profit margin of the firm relative to its industry. As do many famous strategies, this one looks at financial stability, both through liabilities as a ratio to assets and through the amount of free cash flow of the firm. Muhlenkamp also considers labor relations. Although labor relations are a relatively qualitative dimension, data from MSCI KLD or other firms can be used to assess them

John Neff was the portfolio manager of the Vanguard Windsor Fund from 1964 until his retirement in 1995. Neff wrote a book entitled *John Neff on Investing* (1999). For many years prior to his death in 2019, Neff was a member of the distinguished *Barron's* Roundtable.

models can be used to assess them.

Finally, Mulhenkamp believes no model works for all time periods; thus you must adapt your model to the times. This might require reestimating model parameters on some regular basis.

Neff's strategy involves finding low P/E ratio stocks. A company should have strong earnings growth forecasts, but not so strong that the forecasts might represent investor hype (somewhere between 7% and 20%). The Neff strategy finds companies with high DY ratios. Ultimately, Neff wants to find stocks with the lowest dividend-adjusted PEG ratios, the PEG ratio equaling the P/E ratio divided by the stock's rate of earnings growth. Low PEG ratio stocks fit what he called the "cheapo" profile. The strategy also advocates strong sales growth, free cash flow after paying for capital expenditures, and good operating margin relative to the industry. Neff also strongly encourages looking for little-known stocks.

O'Shaughnessy growth James O'Shaughnessy, chairman and co-CIO of O'Shaughnessy Asset Management and founder of NetFolio, a basket-trading firm, gives his insights into stock picking in his book *What Works on Wall Street* (1998). O'Shaughnessy believes that consistency beats the market and that many investors fail to be consistent in their investment approaches.

Within the strategies typically used to pick growth stocks, O'Shaughnessy found that earnings growth and relative strength were the key factors of successful portfolios. He found that persistent earnings growth (e.g., over last five years) is key. Stocks with the highest earnings growth are not necessarily the ones to buy. He also found that stocks with the highest relative strength, which he defined as the largest price increase in the previous year, continue to do well for at least one year. This is more commonly known as a *momentum strategy*. The last growth strategy involves picking stocks in the upper half of the market capitalization spectrum. A possible revised earnings screen thus looks for companies with four consecutive years of growth in

		four consecutive years of growth in earnings. To avoid overvaluation, O'Shaughnessy caps the P/S ratio at 1.5 (sales per share of recent 12 months). Finally, the 50 stocks with the highest relative price growth in the last 12 months are chosen for the portfolio.
O'Shaughnessy value	See above.	O'Shaughnessy's value strategy centers on what he terms the "investable universe." This is a stock universe sufficiently liquid for the portfolio managers' trades. The definition of the investable universe is subject to interpretation and to the size of the portfolio one is trading, but we limit it to the top 50% of stocks, on the exchange by market capitalization. Next, screen for stocks with the lowest P/E, P/B, P/S, P/CF, and highest DY ratios. (Data mining is a worry with this strategy because O'Shaughnessy tested a vast number of factors and chose the ones that worked.) The next element of the strategy is to choose stocks with more shares outstanding than the average stock in the universe, cash flows per share that exceed the universe average, and total sales 1.5 times the universe average. The final step is to select the 50 stocks with the highest dividend yield, excluding utility companies.
Piotroski value	Joseph D. Piotroski is a professor of accounting at Stanford University. He is an expert on financial statement analysis and researches the impacts of financial information on equity prices and valuation, market expectations, and behavioral investment. For more info, see Piotroski (2000).	The idea of the strategy based on Piotroski's work is to find extremely low P/B ratio stocks that have been ignored by analysts or experienced financial distress and about which the flow of information is inefficient. A nine-point scale of financial condition identifies firms with improving financial health. The firms also must meet a minimum profitability standard. Finally, the firms must have positive ROA growth and meet accounting, financial leverage, liquidity, and operating efficiency criteria.
Templeton growth/value	Sir John Templeton founded the famous Templeton Mutual Fund	The foundation of the Templeton strategy is finding value stocks or stocks with low P/B or similar ratios. However, the strategy goes

Templeton Mutual Fund family. He was a student of Benjamin Graham, and his investment approach was rooted in hunting for bargains. Templeton was also known for his contrarian nature, which led him to buy when the majority was selling and vice versa. In his quest to find bargains anywhere, he purchased stocks both domestically and internationally. Books on Templeton's investment philosophy include *Lessons from the Legends of Wall Street* (2000) by Nicki Ross and *Money Masters of Our Time* (2000) by John Train.

similar ratios. However, the strategy goes further in trying to separate "bad" value from "good." This strategy looks for value stocks that have favorable future earnings growth. A company's P/E ratio must sit below its five-year average. One should be wary of companies with abnormally high P/E ratios and perform the standard trimming of outliers (discussed in Appendix 5B). Companies should be selected for positive earnings growth in the last 12 months, as well as the last five years. The strategy also involves choosing firms with positive five-year earnings growth estimates from analysts. In order to capture the competitive advantage idea of Templeton, one might search for stocks that have higher estimated rates of earnings growth in the next five years than the industry average. The strategy calls for firms with positive operating margins over the last 12 months and a 12-month operating margin greater than the 5-year average. Finally, since financial strength was important to Templeton (as it was to Peter Lynch), one might choose firms whose total-assets-to-total-liabilities ratio is higher than the industry average.

¹See Buffett (2020).

²See Buffett (1996).

³See <https://millervaluefunds.com/miller-opportunity-trust/>.

⁴See <https://muhlenkamp.com/>.

TABLE 5A.2

Well-Known Rule-Based Stock Strategies

Screen Name	Background	Description
Analyst upgrades	This strategy relies on the belief that markets are not semistrong efficient and that new information is discounted into the stock not immediately but gradually.	This is a strategy of buying stocks that analysts have upgraded and selling stocks that have been downgraded. The theory is that financial analysts are more knowledgeable about the companies they follow than any other outside investors and that they are the best informed to decide when a stock is not properly valued and deserves action. Although the assumption is that a stock will adjust gradually to an analyst's recommendation, an investor pursuing this strategy must act immediately on the recommendation.
Dogs of the Dow	In 1884, Charles Dow created the first market index by choosing 11 of the leading corporations' stocks (which happened to be mostly railroads at the time) and calculating the simple arithmetic mean of their prices. This was the birth of a now broadly used market barometer, the Dow Jones Industrial Average. The Dow is currently a compilation of 30 stocks chosen by the editors of the <i>Wall Street Journal</i> to represent a cross section of the largest industrial companies in the economy. The Dow is a price-weighted index unlike the S&P 500's market capitalization weighting. The Dogs of the Dow are a selection of 10 stocks in the Dow.	At the beginning of each year, pick the 10 stocks in the Dow with the highest dividend yield and equally weight them.
Earnings momentum	The theory behind this strategy is that large positive (negative) earnings surprises indicate that a company is more likely to exceed (fall short of) future earnings expectations and hence follow an increasing (decreasing) price	This strategy involves buying stocks whose earnings exceed expectations and selling stocks that fail to meet earnings targets. Like the analyst upgrades strategy, it depends on the market's failure to rapidly incorporate public information and an investor's

	trajectory. Chan, Jegadeesh, and Lakonishok (1996) found that, on average, if an earnings surprise occurs in either direction, then the market is likely to be surprised in the same direction over the subsequent two announcements as well.	ability to take advantage of this sluggishness.
Earnings revisions	Earnings revisions occur when management announces new earnings or target earnings and when financial analysts revise their projections upward or downward (not necessarily in response to such announcements). These earnings revisions mean that the outlook for the business has changed.	The earnings revision strategy is to buy and sell according to the direction of revisions. If projections are revised downward, the stock is probably overvalued. If management revises earnings due to inaccuracies in the company's financial statements, the announcement could trigger a crisis of confidence in the company's management and widespread selling of the stock.
GARP	GARP, or growth at a reasonable price, is a combination of both growth and value methods of investing. Many criticize this approach for trying to combine so many elements of other strategies that it fails to serve as a reliable investment tool of its own.	The strategy focuses on uncovering companies with high earnings growth and projections, strong fundamentals, and low P/E and P/B ratios. The aim is to find firms whose stock values (as measured by P/E, P/B, or some other ratio) are lower than their overall rates of earnings growth.
Neglected firms	Investing in neglected firms may be a way to exploit inefficiencies in the market. Areas of the	The neglected-firms strategy involves seeking out stocks that are covered by few, if any, financial analysts and

market that attract media attention, public interest, and sophisticated institutional followings are more likely to be properly priced than areas off the beaten investment track.

attempting to discover sources of value that may have been overlooked by other investors. Neglected firms tend to be small, low-profile companies that have not received much media or analyst coverage. It is likely that neglected firms' stock prices do not reflect all the relevant information available and that their prices will react sluggishly to relevant news. This opens a window of time and opportunity for an astute investor to purchase undervalued, neglected stocks and reap the rewards when the market recognizes the stocks' true values.

TABLE 5A.3

Stock Screens Based on Well-Known Strategies of Individuals

Name	Screening Steps
Buffett	Step 1: Market capitalization is in the top 30% of NYSE, AMEX, and NASDAQ listed equities. Step 2: ROE is greater than 15% for each of the past three years. Step 3: The free cash flow per share is in the top 30% of the database. Step 4: NPM is greater than the industry average. Step 5: The forecasted cash flow per share in five years is greater than the current share price. Step 6: The growth rate in the market value is greater than the growth rate in the book value.
Dreman	Step 1: Market capitalization is in the top 80% of the S&P 1500. Step 2: P/E ratio is in the bottom 40% of the S&P 1500. Step 3: D/E ratio is less than 1. Step 4: The ratio of total liabilities to total assets is below the industry median. Step 5: The growth rate of EPS is greater than that of S&P 1500. Step 6: The growth rate of EPS is greater than the industry median. Step 7: The growth rate of EPS this year is greater than that of last year. Step 8: Forecasted growth rate of EPS next year is greater than the forecasted growth rate of EPS this year. Step 9: Dividend yield is greater than the S&P 1500 median. Step 10: The growth rate of net income is greater than 8%.
Driehaus	Step 1: The percentage change in EPS is growing for the past five fiscal years. Step 2: The percentage change in EPS current year is positive. Step 3: Percentage difference between announced earnings and consensus earnings is greater than 10%. Step 4: The percentage change in EPS current year is greater than the industry average. Step 5: Share price is higher than the 200-day moving average. Step 6: The return of the stock's industry is greater than the return of the S&P 500 for the last quarter. Step 7: 20-day on-balance volume is positive. Step 8: The number of analysts following the stock is less than or equal to five. Step 9: Size is in the bottom 85% of all stocks listed in the NYSE, AMEX, and NASDAQ.
Fisher	Step 1: Sales have been growing for the past three years. Step 2: The growth rate of sales over the last three years is greater than the industry median. Step 3: NPM is greater than the industry median for each of the last three fiscal years. Step 4: PEG ratio is in the bottom 10% of the industry. Step 5: Research-and-development-to-sales ratio (RNDS) is greater than the industry median. Step 6: The growth rate in sales is greater than the growth rate in the research and development expenditures. Step 7: The dividend is zero.
Lakonishok	Step 1: Market capitalization is in the top 30% of NYSE, AMEX, and NASDAQ equities. Step 2: P/E ratio is less than the industry median. Step 3: P/B ratio is less than the industry median. Step 4: The forecasted EPS for the next year is greater than the forecasted EPS for the current year. Step 5: The EPS forecasts have been revised upward at least once in the last month, and there have been no downward revisions. Step 6: The return of the stock has been greater than the return of the S&P 500 over the last 6 months.
Lynch	Step 1: P/E ratio is less than the industry average. Step 2: PEG ratio is

less than 1. Step 3: The ratio of P/E to DY is less than 4. Step 4: Current P/E ratio is less than the average P/E ratio of the past five years. Step 5: The ratio of long-term debt to equity is less than the industry average. Step 6: The ratio of long-term debt to equity is less than 1. Step 7: The ratio of net cash per share to share price is greater than 0.2. Step 8: The growth rate in EPS is between 0% and 50%. Step 9: The ratio of the current-year insider buying to insider selling is greater than 1.5. Step 10: Institutional ownership is less than 50%. Step 11: The market capitalization is less than \$5 billion.

Miller Step 1: The ratio of market capitalization to free cash flow is less than 3. Step 2: The current-year free cash flow is greater than the previous-year free cash flow. Step 3: PEG ratio is less than 1.5. Step 4: GPM is greater than the industry median. Step 5: ROE is greater than the industry average. Step 6: Sales have been growing for each of the last five years. Step 7: The ratio of the long-term debt to equity is less than the industry average. Step 8: The size is in the top 75% of NYSE, AMEX, and NASDAQ equities.

Muhlenkamp Step 1: ROE is greater than the industry average. Step 2: The average ROE of the past five years is greater than the industry average. Step 3: P/E ratio is less than the industry average. Step 4: P/B ratio is less than 2. Step 5: The compounded annualized growth rate of EPS over the last five years is greater than 0. Step 6: The compounded annualized percentage growth rate in EPS is greater than the industry average. Step 7: NPM is greater than the industry average. Step 8: The ratio of total liabilities to total assets is less than the industry average. Step 9: The cash ratio is greater than 1. Step 10: Free cash flow is positive.

Neff Step 1: P/E ratio is less than the industry average. Step 2: The percentage growth rate in EPS is between 7% and 20%. Step 3: PEG ratio is less than 1. Step 4: Sales have been growing for each of the past five years. Step 5: OPM of the current year is greater than OPM of the previous year. Step 6: OPM is greater than the industry average. Step 7: Free cash flow per share has been growing for each of the past three fiscal years.

O'Shaugnessy growth	Step 1: The market capitalization is in the top 25% of the S&P 1500. Step 2: P/S ratio is less than 1.5. Step 3: The percentage growth in EPS is greater than 0. Step 4: The percentage change in EPS for the past five years is greater than that of the S&P 500. Step 5: P/E ratio is less than the industry average. Step 6: Choose the stocks with the largest one-year momentum (i.e., 12-month return) of the remaining stocks.
O'Shaugnessy value	Step 1: The market capitalization is in top 50% of NYSE, AMEX, and NASDAQ stocks. Step 2: P/E ratio is less than the industry average. Step 3: The ratio of price to cash flow is less than the industry average. Step 4: Sales are greater than 1.5 times the industry average. Step 5: P/S ratio is less than 1.25. Step 6: DY ratio is greater than that of the S&P 500. Step 7: The percentage change in EPS is between 0% and 50%. Step 8: The number of shares outstanding is greater than the average number of shares outstanding of the stocks within the database. Step 9: DY ratio is highest among the remaining stocks excluding utilities.
Piotroski	Step 1: The market capitalization is in the top 30% of NYSE, AMEX, and NASDAQ stocks. Step 2: P/B ratio is in the lowest 30% of NYSE, AMEX, and NASDAQ stocks. Step 3: ROA is positive. Step 4: D/E ratio is less than 1. Step 5: The ratio of long-term debt to equity in the previous year is less than the ratio two years ago. Step 6: NPM of the current year is greater than NPM of the previous year. Step 7: Cash flow from operations is greater than net income. Step 8: The cash ratio of the current year is greater than the cash ratio of the previous year. Step 9: The ratio of cash to current liabilities in the current year is greater than the ratio in the previous year. Step 10: The ratio of sales to total assets is greater than the industry average.
Templeton	Step 1: P/B ratio is in the lowest 40% of the stock database. Step 2: P/E ratio is less than the average P/E ratio of the past five years. Step 3: The percentage changes in earnings over the trailing 12 months and the past 5 years are positive. Step 4: The percentage change in EPS is greater than the industry average. Step 5: OPM is greater than the average OPM of the past five years. Step 6: The ratio of the long-term debt to equity is less than the industry average. Step 7: The ratio of total assets to total liabilities is greater than the industry average. Step 8: ROE is greater than the industry average.

Note: D/E ratio is the debt-to-equity ratio; DY ratio is the dividend yield; EPS is earnings per share, GPM is gross profit margin; NPM is net profit margin; OPM is operating profit margin; P/B ratio is the price-to-book ratio; P/E ratio is the price-to-earnings ratio; PEG ratio is the price-to-earnings-to-growth ratio; ROE is the return on equity; and RSI is the relative strength index.

TABLE 5A.4

Stock Screens Based on Well-Known Rule-Based Strategies

Name	Screening Steps
Analyst upgrades	Step 1: Select the 50 stocks whose average analyst rating has increased the most over the last month.
Dogs of the Dow	Step 1: Exclude all companies not in the Dow Jones Industrial Average. Step 2: DY ratio is in the top 33% of Dow stocks.
Earnings momentum	Step 1: The percentage change in EPS for the past five years is positive. Step 2: The stocks are within the top 20% of the rankings by the percentage change in EPS for the past five years. Step 3: The forecasted percentage change in EPS for the next year is greater than the percentage change in EPS for the past five years.
Earnings revisions	Step 1: Screen for the largest percentage revision in forecasted EPS for the next year during the last two months according to IBES. Step 2: Choose the top 20% in terms of the percentage change in EPS for the past five years. Step 3: The percentage change in EPS is greater than the industry average. Step 4: Choose those with the largest earnings surprises.
GARP	Step 1: PEG ratio is less than or equal to 1. Step 2: The percentage change in EPS for the past five years is greater than that of S&P 500. Step 3: The percentage change in EPS for the past three years is greater than the industry average. Step 4: The ratio of total liabilities to total assets is less than the industry average. Step 5: D/E ratio is less than 1.
Neglected firms	Step 1: Market capitalization is in the top 75% of NYSE, AMEX, and NASDAQ securities. Step 2: The number of analysts following the stock is less than or equal to two. Step 3: The compounded annualized percentage change in EPS is greater than the industry average. Step 4: P/E ratio is greater than the industry average. Step 5: NPM is greater than the industry average. Step 6: P/B ratio is less than or equal to 3. Step 7: The average ROE of the past five years is greater than the industry average.

Note: D/E ratio is the debt-to-equity ratio; DY ratio is the dividend yield; EPS is the earnings per share; NPM is the net profit margin; P/B ratio is the price-to-book ratio; P/E ratio is the price-to-earnings ratio; ROE is the return on equity; and PEG ratio is the price-to-earnings-to-growth ratio.

On Outliers

5B.1 GENERAL CONCEPTS

An *outlier* is an observation in a distribution that is extreme and looks very inconsistent with the rest of the data. These outliers could arise due to data errors by the data supplier or measurement error or could be correct values representing novel phenomena. If they arise due to data errors, it is a good idea for the portfolio manager to remove them entirely. If they are accurate and represent novel phenomena, there is no need to remove them. In practice, however, the portfolio manager may be unable to determine whether he or she is dealing with data errors or with novel phenomena. Thus, the portfolio manager in this case may wish to modify their values to some reasonable values just for safety. The outliers will still be strong signals vis-à-vis other stocks.

Formally speaking, we consider two possibilities. The first possibility (which might be called a *null hypothesis*) is that all the observations are random draws from distribution A . In this case, no outliers exist. The second possibility (which might be called an *alternative hypothesis*) is that most observations are random draws from distribution A , but some observations are from distribution B . Then we might label the observations from distribution B as outliers. In order to apply this categorization of outliers to real data, we need to know or assume the A distribution (of the null hypothesis). If we do not know distribution A or we are very uncertain about it, detecting outliers becomes theoretically difficult, and excluding suspect observations from the analysis might not be justified.

Popular outlier detection techniques assume that the underlying distribution is a normal distribution. Grubb's (1950) test, Dean-Dixon's (1951) test, and Rosner's (1975) test are three examples. There are also procedures applicable to non-normal distributions. Iglewicz and Hoaglin (1993) is a good reference for further discussion.

Dealing with outliers involves two steps. In the first step, we identify certain observations as outliers. This is the *labeling step*. In the second step, we decide what to do with identified outliers. We may remove these observations (*trimming* or *truncation*), replace their values with something less extreme (*winsorization*), or we may use a more creative approach.

5B.2 SPECIFIC PRACTICAL TECHNIQUES

5B.2.1 Utilizing the Z-Score

One method to deal with potential outliers is simply to label stocks with the absolute value of the Z-score greater than 3 as outliers. While easy, this is not a reliable method because the extreme values were used to compute means and standard deviations. A second method is the modified Z-score method, where, rather than using the standard deviation of the factor values, the median is used to compute the modified Z-score. The modified Z-score,

$\tilde{z}$, is defined as $\frac{(\beta_{i,j} - \bar{\beta}_j)}{M(\beta_{i,j})}$, where $M(\beta_{i,j})$ is the median value of factor j

across stocks. All observations with the absolute value of the modified Z-score greater than or equal to 3 are considered outliers. Once the outliers are identified, they can be eliminated or their values can be adjusted.

Winsorization is the process of replacing outliers with other values in order to prevent the outliers from unduly influencing the estimates. One form of winsorization we discussed in the main text is to convert all Z-scores with a value greater than 3 to the value 3 and those with a value less than -3 to -3 .

Trimming is the process of removing the n values consisting of $n/2$ of the largest values and $n/2$ of the smallest values. The Z-scores are then recomputed for the remaining data. A version of this is suggested in the

text, which is to simply remove all Z-scores whose absolute values are greater than some determined value, such as 3, 4, or 5.

5B.2.2 The Interquartile Method

One effective method to deal with outliers when building QEPM models is the *interquartile method*.¹ To implement the interquartile method, compute for each factor the interquartile range of the factor for the cross section of factors amongst stocks. This will be the value that is a measure of the middle 50% of factor values. This is computed by finding the third-quartile entry of every factor (Q3) and the first-quartile entry of every factor (Q1) and then subtracting them (Q3 – Q1) to find the interquartile range (IQR). At this point, a decision must be made on what deviation from the 75th percentile and 25th percentile constitutes an outlier. This is known as the *IQR coefficient*. For example, if the IQR coefficient is equal to 3, then upper and lower bounds of the factor are computed as²

$$UB = Q3 + 3IQR \quad (5B.1)$$

$$LB = Q1 - 3IQR \quad (5B.2)$$

Then consider all stocks with factor values above the upper bound and below the lower bound to be outliers.

For these, set their values to missing and compute the Z-scores for the remaining stocks. For the outlier stocks, fix the Z-scores at the maximum and minimum of the nonoutlier stocks' Z-scores.

A Numerical Example

In December 2020, the raw data for our beta calculations based on excess market returns and company stock returns for the universe of the top 3,000 stocks in the United States by market capitalization ranged from –5.78 to 7.48. These corresponded to the stocks AVGO and IBIO. Based on an IQR coefficient of 3, the values for the IQR procedure were Q3 = 1.7391, Q1 = 0.8721, IQR = 0.8669, UB = 4.3398, and LB = –1.7287. This removed 0.83% of all of our cross-sectional data, or 25 observations, but left very stable Z-score values for 99.17% of our stock data. The range of Z-scores was also less extreme. Using the traditional Z-score method, the maximum

and minimum Z-scores were 7.46 and -8.75 , respectively, compared to 4.10 and -3.97 using the IQR method.

5B.2.3 The Ranking Method

Another reasonable method to remove outliers is to use a ranking method. Rather than use Z-scores for the individual factors, the portfolio manager could use cross-sectional ranks. That is, every evaluation period, order the stocks according to the stock's factor value, and rank the stocks from highest to lowest. Then compute Z-scores on the actual rank, rather than the factor values. This resolves the outlier problem; however, it also loses information contained in the distance between different stocks' factor values, keeping just the relative value of Z-score between stocks. The ranking Z-score value for each stock in a given month is given by

$$Z_{it}^{\text{rank}} = \frac{\text{rank}(f_{it}) - \frac{\sum_{i=1}^N \text{rank}(f_{it})}{N}}{\sigma_{\text{rank}(f_{it})}} \quad (5B.3)$$

where rank is the cross-sectional numerical rank of each stock in the universe from high (better attribute) to low.

A Numerical Example

Using the same example as in Section 5B.2.2 with betas in December 2020 with the rank method, the maximum and minimum rank Z-scores were 1.7312 and -1.7312 , respectively.

5B.2.4 The Percentile Ranking Method

A method very similar to the ranking method is the *percentile ranking method*. Once the N stocks are ranked from highest to lowest, the following calculation is made for each stock:

$$Z_{it}^{\text{per}} = \frac{N - \text{rank}(f_{it})}{N} \quad (5B.4)$$

Using the same example as in Section 5B.2.2 with betas in December 2020 using the percentile method, the highest-beta stock, AVGO, would have a 100th percentile and the lowest-beta stock, IBIO, would have a 0th percentile. With multiple factors, the factors can be combined by percentile rank—and this can be normalized further to obtain an aggregate percentile rank that ranges from zero to one.

¹ See Chincarini (2017, 2018) for more information. This is a better method than simple winsorization, which has a major drawback: after the winsorization method has removed certain stocks and the Z-scores are recomputed, there may be still a group of outliers. The procedure must then be repeated again, leading to an iterative process that takes more time and might be less efficient.

² Another common value for the IQR coefficient is 1.5. Other values can be chosen as well.

Converting Z-Scores to Returns

In some instances, it will be useful to convert Z-scores to returns, as we have already mentioned in this chapter. For example, when computing after-tax returns or after transactions costs returns, it will be more pertinent to convert one's Z-scores to returns. There are various ways to do this that are more accurate, which we discussed in Sections 5.5 and 5.6. In this appendix, we will give an example using the second method described in Section 5.6 and based on Eq. (5.10), but it could easily be extended to a method also controlling for other risk factors. In particular, Eq. (5.10) shows that when regressing Z-scores on future stock returns, the expected return of a stock conditional on the Z-score is given by $E[r_{i,t}|z_{i,t-1}] = E[r_t] + \delta [z_{i,t-1} - E(z_{t-1})] = E[r_t] + \rho(r_t, z_{t-1})S(r_t)z_{i,t-1}$. When estimated as a cross-sectional regression (or a panel regression), the estimates will be the best predictor of the realized return, that is, $\hat{r}_{it} = \hat{r}_t + \hat{\rho}(r_t, z_{t-1}) \hat{S}(r_t)z_{i,t-1}$.

Suppose you are looking forward and have computed the Z-scores for all stocks in period t . You can then use the distribution of stock returns in period t to translate your Z-scores to either α or expected stock returns. If you do not perform estimations and assume that the correlation between stock returns and the Z-score is 1, then the transformation is simple: convert the Z-score distribution and maintain the relative Z-score values between stocks, but convert the distribution to resemble the first two central moments of the cross section of stock returns. Thus, the Z-scores for each stock would be modified as follows:

$$\tilde{Z}_{it}^1 = Z_{it} \sigma_r + \mu_r \quad (5C.1)$$

where $\tilde{Z}_{it}$ is the adjusted Z-score to be used as the expected return for each stock, and σ_r and μ_r are the cross-sectional standard deviation and mean of stock returns.

However, this is unrealistic and probably exaggerated, since it would be rare for the Z-score and stock returns to be perfectly correlated. If instead a regression had been run or an empirical estimate of the correlation between stock returns and the Z-score had been estimated, then you could transform the Z-scores to the stock return distribution as follows:

$$\tilde{Z}_{it}^2 = \hat{\rho}(r_t, z_{t-1}) Z_{it} \sigma_r + \mu_r \quad (5C.2)$$

where the $\hat{\rho}(r_t, z_{t-1})$ has been estimated on recent data.¹ This would have the effect of scaling down the distribution to reflect the predictive component of the Z-scores in historical data. Of course, all of these examples involve using past data to extrapolate to the future, which will result in a margin of error. Even when historical correlations are taken into account, if the correlation is low, the result only represents a partial explanation of stock returns. Nevertheless, this is a technique that can aid practitioners in transforming Z-scores to stock returns to be used in optimization models that require assessment of transactions costs or tax costs.

5C.1 A NUMERICAL EXAMPLE

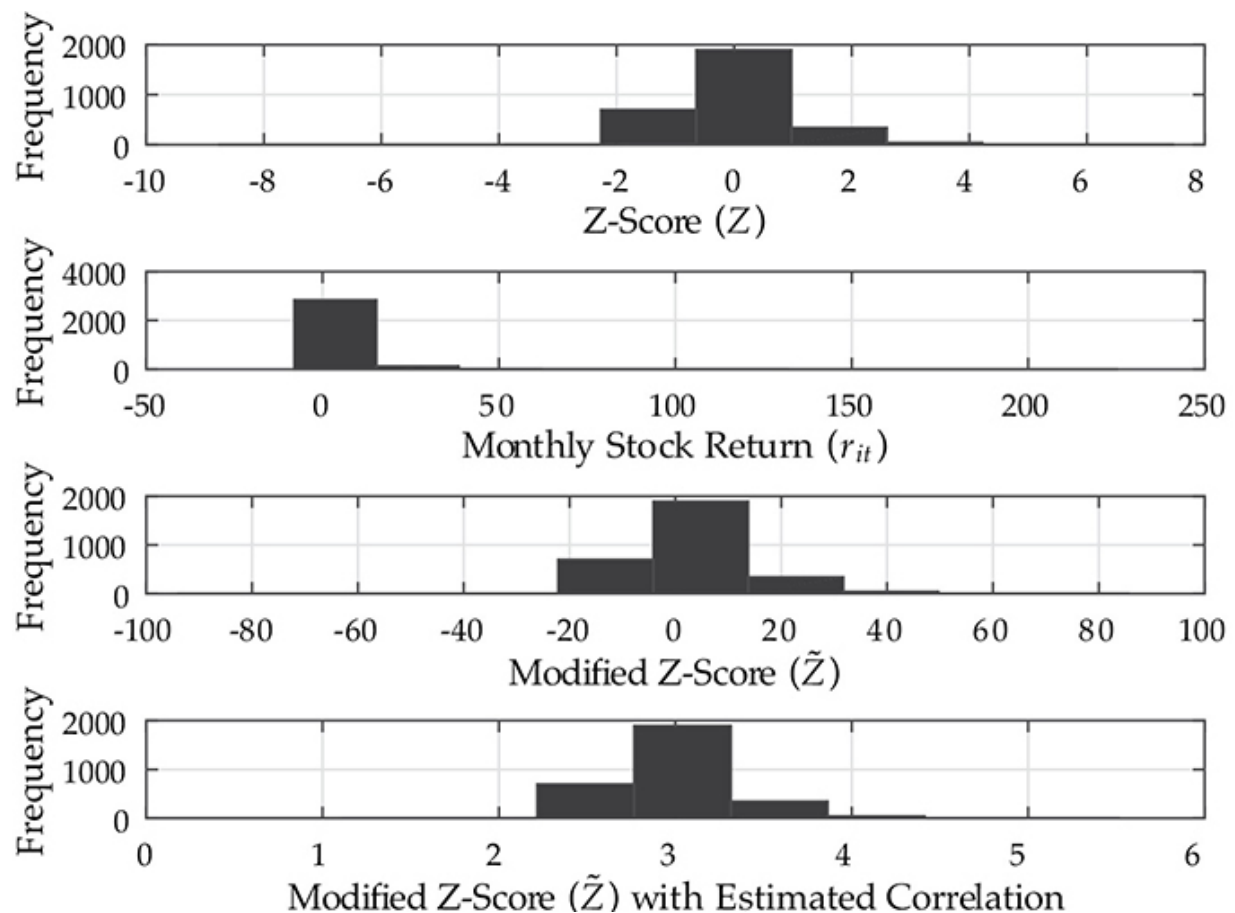
An example of this conversion of Z-scores is shown in [Fig. 5C.1](#) for December 2020. The first graph in [Figure 5C.1](#) shows the distribution of the Z-scores among 3,000 stocks according to a particular Z-score model in December 2020. The second graph shows the distribution of monthly stock returns for the same stocks. The third graph shows the distribution of the modified Z-scores adjusted for the mean and standard deviation of the cross section of stock returns assuming unrealistically a perfect correlation between stock returns and Z-scores. In this particular example, the mean and standard deviation of the cross section of stock returns was 2.98% and

11.07%. The mean of the Z-score was 0 and 1, and the modified Z-score has the same mean and standard deviation as the stock returns. In the fourth graph, we show the modified distribution of Z-scores using the actual historical correlation between Z-scores and stock returns of 0.0306. One can see that this reduces the return distribution based on Z-scores to something more in line with the Z-score predictive power.

FIGURE 5C.1

Conversion of Z-scores to modified Z-scores using data from December 2020 in a universe of 3,000 stocks. In this figure, $\mu_r = 2.98$, $\sigma_r = 11.07$, $\mu_Z = 0$, $\sigma_Z = 1$,

$\hat{\rho}(r_t, z_{t-1}) = 0.0306$, $\mu_{\tilde{Z}} = 2.98$, and $\sigma_{\tilde{Z}} = 11.07$. The means and standard deviations are from a cross section of monthly stock returns.



QUESTIONS

5.1. What is stock screening used for in QEPM?

- 5.2. Compare our Lakonishik-inspired screen with our Piotroski-inspired screen. How are they similar? How are they different?
- 5.3. How might we screen for management's view of the future of a company?
- 5.4. There are many ways for a business to increase its earnings. One way is to increase the net profit margin. Another way is to increase the number of stores in operation. How might one measure a company's market saturation?
- 5.5. What types of factor screening are very common among the famous portfolio managers whom we discuss in this chapter?
- 5.6. What are the advantages of simultaneous stock screening over sequential stock screening?
- 5.7. What is the Z-score, and how is it related to simultaneous stock screening?
- 5.8. (a) What is meant by an ad hoc aggregate Z-score?
(b) What is the danger with using this method of weighting?
(c) Why might some people say that this method is *not really* QEPM?
- 5.9. What is the optimal aggregate Z-score? How do you come up with it?
- 5.10. A portfolio manager is building his portfolio of stocks. His benchmark is the S&P 500. He will focus on three factors to build his portfolio of stocks. These factors are P/B ratio, P/E ratio, and 12-month momentum (M12M). He generally believes that high-momentum stocks will continue to perform well in the future. For the other factors, he bases his judgment on the documented anomalies.

Company Name	Ticker	P/E	P/B	M12M	$z_{P/E}$	$z_{P/B}$	z_{M12M}	$\bar{z}$
Eli Lilly	LLY	28.98	33.46	31.06				
AbbVie Inc.	ABBV	25.63	12.39	27.61				
Perrigo Company	PRGO	-110.18	1.05	-11.92				
Gilead Sciences	GILD	57.60	4.19	-6.62				
Mean S&P 500	21.21	7.87	15.24					
SD S&P 500	908.21	64.73	46.88					

- (a) In the preceding table, for each stock, compute the Z-scores of each factor ($z_{P/E}$, $z_{P/B}$, z_{M12M}).
 - (b) The manager believes that momentum is the most important item in determining stock returns. Thus he wishes to assign it a 50% weighting while weighting the other factors equally. In the preceding table, compute the aggregate Z-score for each stock.
 - (c) If the portfolio manager were to create a long-only portfolio of two stocks weighted by their relative Z-scores, which stocks would go in the portfolio, and what would their weights be?
- 5.11.** Suppose a portfolio manager has created Z-scores for all stocks in the universe for a given factor. What is the probability that a stock with a Z-score of 2 or greater is observed in the population? How is your answer dependent on whether the distribution is normal or non-normal?
- 5.12.** Given the similarity in the signal between the factor group aggregate Z-score and the regular Z-score, why would a quantitative portfolio manager ever use the former?
- 5.13.** A portfolio manager has the data available in the following table. She believes in using a factor group approach to creating aggregate Z-scores. She has decided to construct four groups (i.e., $M = 4$). Group 1 is titled the valuation group (V), group 2 is titled the technical group (T), group 3 is titled the profitability group (P), and group 4 is the alternative group (A). She also has seven factors that she has identified in predicting stock returns. They are the book-to-price (B/P) ratio, the earnings-to-price (E/P) ratio, the cash-flow-to-price (CF/P) ratio, the 12-month past stock return (M12M), standardized unanticipated earnings (SUE), current accruals (CACC), and net profit margin (NPM).

Ticker	B/P	E/P	CF/P	M12M	SUE	CACC	NPM	z_V	z_P	z_T	z_A	$\bar{z}$
CL	0.009	0.037	0.051	27.159	0.500	-0.065	0.167					
KMB	0.007	0.051	0.082	0.999	3.667	-0.089	0.125					
PG	0.140	0.040	0.052	14.150	3.750	-0.039	0.189					
EL	0.074	0.010	0.047	29.777	8.000	-0.043	0.044					
CLX	0.044	0.045	0.065	34.485	2.133	-0.078	0.161					
CHD	0.148	0.036	0.048	25.475	1.000	-0.038	0.164					
Universe												
Mean	0.275	0.041	0.080	9.095	2.067	-0.045	0.108					
SD	0.339	0.026	0.068	17.009	3.017	0.026	0.082					

- (a) In which group would each of these factors most likely be placed?
 - (b) Compute the Z-score of each factor and each stock.
 - (c) Suppose that factors within each group are equally weighted. Compute the aggregate Z-score for every factor group.
 - (d) After careful analysis, the portfolio manager has found that the optimal weights of the factor groups are 20%, 30%, 20%, and 30%, respectively, for the four groups. Compute the aggregate Z-score for each stock.
 - (e) Suppose that the δ in [Equation \(5.8\)](#) is 1. Suppose that the portfolio manager constructs a portfolio with the three stocks with highest Z-scores equally weighted. What would be the difference between the expected return of this portfolio and the return of a portfolio with all six stocks equally weighted?
- 5.14.** Quantitative portfolio managers generally manage their portfolios versus a benchmark. Thus they usually require a set of expected returns and risks to construct a portfolio.
- (a) What problem does this requirement pose when using solely the aggregate Z-score to rank favorable stocks?
 - (b) What are three methods to convert an aggregate Z-score of a stock into an expected return or α for use in portfolio construction?
 - (c) Describe the strengths and weaknesses of each method.
- 5.15.** Suppose that a portfolio manager is interested in whether an aggregate Z-score is useful in forecasting the return of stocks in the next period. Suppose that he takes a month of stock returns and the

aggregate Z-score of the individual stocks and runs the following regression: $r_{i,t} = \gamma + \delta z_{i,t-1} + \epsilon_{i,t}$. What hypothesis should he test?

- 5.16.** Some people express the regression equation $r_{i,t} = \gamma + \delta z_{i,t-1} + \epsilon_{i,t}$ as something called the *forecasting rule of thumb*, which states that the expected return of the stock conditional on the Z-score information minus the unconditional return of the stock is equal to $IC \cdot \text{volatility} \cdot \text{score}$.
- (a) What does it mean when $IC = 0$, and what does it imply for the conditional excess expected return? What would the regression coefficient $\hat{\delta}$ be equal to if this were true?
 - (b) What does it mean when volatility = 0, and what does it imply for the conditional excess expected return? What would the regression coefficient $\hat{\delta}$ be equal to if this were true?
 - (c) What does it mean when score = 0, and what does it imply for the conditional excess expected return? What would the regression coefficient $\hat{\delta}$ be equal to if this were true?
- 5.17.** Suppose that a portfolio manager is examining two sectors of the U.S. economy in which to invest. She constructs Z-scores relative to the sector for each stock. The cross-sectional volatility of stock returns for the latest month is 20% and 40%, and the correlation between the Z-score and returns of stocks is 0.2 and 0.1 for stocks in sectors *A* and *B*, respectively.
- (a) In which sector will a Z-score be relatively more important at forecasting returns?
 - (b) What are $\hat{\delta}_A$ and $\hat{\delta}_B$ of Eq. (5.8)? Is your conclusion the same as in part (a)?
- 5.18.** When is the Z-score model, as used in Eq. (5.8), equivalent to the fundamental factor model? What does this mean?
- 5.19.** (a) Why do portfolio managers who use the Z-score typically eliminate Z-scores with absolute values greater than 3?
- (b) What is an alternative way of handling such Z-scores? What is the common name given to this method?
 - (c) Name a third way of dealing with these Z-scores that is mentioned in the chapter.
- 5.20.** For each test below, describe what it is used for and to what type of data it applies.

- (a) Rosner's test
- (b) Grubb's test
- (c) Dixon's test

¹ The conversion of Z-scores to returns in [Eq. \(5C.2\)](#) is exactly the same as using the estimated [Eq. \(5.8\)](#) to get the predicted values. That is, $\tilde{Z}_{it}^2 = \hat{\gamma} + \hat{\delta}Z_{it}$.

CHAPTER 6

Fundamental Factor Models

In nature there is fundamental unity running through all the diversity we see about us.

—Mohandas Gandhi

6.1 INTRODUCTION

Recall from [Chapter 3](#) the central idea of modern financial economics: *The average return of a stock is the payoff for taking risk.* In a factor model, the factor exposure represents the exposure of a stock to some kind of risk. The factor premium quantifies the payoff to an investor who takes on that risk by buying the stock. The average stock return therefore equals the product of the factor exposure and the factor premium:

$$\text{Average stock return} = \text{factor exposure} \times \text{factor premium} \quad (6.1)$$

In the fundamental factor model, the factor exposure is known. It is some observable fundamental characteristic of the stock, such as its market capitalization or book-to-market ratio. The factor premium, on the other hand, is not known. It is the proportionality between the average stock return and the factor exposure, and it must be estimated empirically.

The formula for the average stock return given above can be extended easily to a more realistic case of multiple factors. When there are K factors, the factor exposures of stock i can be written as $\beta_{i1}, \dots, \beta_{iK}$, and the factor

premium can be written as $f_1, \dots, f_K$.¹ Note that the factor premium does not vary across stocks and therefore does not require the subscript i . The return on stock i , r_i , can be written as

$$r_i = \alpha + \beta_{i1}f_1 + \dots + \beta_{iK}f_K + \epsilon_i \quad (6.2)$$

where α is the constant term and ϵ_i is the error term (i.e., the part of the stock return that does not depend on the K factors).²

The average of ϵ_i is zero, so the average stock return is the cross-sum of the factor exposure and the factor premium³:

$$E(r_i) = \underbrace{E(\alpha)}_{\alpha} + \beta_{i1}E(f_1) + \dots + \beta_{iK}E(f_K) \quad (6.3)$$

For the convenience of exposition, we will define K -dimensional column vectors $\mathbf{f}$ and $\boldsymbol{\beta}_i$ as

$$\mathbf{f} = (f_1, \dots, f_K)' \quad (6.4)$$

$$\boldsymbol{\beta}_i = (\beta_{i1}, \dots, \beta_{iK})' \quad (6.5)$$

Then the preceding equation becomes

$$r_i = \alpha + \boldsymbol{\beta}_i' \mathbf{f} + \epsilon_i \quad (6.6)$$

Using this notation, the average stock return becomes

$$E(r_i) = \alpha + \boldsymbol{\beta}_i' E(\mathbf{f}) \quad (6.7)$$

We can see that the average stock return is simply the product of the factor exposure ($\boldsymbol{\beta}_i$) and the factor premium ($\mathbf{f}$) as before.⁴

The average stock return is the payoff for taking risk—but what is this risk exactly? The risk of a stock has two components, diversifiable risk and nondiversifiable risk.

$$\text{Total risk} = \text{nondiversifiable risk} + \text{diversifiable risk} \quad (6.8)$$

Since investors can eliminate diversifiable risk from their portfolios through diversification, the market only rewards exposure to nondiversifiable risk. Thus we may restate the central idea of modern financial economics as: *The average stock return is the payoff for taking nondiversifiable risk.*

The nondiversifiable risk can be expressed as the product of the factor exposure squared and the risk included in one unit of exposure.⁵ We will call the risk of one unit of exposure the *factor risk*. Then we may write

$$\text{Nondiversifiable risk} = \text{factor exposure}^2 \times \text{factor risk} \quad (6.9)$$

Within the framework of the fundamental factor model, the total risk of a stock (nondiversifiable risk plus diversifiable risk) can be measured by the variance:

$$\begin{aligned} V(r_i) &= V(\alpha + \beta_i f_1 + \dots + \beta_{iK} f_K) + V(\epsilon_i) \\ &= V(\beta_i f_1 + \dots + \beta_{iK} f_K) + V(\epsilon_i) \end{aligned} \quad (6.10)$$

Using the same vector notation that simplified the formula for the stock return, the variance formula becomes

$$V(r_i) = \beta_i' V(\mathbf{f}) \beta_i + V(\epsilon_i) \quad (6.11)$$

where $V(\mathbf{f})$ is a $K \times K$ -dimensional variance-covariance matrix. In this model we can see each of the components of the total risk. The nondiversifiable component—the part that the market rewards—is the product of the factor exposure squared ($\beta_i' \dots \beta_i$) and the factor premium risk [$V(\mathbf{f})$]. The diversifiable component, which an investor can diversify out of his or her portfolio, is the term $V(\epsilon_i)$.

The fundamental factor model can be used to predict both the returns and the risks of stocks, and these predictions can guide a portfolio manager in choosing the best stocks for his or her portfolio. In general, formulating a fundamental factor model involves four broad steps. First, before observing the factor exposure and estimating the factor premium, there is some preliminary work to do, including deciding on the factors to include in the

model, the treatment of the risk-free rate, the time interval and time period of the data, and the investment universe. Second, the factor exposures of individual stocks (β_i) must be determined. The third step is to estimate the factor premium ($\mathbf{f}$) and the constant α from the factor exposure and the return. Finally, an assessment of the total risk should be made by estimating both the factor risk [$\beta_i' V(\mathbf{f})\beta_i$, a.k.a. the *nondiversifiable risk*] and the diversifiable risk [$V(\epsilon_i)$]. We explain these steps in the following sections, and [Table 6.1](#) summarizes the specifics of the modeling process.

TABLE 6.1

Creating a Fundamental Factor Model

Preliminary work:

1. Choose the factors for the model.
2. Determine the treatment of the risk-free rate.
3. Define the investment universe.
4. Decide on the time interval and time period of the data.

Main steps for modeling total stock return (for nonbenchmark managers):

1. Stock returns are estimated with the following equation: $r_i = \alpha + \beta_1 f_1 + \dots + \beta_K f_K + \epsilon_i$.
2. Collect stock-return and factor exposure data for the time period at each time interval.
3. Estimate the factor premium from a panel regression of the stock return on the factor exposure using either the OLS, MAD, or the GLS method.
4. Check the robustness of the factor premium estimation by splitting the data into subsets and comparing the estimates for each subset. If the estimates are similar across subsets, the estimation is robust.
5. If the estimation is not robust, try a different estimation method.
6. Calculate each stock's average return as the product of the factor premium and the vector containing the stock's factor exposures.
7. Decompose the risk of the stock return into its diversifiable and nondiversifiable components.
8. Calculate the correlation between the returns of the stocks in the investment universe.

Main steps for modeling residual stock return (for benchmark managers):

1. Stock returns are estimated with the following equation: $r_i = \tilde{\alpha}_i + \tilde{\beta} r_B + \tilde{\epsilon}_i$.
 2. Collect stock return and benchmark return data for the given time period at the given time intervals.
 3. After estimating $\tilde{\alpha}$ and $\tilde{\beta}$, calculate the residual return as $\tilde{r}_i \equiv \tilde{\alpha}_i + \tilde{\epsilon}_i$.
 4. Estimate residual return with the following equation: $\tilde{r}_i = \alpha + \beta_1 f_1 + \dots + \beta_K f_K + \epsilon_i$.
 5. Follow steps 2 through 8 of the main steps for modeling total stock return, substituting the residual return for the total stock return.
-

6.2 PRELIMINARY WORK

A sound model begins with good planning. The preliminary work of the portfolio manager is to make a series of decisions that will serve as a blueprint for construction of the model. The primary decision is the choice of factors. Other preliminary decisions determine the shape of the data set (including the nature of the stock returns, the time interval between data points, and the overall time horizon of the data) and the scope of stocks to

consider for the portfolio. These general decisions come into play in building any factor model.

6.2.1 Choosing Factors

Factors represent risk. There are all kinds of factors, some of which describe stock characteristics and some of which describe conditions in the overall market.⁶ Fundamental factors are observable characteristics of the stock itself, and they usually can be read (or easily calculated) from financial statements. Some technical factors and analyst factors also count as fundamental factors. For reference, here are some representative fundamental factors:

1. *Fundamental valuation factors*: dividend yield, book-to-price ratio, earnings-to-price ratio, and the sales-to-price ratio
2. *Fundamental size factors*: log of market capitalization
3. *Fundamental solvency and financial risk factors*: debt-to-equity ratio, current ratio, and the inverse interest coverage ratio
4. *Fundamental operating profitability factors*: net profit growth, return on assets, and the return on common equity
5. *Technical factors*: 12-month momentum, trading volume, and the short interest ratio
6. *Analyst factors*: analyst rating changes and earnings revisions

Note that economic factors [such as gross domestic product (GDP) and inflation] and behavioral factors (such as consumer sentiment) do not fit in the fundamental factor model because neither of these types of factors really can be called a characteristic of the stock itself. Rather, they measure risks in the market that affect all stocks. Statistical factors, which must be derived from stock-return series, do not belong in the fundamental factor model either.

The astute reader will notice that when listing some of the more common factors, we describe their inverse. This is different than how we used these factors in [Chapter 4](#). For example, most investors refer to the price-to-earnings ratio of a company rather than the earnings-to-price ratio. The reason for using the inverse is very practical. When companies have negative earnings, their P/E ratio doesn't really have any meaning. As a result, it does not make sense to argue that a low P/E ratio company is

“cheaper” than a high P/E ratio company when companies with zero or negative earnings are included. However, if we calculate the E/P ratio of companies, we have a ratio that can be compared across companies, which makes a lot more sense. For example, a company with a very high P/E ratio will have a very small E/P ratio but will still be considered relatively “cheaper” than a company with a negative E/P ratio. In practice, some practitioners use P/E and exclude stocks with negative earnings, while others use E/P and have a larger set of stocks to choose from.⁷ The key point is that by using the inverse of certain common factors, we are able to analyze many more stocks, which offers a much richer view for stock selection. In the remainder of this book we shall use factors that allow us to evaluate many more stocks, even if the definitions are slightly altered from what the typical investor is accustomed to.

6.2.2 Treatment of the Risk-Free Rate

The factor premium in a fundamental factor model has to be estimated by looking at the relationship between stock returns and factor exposures. The reliability of the model therefore depends greatly on the quality of the stock return data that are used to estimate this relationship. Before gathering stock-return data, the portfolio manager must confront a larger theoretical question: how much of a stock’s return comes from the stock itself, and how much could have been earned on any investment? The investor earns a portion of the average stock return for free—not as a reward for taking on risk. The reward for taking on risk by buying a stock is therefore the average stock return in *excess of a risk-free rate*.

In order to focus on the portion of the stock return that actually rewards risk, some portfolio managers subtract a risk-free rate from the stock return before implementing the fundamental factor model. While a theoretically sound move, this is not easy to do in practice. The problem is that it is difficult to identify a truly risk-free asset, the return on which could be considered the risk-free rate. Financial economists tend to use the return on short-term U.S. Treasury bills as the risk-free rate because U.S. Treasury bills are considered a very safe investment. Yet they are not completely risk-free assets. Money market funds (MMFs), another substitute for the theoretical risk-free asset, certainly contain a level of risk as well.⁸

Given the difficulty of identifying the true risk-free rate, subtracting an estimated risk-free rate might seem like an arbitrary practice. We believe, however, that a model that adheres to risk-reward theory by adjusting for the risk-free rate will provide more accurate results than one that does not. Given the return of stock i at time t , r_{it} , and the risk-free rate at time t , r_{ft} , we define the excess return r_{it}^* as

$$r_{it}^* = r_{it} - r_{ft} \quad (6.12)$$

In subsequent calculations, we can replace r_{it} with r_{it}^* to adjust for the risk-free rate.

6.2.3 Choosing the Time Interval and Time Period

Two other interrelated questions also affect the quality of the data set: what time interval should fall between the data points, and what overall time horizon should the data set cover? Ideally, the time interval should reflect the investment horizon (i.e., the rebalancing frequency). If the portfolio is rebalanced every month, a monthly interval should be used. If the portfolio is rebalanced every year, an annual interval should be used. This will make things easier in the later stages of analysis.

However, consideration of the estimation precision may force the portfolio manager to choose a time interval different from the investment horizon. The fundamental factor model assumes that the factor exposure of a stock determines its average return. The statistical estimation of the relationship between the factor exposure and the stock return will be imprecise unless the relationship is stable over the given time interval. For example, the stock return could have a stable relationship with the factor exposure for one month but not for an entire year. In that case, it would hardly make sense to set the time interval to a year, even if rebalancing occurs annually. At the same time, if the time interval is too short, the precision of the estimation again may decline because short-term fluctuations in the stock return may be random and unrelated to the factor exposure. Nonetheless, it is generally best to err on the side of a shorter time interval rather than a longer one. One can forecast the results of a longer time interval from the results of a shorter time interval, but not vice

versa.⁹ In practice, financial economists tend to prefer a monthly interval and, to a lesser degree, a weekly interval. Daily and annual intervals are rather uncommon.

The overall time period that the data cover (in statistical jargon, the *sample period*) should be decided in light of the estimation precision as well. If the time period is too short, so that the data encompass only a small number of time intervals, then the estimation precision may suffer. The general rule of estimation precision is: the larger the sample, the higher is the precision. However, if the time period covers many years, then the relationship between the stock return and the factor exposure may change over the course of the period. It is unlikely that the premium on any given factor will remain constant for an entire decade, for example. A very long time period also means that the data will cover more time intervals. The more time intervals, the more data points there are to collect, and the bigger will be the problem of finding complete data for a stock. Financial economists tend to include between 36 and 60 time intervals in their analyses when using a monthly interval (so that the total time period is three to five years). When using a weekly interval, the number of time intervals tends to be greater than 60 (while the overall time period tends to be shorter than three years). When using a quarterly or annual interval, the number of time intervals tends to be much smaller (while the overall time period is generally longer than five years).

We will use the lowercase letter t to indicate one time interval. For example, r_{it} represents the return to stock i at time t , whereas β_{it} denotes the factor exposure of stock i at time t . We will use capital letter T to indicate the number of time intervals used in the analysis. Thus t will take an integer value from 1 to T . We write the relationship between the stock return and the factor exposure as

$$r_{it} = \alpha + \beta_{it}'\mathbf{f} + \epsilon_{it} \quad t = 1, \dots, T \quad (6.13)$$

where, as before, β_{it} and $\mathbf{f}$ are K -dimensional column vectors, and r_{it} and ϵ_{it} are scalars.¹⁰

6.2.4 Choosing the Universe of Stocks

In addition to questions of the nature and time span of the data, the portfolio manager faces the question of how many stocks to consider when applying the model. There already may be external restrictions on the investment universe of the portfolio. The portfolio may be limited, for instance, to technology stocks, value stocks, or stocks of some other stripe. Whether or not that is the case, the portfolio manager may have his or her own preliminary screening strategy that will narrow the list of potential investments. As we discussed in [Chapter 5](#), many well-known portfolio managers put their stamp on a portfolio with a signature stock screen.

In the absence of external restrictions on the investment universe, there is no need (from a technological standpoint) to shrink the investment universe preliminarily. Given the current state of computing technology, the number of stocks is not really an issue. Even a personal computer can handle thousands of stocks without much trouble.

The size of the stock universe does affect some aspects of implementation, though. A very large investment universe provides a large pool of stocks from which to select. The bigger the selection pool, the more likely it is that there are high-return stocks somewhere in the pool. Yet calculation of stock correlations loses precision as the number of stocks in the pool increases, making the model less and less reliable. Thus, even though plenty of good stocks may be floating around, it is harder to identify them. For this reason, we do not recommend starting with an investment universe of more than a few thousand stocks.

6.3 BENCHMARK AND α

Many portfolio managers have a specific benchmark against which they measure the performance of their portfolios. If the portfolio manager aims to outperform a benchmark while minimizing the portfolio's tracking error, then the benchmark must play a role in the model.

One approach to incorporating the benchmark into the fundamental factor model is to use the model to predict only the residual return rather than the entire stock return. The residual return is the part of the stock return not correlated with the benchmark. To transform the model into one of residual return, we need to run a regression of the stock return on the benchmark.

Given the stock return r_i and the benchmark return r_B , we estimate the following equation:

$$r_i = \tilde{\alpha}_i + \tilde{\beta}_i r_B + \tilde{\epsilon}_i \quad (6.14)$$

The typical way to estimate this equation is to use a time-series regression. We find the stock return and the benchmark return over many time periods, say, $r_{i1}, \dots, r_{iT}$ and $r_{B1}, \dots, r_{BT}$, where the second subscript refers to time periods. After estimating $\tilde{\alpha}$ and $\tilde{\beta}$, we define the residual return $\tilde{r}_i$ as

$$\tilde{r}_i \equiv \tilde{\alpha}_i + \tilde{\epsilon}_i \quad (6.15)$$

Once the residual return is defined, the portfolio manager can use the fundamental factor model to predict the value of the residual return. To do so, he or she simply substitutes the residual return for the stock return given in the fundamental factor model in [Eq. \(6.2\)](#).

The portfolio manager's value-added is wrapped up in the term $\tilde{\alpha}$ (also known as α^B). Note that the expected return and the risk of the last term $\tilde{\epsilon}$ are already known from the preceding estimation. The term with flexibility to boost the residual return is $\tilde{\alpha}$. What the portfolio manager does with this fundamental factor model is therefore all about $\tilde{\alpha}$.¹¹

An alternative way to account for the benchmark, rather than predicting the residual stock return, is to add the benchmark to the model as one of the factors. This approach fits better with the economic factor model, however, so we will delay discussing it until the next chapter.¹² Accounting for the benchmark at this point in the modeling process, though useful, is not necessary. Some portfolio managers prefer to deal with the benchmark at a later stage in the process by, for instance, controlling for the tracking error in construction of the portfolio. This approach may make the most sense if the composition of the benchmark is known clearly.¹³

6.4 FACTOR EXPOSURE

The factor exposure quantifies the exposure of a stock to risk. In the fundamental factor model, the factor exposure is the value of some

observable (or easily calculable) characteristic of the stock. In some cases, the factor exposure shows up right in a company’s financial statements or on a price-volume chart. In other cases, a straightforward calculation drawing on figures from financial statements is enough. In other words, once the factors are chosen, determining the factor exposure is rarely a challenging task.

Table 6.2 presents the factor exposures of selected stocks for five commonly used factors. The E/P ratio is the annual earnings per share divided by the share price. The B/P ratio is the value of common equity reported in the company’s balance sheet divided by the share price.¹⁴ The D/E ratio is the total value of liabilities divided by the total value of equity reported in the balance sheet. LOGSIZE is the natural log of the market capitalization of the company (in millions of dollars), where market capitalization is defined as the share price multiplied by the number of shares outstanding. Momentum (M12M) is the average monthly return (as a percentage) over the previous 12 months.

TABLE 6.2

Factor Exposure of Selected Stocks

Ticker	E/P	B/P	D/E	LOGSIZE	M12M
AAPL	0.0289	0.0357	3.3904	14.5206	0.7973
MSFT	0.0274	0.0731	1.5469	14.2970	0.4291
AMZN	0.0083	0.0464	2.5036	14.2790	0.7592
GOOGL	0.0285	0.1871	0.3433	13.9183	0.3493
FB	0.0353	0.1659	0.2648	13.4088	0.3736
BRK.B	0.0414	0.7329	0.9840	13.1935	0.0391
WMT	0.0413	0.1739	1.9235	12.9784	0.3056
JNJ	0.0399	0.1654	1.5148	12.8502	0.0812
JPM	0.0700	0.6523	11.1494	12.7920	-0.0727
V	0.0330	0.0849	1.1924	12.7827	0.1471

Note: Factor exposure is as of the end of November 2020. E/P ratio is the earnings-to-price ratio, B/P ratio is the book-to-price ratio, D/E ratio is the debt-to-equity ratio, LOGSIZE is natural log of market capitalization in millions of dollars, and M12M is the cumulative returns over the past 12 months in decimal form (i.e., 0.79 equals 79%). Ticker symbols are for the following companies: Apple Inc. (AAPL), Microsoft Corporation (MSFT), Amazon (AMZN), Alphabet Inc. or Google (GOOGL), Facebook Inc. (FB), Berkshire Hathaway B shares (BRK.B), Johnson & Johnson (JNJ), Walmart Inc. (WMT), J.P. Morgan Chase & Co. (JPM), and Visa Inc. (V). All values were computed as of December 2020 with appropriate reporting lags. Since many readers are more familiar with P/E and P/B ratios, the P/Es of the companies are 34.60, 36.50, 120.48, 35.09, 28.33, 24.15, 24.21, 25.06, 14.29, and 30.30, respectively, and the P/Bs of the companies are 28.01, 13.68, 21.55, 5.34, 6.03, 1.36, 5.75, 6.05, 1.53, and 11.78, respectively.

From the table we can see that Amazon (AMZN) has a low earnings-to-price ratio of 0.0083 (i.e., a high price-to-earnings ratio of about 120) likely owing to expectations of rapid earnings growth. J.P. Morgan Chase (JPM), on the other hand, has a relatively high earnings-to-price ratio. If the factor premium for the earnings-to-price ratio were negative, then—all other things being equal—we would expect Amazon to have a higher return than J.P. Morgan Chase. Conversely, if the factor premium for the earnings-to-price ratio were positive, we would expect J.P. Morgan Chase to perform better than Amazon.

Looking at another factor, we see that momentum is very high for Apple and Amazon (reflecting their exceptional returns in 2020), whereas it is negative for J.P. Morgan Chase (reflecting its losses in value during the year). We may be able to predict which of these stocks will have the highest return in the following year if we determine the factor premium for momentum. If the factor premium were positive, we would expect Apple and Amazon to perform better than J.P. Morgan Chase.

Factor exposures change over time, and we need to be sure to assign the correct values to the correct dates. For monthly time intervals, it is standard practice to use the factor exposure recorded at the beginning of the month (or, alternatively, the value recorded at the end of the previous month). In calculating the earnings-to-price ratio (E/P), the book-to-price ratio (B/P), and the log of market capitalization (LOGSIZE) for this table (which shows factor exposures at the beginning of December 2020 or, alternatively, at the end of November 2020), we used the closing stock prices and shares outstanding as of November 30, 2020. In calculating momentum, we used the stock returns from December 2019 through November 2020.¹⁵

Extra care needs to be taken in applying data from a company's financial statements. Companies release their financial statements quarterly but not immediately when the fiscal quarter ends. As of the beginning of December 2020, the most recently released financial statements might represent the second or third quarter of 2020. Since we use quarterly data in calculating the earnings-to-price ratio, we used the earnings for the 12-month period ending in August, July, or June, depending on when a company's fiscal quarter ended. If the fiscal quarter was over in September, the earnings for the July–September quarter might not be known by the beginning of December, so we did not use them to calculate the earning-to-price ratio. For the same reason, we used only figures for fiscal quarters ending in or before August to calculate the book-to-price and debt-to-equity ratios. Generally, it is reasonable to allow two to three months' lag between the end of a fiscal period and the reporting of the variable. [Table 6.3](#) summarizes this timing scheme.

TABLE 6.3

Assigning the Correct Values to the Correct Time for December 2020

Data Frequency	If the Fiscal Quarter Ends in:	If the Fiscal Year Ends in:	Use the Data for:
Daily data			End of Nov 2020
Monthly data			Nov 2020
Quarterly data	Mar, Jun, Sep, Dec		Apr 2020–Jun 2020
	Feb, Mar, Aug, Nov		Jun 2020–Aug 2020
	Jan, Apr, Jul, Oct		May 2020–Jul 2020
Annual data		Dec	Jan 2019–Dec 2019
		Nov	Dec 2018–Nov 2019
		Oct	Nov 2018–Oct 2019
		Sept	Oct 2018–Sep 2019
		Aug	Sep 2019–Aug 2020
		⋮	⋮

Note: Factor exposure for month t is calculated from the latest available information as of the beginning of month t (i.e., the end of month $t - 1$). Thus, for December 2020, we use the values known at the end of November 2020. This is what we do for daily and monthly data. For quarterly data, we assume a reporting gap of three months. Thus, if a fiscal quarter ends in March, June, September, or December, then the last fiscal quarter whose information is available as of the end of November 2020 is the fiscal quarter ending in June 2020 (December fiscal quarter information is obviously not yet available, and the September fiscal quarter information does not meet the three-month gap requirement). Similarly, for annual data, we assume a reporting gap of three months. Thus, if a fiscal year ends in December, then the last fiscal year whose information is available as of the end of November 2020 is the fiscal year ending in December 2019.

After all the factor exposures are determined, we collect the numbers and save them as a set of vectors $\{(\beta_{11}, \dots, \beta_{N1}), \dots, (\beta_{1T}, \dots, \beta_{NT})\}$ where β_{it} is the factor exposure of stock i for time t . For example, the factor exposure of Apple for December 2020 is (from [Table 6.2](#))

$$\begin{aligned}\beta_{AAPL, \text{Dec20}} &= (\beta_{AAPL, E/P, \text{Dec20}}, \dots, \beta_{AAPL, M12M, \text{Dec20}})' \\ &= (0.0289, 0.0357, 3.3904, 14.5206, 0.7973)' \quad (6.16)\end{aligned}$$

We can read similar information for other stocks (different i 's) and for other time periods (different t 's) in a similar way.

6.5 THE FACTOR PREMIUM

The factor premium is the payoff for each unit of factor exposure, or exposure to risk, that the stock possesses. In the fundamental factor model,

the factor premium is estimated from the pooled cross-sectional regression (i.e., panel regression) of the stock return on the factor exposure. Estimation of the factor premium with a regression is possible because the premium likely remains constant over several years¹⁶ and across stocks.

Given the returns of N stocks over T time periods, $\{(r_{11}, \dots, r_{N1}), \dots, (r_{1T}, \dots, r_{NT})\}$, and the factor exposures of N stocks over T time periods, $\{(\beta_{11}, \dots, \beta_{N1}), \dots, (\beta_{1T}, \dots, \beta_{NT})\}$, we can estimate the following equation:

$$r_{it} = \alpha + \beta'_{it} \mathbf{f} + \epsilon_{it} \quad (6.17)$$

There are a number of ways to estimate this equation. The simplest is the ordinary least squares (OLS) approach. While the OLS estimator is simple to obtain, it may not be the most reliable estimator. We suggest that portfolio managers do a number of robustness checks on it and then decide whether to use a more sophisticated technique.

6.5.1 OLS Estimator of the Factor Premium

The OLS estimator of the factor premium $\mathbf{f}$ is given as¹⁷

$$\hat{\mathbf{f}} = \left[\sum_{t=1}^T \sum_{i=1}^N (\beta_{it} - \bar{\beta})(\beta_{it} - \bar{\beta})' \right]^{-1} \sum_{t=1}^T \sum_{i=1}^N (\beta_{it} - \bar{\beta})(r_{it} - \bar{r}) \quad (6.18)$$

where

$$\bar{\beta} = \frac{1}{NT} \sum_{t=1}^T \sum_{i=1}^N \beta_{it} \quad (6.19)$$

$$\bar{r} = \frac{1}{NT} \sum_{t=1}^T \sum_{i=1}^N r_{it} \quad (6.20)$$

The constant $\hat{\alpha}$ is calculated as $\bar{r} - \bar{\beta}' \hat{\mathbf{f}}$. The standard error for the factor premium is the square root of (the diagonal elements of) the following variance:

$$\hat{V}(\hat{\mathbf{f}}) = \hat{\sigma}^2 \left[\sum_{t=1}^T \sum_{i=1}^N (\beta_{it} - \bar{\beta})(\beta_{it} - \bar{\beta})' \right]^{-1} \quad (6.21)$$

where $\hat{\sigma}^2$ is the estimated variance of ϵ_{it} , that is,

$$\hat{\sigma}^2 = \frac{1}{NT} \sum_{t=1}^T \sum_{i=1}^N (r_{it} - \beta_{it}' \hat{\mathbf{f}})^2 \quad (6.22)$$

Table 6.4 presents the factor premium estimates and their standard errors. We made the estimations for Standard & Poors (S&P) 500 stocks in the period from January 2016 to December 2020. If we look at the E/P ratio factor premium for a moment, we see that it is -3.425 . A negative premium indicates that a high factor exposure hurts stock returns. In the case of the E/P ratio factor, the expected stock return will drop by 3.425% for every unit of exposure. To some, these results may seem counterintuitive, since we are familiar with the value anomaly. However, over a short horizon, as with this period, the realized premium may be different than the long-term premium, and this failure of “value” consumed much of the talk in the quant world from 2016 to 2020.

TABLE 6.4

The Factor Premium

E/P	B/P	D/E	LOGSIZE	M12M
-3.4248 (0.4051)	0.2229 (0.1652)	-0.0005 (0.0008)	0.0470 (0.0549)	-2.4032 (0.2022)

Note: Estimated for the period from January 2016 to December 2020. Standard errors are inside parentheses. Factors are as in Table 6.2.

Standard errors indicate how precise the estimates are. Small standard errors suggest that the estimates are very precise. For the E/P ratio and 12-month momentum, the standard errors are small enough to make the factor premiums “significant” in a statistical sense. For example, we can say with 95% certainty that the “true” value of the E/P ratio premium is

approximately between -4.235 and -2.615 , whereas the “true” value of the M12M premium is approximately between -2.807 and -1.999 .¹⁸

6.5.2 Robustness Check

A model is not usually an exact description of reality, only a good approximation of it. In statistics, shortcomings in the model are called *specification errors*.¹⁹ Specification errors arise in any regression, and estimation of the factor premium is no exception. Yet we should strive to build models that reflect persistent and stable patterns, as described by tenet 6 of quantitative equity portfolio management (QEPM). With a robustness check, we can gauge whether the factor premium estimates are stable when small details of the estimation change. If the current estimation is not robust, we ought to try an alternative estimation technique.

To check the robustness of the estimation, split the data set into a few subsets and see whether the estimates are very different across the subsets. They should not be too different if the estimation is robust. Subsets can be created along the time dimension. For example, we can estimate the factor premiums for the period from January 2016 to June 2018 and the period from July 2018 to December 2020, as reported in [Table 6.5](#). Subsets also can be divided along the cross-sectional dimension. For example, we can estimate the factor premiums for different sectors, as shown in [Table 6.6](#). These two tables suggest a certain level of stability across time and sectors.²⁰

TABLE 6.5

The Factor Premium for Subperiods

Period	E/P	B/P	D/E	LOGSIZE	M12M
Jan 2016–Jun 2018	-2.5101 (0.4041)	0.3373 (0.1960)	-0.0005 (0.0007)	-0.0319 (0.0628)	-0.9915 (0.2225)
Jul 2018–Dec 2020	-5.1183 (0.7909)	-0.2026 (0.2657)	-0.0044 (0.0047)	0.1601 (0.0902)	-4.1142 (0.3497)

Note: Standard errors are inside parentheses. Factors are as in Table 6.2.

TABLE 6.6

The Factor Premium for Various Economic Sectors

Sector	E/P	B/P	D/E	LOGSIZE	M12M
10	−4.8850 (0.9017)	2.0557 (0.7182)	0.0455 (0.1483)	0.1761 (0.3644)	−5.6620 (1.1626)
15	−10.8167 (2.0278)	−0.0625 (0.8931)	−0.0157 (0.0828)	−0.4396 (0.3454)	−3.1329 (0.9029)
20	−1.6735 (2.6884)	0.7879 (0.8404)	−0.0038 (0.0046)	−0.0841 (0.1517)	−3.0761 (0.5620)
25	−4.2285 (2.1720)	1.5400 (0.6386)	−0.0003 (0.0009)	0.3712 (0.1708)	−2.4307 (0.5544)
30	5.5367 (2.2242)	−1.2432 (0.6087)	−0.0023 (0.0038)	−0.0827 (0.1641)	−3.4636 (0.7859)
35	5.0186 (3.1616)	−1.6256 (0.6732)	0.0202 (0.0332)	−0.1480 (0.1437)	−1.7809 (0.4803)
40	−3.2311 (3.3614)	0.4321 (0.3431)	0.0122 (0.0234)	−0.1228 (0.1343)	−3.9668 (0.6167)
45	−4.5227 (2.2509)	−0.7912 (0.6619)	0.0023 (0.0062)	0.1103 (0.1250)	−1.8753 (0.4332)
50	−0.4454 (3.3899)	0.5284 (1.1857)	−0.0977 (0.1504)	0.6444 (0.2827)	−6.7083 (1.6776)
55	−0.6537 (0.9968)	0.3741 (0.9778)	0.3129 (0.1422)	−0.2492 (0.2326)	−2.6367 (0.8356)

Note: Estimated for the period from January 2016 to December 2020. Standard errors are inside parentheses. Factors are as in Table 6.2. Sector names are based on the Global Industry Classification Standard (GICS), i.e., Energy (10), Materials (15), Industrials (20), Consumer Discretionary (25), Consumer Staples (30), Health Care (35), Financials (40), Information Technology (45), Telecommunication Services (50), and Utilities (55).

6.5.3 Outliers and MAD Estimator of Factor Premium

The weakness of the OLS method is that in trying to minimize the sum of the squared residuals, it is highly sensitive to outliers. If a robustness check indicates that the OLS estimator is very unstable, it may be necessary to use an alternative estimation procedure that is less sensitive to outliers.

The *minimum absolute deviation* (MAD) estimation, also known as *median estimation*, is one such alternative. The MAD minimizes the sum of the absolute value of residuals rather than the squared residuals. Since this approach avoids squaring the residuals, any outliers have a much smaller effect on the estimate than they do in the OLS approach. Standard statistical software supports the MAD approach.

Table 6.7 shows the MAD estimates of the factor premiums in our example. We again performed an estimation using S&P 500 stocks for the

period January 2016 to December 2020. Note that the earnings-to-price ratio (E/P) changed its sign from being significantly negative to significantly positive when we shifted from the OLS to the MAD approach. This implies that primarily outliers are driving the E/P ratio estimate here.

TABLE 6.7

The MAD Factor Premium

E/P	B/P	D/E	LOGSIZE	M12M
0.7784 (0.3639)	-0.2912 (0.1484)	-0.0004 (0.0007)	-0.0085 (0.0493)	-2.6169 (0.1817)

Note: Estimated for the period from January 2016 to December 2020. Standard errors are inside parentheses. Factors are as in Table 6.2.

6.5.4 Heteroscedasticity and Autocorrelation-Consistent Estimation of the Standard Error

While OLS estimation often produces a not-so-reliable output, it has been the standard empirical tool for many generations of empirical analysts. Moreover, adopting an alternative estimation technique risks introducing additional errors, since every estimation technique is based on its own set of assumptions—and these assumptions may not be any more realistic than the assumptions driving OLS. Given this consideration, one popular approach is to continue to report the OLS estimates but at the same time to make adjustments for the standard errors so that certain assumptions of the OLS estimation are relaxed. More specifically, when calculating standard error, we may wish to relax two of the OLS assumptions: (1) that all the error terms of different firms have the same variance (the assumption of homoscedastic errors), and, (2) that the error terms of different time periods are independent (the assumption of no autocorrelation). The standard errors that we obtain after relaxing these two assumptions are called *heteroscedasticity- and autocorrelation-consistent standard errors* or, in short, *HAC standard errors*. Even for HAC standard errors, we do maintain certain restrictions regarding the error terms. For example, with HAC, we do not allow error terms of different firms to be correlated (though we allow error terms of one firm for different periods to be correlated). Also, HAC requires error terms of one firm for different periods to have the same

variance (though we allow error terms of different firms to have different variances). The maintained assumptions guarantee that the OLS estimates are still consistent (i.e., correct if the sample is large enough), though they may not be the most efficient (i.e., there may be other estimates with greater precision). Under the assumption of heteroscedastic and autocorrelated errors, the variance of the factor premium can be calculated as

$$\hat{V}\left[\begin{pmatrix} \hat{\alpha} \\ \hat{\mathbf{f}} \end{pmatrix}\right] = \left[\sum_{t=1}^T \sum_{i=1}^N \begin{pmatrix} 1 \\ \beta_{it} \end{pmatrix} \begin{pmatrix} 1 \\ \beta_{it} \end{pmatrix}' \right]^{-1} \left\{ \sum_{i=1}^N \left[\sum_{t=1}^T \hat{\epsilon}_{it} \begin{pmatrix} 1 \\ \beta_{it} \end{pmatrix} \right] \left[\sum_{t=1}^T \hat{\epsilon}_{it} \begin{pmatrix} 1 \\ \beta_{it} \end{pmatrix} \right]' \right\} \quad (6.23)$$

$$\left[\sum_{t=1}^T \sum_{i=1}^N \begin{pmatrix} 1 \\ \beta_{it} \end{pmatrix} \begin{pmatrix} 1 \\ \beta_{it} \end{pmatrix}' \right]^{-1}$$

where the residual $\hat{\epsilon}_{it}$ is calculated from the OLS estimate as

$$\hat{\epsilon}_{it} = r_{it} - \beta_{it}' \hat{\mathbf{f}} \quad (6.24)$$

Compared to the simple variance formula in [Eq. \(6.21\)](#), we now have a rather complicated formula. Looking closely, however, one may note that the first and last terms are identical. (For this reason, the formula is sometimes referred to as a “sandwich formula.”) The first and last terms represent the variation of factor exposure β_{it} whereas the middle term represents the variance of $\hat{\epsilon}_{it} \beta_{it}$. The HAC standard errors are simply the square root of the diagonal elements of the variance matrix shown above. [Table 6.8](#) shows the HAC standard errors together with the OLS estimates of factor premiums. The earnings-to-price ratio and the book-to-price ratio have noticeably higher standard errors compared to the unadjusted standard errors shown in [Table 6.4](#). That is, the premiums for these two factors are not precisely estimated, and the standard errors shown in [Table 6.4](#) exaggerate the precision for these two premiums.

TABLE 6.8

The HAC Standard Errors along with the OLS Estimates

E/P	B/P	D/E	LOGSIZE	M12M
-3.4248 (1.1315)	0.2229 (0.2268)	-0.0005 (0.0003)	0.0470 (0.0612)	-2.4032 (0.3552)

Note: Estimated for the period from January 2016 to December 2020. Standard errors are inside parentheses. Factors are as in Table 6.2.

6.6 DECOMPOSITION OF RISK

Given the factor exposure β_i , the factor premium estimate $\hat{f}$, and the constant estimate $\hat{\alpha}$, we can calculate the average stock return as the product of these two plus the constant, $\hat{\alpha} + \beta'_i \hat{f}$. To construct the optimal portfolio, however, we need to know the risk of the stock return. In the fundamental factor model, the risk of the stock return has two components: nondiversifiable risk captured by $\beta'_i \hat{f}$ and diversifiable risk captured by $\hat{\epsilon}_i$. The total risk of the stock return is simply the sum of these two risks. Let's look at each of these two in turn.

Nondiversifiable risk arises from the randomness of the factor premium. Within the estimation sample, however, both factor exposure β_i and factor premium $\hat{f}$ vary over time, and it is not necessary to separate these two components. For the purpose of risk decomposition, we carry out a series of cross-sectional regressions, one regression for each month t , rather than a single pooled regression. From the regression for month t , we obtain factor premium and constant estimates $\hat{f}_t$ and $\hat{\alpha}_t$. By repeating the regressions for all the sample months, we obtain a time series of factor premium and constant estimates, $\hat{f}_1 \dots \hat{f}_T$ and $\hat{\alpha}_1 \dots \hat{\alpha}_T$. By multiplying the factor premium by the factor exposure and adding the constant, we also obtain a time series of $\hat{\alpha} + \beta'_i \hat{f}$, i.e., $\hat{\alpha}_1 + \beta'_{i1} \hat{f}_1, \dots, \hat{\alpha}_T + \beta'_{iT} \hat{f}_T$. We can calculate the sample variance from this time series as

$$\frac{1}{T} \sum_{t=1}^T \left[\hat{\alpha}_t + \beta'_{it} \hat{f}_t - \frac{1}{T} \sum_{t'=1}^T (\hat{\alpha}_{t'} + \beta'_{it'} \hat{f}_{t'}) \right]^2 \quad (6.25)$$

This variance, or the sample standard variation obtained by taking the square root of the variance, is our measure of nondiversifiable risk.

The diversifiable risk represents the part of the variation in the stock return that the variation in the model's factors cannot explain. This part of the stock's variation appears as the error term ϵ_i , that is, σ_i^2 . The procedures outlined in the previous paragraph provide an estimate of the variance of the error term:

$$\hat{\sigma}_i^2 = \frac{1}{T} \sum_{t=1}^T \left(r_{it} - \hat{\alpha}_t - \hat{\beta}'_{it} \hat{\mathbf{f}}_t \right)^2 \quad (6.26)$$

This variance is our measure of the diversifiable risk. One can alternatively use the square root of the variance (standard deviation) as a measure of risk. The total risk of stock i is simply the sum of the two risk components:

$$\frac{1}{T} \sum_{t=1}^T \left[\hat{\alpha}_t + \hat{\beta}'_{it} \hat{\mathbf{f}}_t - \frac{1}{T} \sum_{t'=1}^T \left(\hat{\alpha}_{t'} + \hat{\beta}'_{it'} \hat{\mathbf{f}}_{t'} \right) \right]^2 + \frac{1}{T} \sum_{t=1}^T \left(r_{it} - \hat{\alpha}_t - \hat{\beta}'_{it} \hat{\mathbf{f}}_t \right)^2 \quad (6.27)$$

Table 6.9 shows the decomposition of risk for selected stocks estimated from January 2016 to December 2020. We report the risk in terms of the variance and the standard deviation.

TABLE 6.9

The Risk Decomposition for Selected Stocks

Ticker	Total Risk		Nondiversifiable Risk		Diversifiable Risk	
	V	SD	V	SD	V	SD
AAPL	824.50	28.71	170.47	13.06	654.03	25.57
MSFT	364.38	19.09	149.52	12.23	214.85	14.66
AMZN	667.40	25.83	183.24	13.54	484.17	22.00
GOOGL	438.11	20.93	167.99	12.96	270.12	16.44
FB	682.49	26.12	179.62	13.40	502.86	22.42
BRK.B	368.75	19.20	254.25	15.95	114.51	10.70
JNJ	336.23	18.34	178.93	13.38	157.30	12.54
WMT	450.71	21.23	159.11	12.61	291.60	17.08
JPM	456.13	21.36	272.33	16.50	183.80	13.56
V	275.62	16.60	158.06	12.57	117.55	10.84

Note: Based on the period from January 2016 to December 2020. V refers to the variance, and SD refers to the standard deviation. Both variance and standard deviation are annualized and expressed in percentage. Company names for ticker symbols are reported in Table 6.2.

When we get to the stage of finding the optimal portfolio, we also need to know the correlations between stock returns. The total risk of a stock is composed of nondiversifiable risk and diversifiable risk. Likewise, the total return on a stock is composed of nondiversifiable and diversifiable components. The correlation between the returns on a pair of stocks consists of the correlation between their nondiversifiable components and the correlation between their diversifiable components. Instead of correlation, we will use covariance; the covariance between r_{it} and r_{jt} is

$$C(r_{it}, r_{jt}) = C(\hat{\beta}'_{it}\hat{\mathbf{f}}_t, \hat{\beta}'_{jt}\hat{\mathbf{f}}_t) + C(\hat{\epsilon}_{it}, \hat{\epsilon}_{jt}) \quad (6.28)$$

In practice, we estimate only the correlation between the nondiversifiable components of two stocks' returns. As for the diversifiable components, the correlation between them can, in principle, be estimated in a straightforward way using the standard estimation technique. In practice, though, it is rarely done this way. One reason is that there are often simply too many parameters to estimate precisely. If there are N stocks, then $[N(N-1)]/2$ covariances have to be estimated. These covariances cannot be

estimated precisely without using a great many time intervals for the estimation. The second reason that the standard estimation technique is hardly used to estimate the diversifiable component of the return is that the diversifiable return, like diversifiable risk, has little bearing on the stock return to the investor (or so says modern financial economic theory). For both these reasons, the convention is to assume that diversifiable risk $C(\epsilon_{it}, \epsilon_{jt})$ equals zero.

6.7 CONCLUSION

In this chapter we explored one of the most common models of stock returns, the fundamental factor model. Factor models relate the return of a stock to the stock's exposure to risks (a.k.a. *factor exposures*) and the market's payoff to investors for taking on those risks (a.k.a. *factor premiums*). We showed how the fundamental factor model can act as a model of overall stock returns or, for managers who track an index, as a model of benchmark α . We explained how to obtain a stock's fundamental observable factor exposures and how to estimate the factor premiums that the market awards them. With the factor exposures and factor premiums in place, the fundamental factor model can predict the expected return and expected risk of any stock in a portfolio manager's investment universe. The model also can decompose the risk of a stock into its diversifiable and nondiversifiable components. The fundamental factor model is one basic tool for constructing a portfolio. In the next chapter we will take a look at an alternative tool, the economic factor model, that the portfolio manager also can use to predict and analyze stock returns and risks.

QUESTIONS

- 6.1. Describe the main steps in creating and estimating a fundamental factor model.
- 6.2. Explain why economic factors such as GDP and behavioral factors such as consumer sentiment cannot be included in a fundamental factor model.
- 6.3. One strength of the fundamental factor model over the economic factor model is that the data requirement is relatively small. What is

the minimum data requirement to estimate the fundamental factor model? Is it possible to estimate the fundamental factor model from a single cross section (i.e., the observation of many stocks in one time period)?

- 6.4.** It is possible to derive returns of many periodicities from returns of one periodicity. Suppose that we have estimated the following fundamental factor model using daily data:

$$r_i = \alpha + \beta_i f + \epsilon_i \quad \epsilon_i \sim N(0, \sigma^2)$$

Let us denote the estimates by $\hat{\alpha}$, $\hat{f}$, and $\hat{\sigma}^2$.

- (a) What would be the expected daily return and the variance of the daily return?
 - (b) Assuming that error ϵ_i is serially uncorrelated over time, what would be the expected weekly return and the variance of the weekly return?
 - (c) Under the same assumption as in part (b), what would be the expected monthly return and the variance of the monthly return?
 - (d) Is the assumption made in parts (b) and (c) realistic?
- 6.5.** It makes a difference whether one includes the risk-free rate in the model. Consider two versions of a fundamental factor model:

$$r_i = \alpha^{(1)} + \beta_i f^{(1)} + \epsilon_i \quad \epsilon_i \sim N[0, (\sigma^{(1)})^2]$$

$$r_i - r_f = \alpha^{(2)} + \beta_i f^{(2)} + \epsilon_i \quad \epsilon_i \sim N[0, (\sigma^{(2)})^2]$$

where r_f is the risk-free rate.

- (a) If r_f is constant for the estimation period, what is the relationship between the estimates of $\alpha^{(1)}$, $f^{(1)}$, and $\sigma^{(1)}$ and those of $\alpha^{(2)}$, $f^{(2)}$, and $\sigma^{(2)}$?
 - (b) Would your answer to part (a) be different if r_f is not constant but is not correlated with any of the variables in the model?
 - (c) Under what conditions does the inclusion of the risk-free rate in the model not change the estimates?
- 6.6.** Consider a fundamental factor model with many factors:

$$r_i = \alpha + \beta_{i1}f_1 + \cdots + \beta_{iK}f_K + \epsilon_i$$

The goodness of fit of the regression can be measured by the adjusted R^2 :

$$\bar{R}^2 = 1 - \frac{\frac{N-2}{N-K-2} V(\hat{\epsilon}_i)}{V(r_i)}$$

where N is the number of observations, K is the number of factors, $\hat{\epsilon}_i$ is the residual from the regression (i.e., the part of r_i that was not explained by the regression), and $V(\cdot)$ refers to the sample variance.

- (a) Calculate $\bar{R}^2$ when the number of observations is 100, the number of factors is 2, the sample variance of the return is 5%, and the sample variance of the residual is 3%.
 - (b) What will happen to $\bar{R}^2$ if we keep the number of observations constant and increase the number of factors to 4? Assume that the sample variance of the return and the residual did not change.
 - (c) What would be the required number of observations to keep $\bar{R}^2$ unchanged when the number of factors increases to 4?
- 6.7.** Define the residual return. What are the main components of it?
- 6.8.** Suppose that we estimated a fundamental factor model

$$r_i = \alpha + \beta_i f + \epsilon_i$$

where β_i is the size exposure measured in dollars. Would the estimates of α and f be different if the size exposure were measured in thousand of dollars instead?

- 6.9.** After estimating a fundamental factor model, we obtained the following estimates:

$$r_i = 1.5 + 0.02\beta_{i1} + 0.3\beta_{i2} + \epsilon_i$$

where β_{i1} is the size exposure of firm i , and β_{i2} is the price-to-earnings ratio exposure.

- (a) What is the expected return of firm A whose size is 100 (million dollars) and that has a price-to-earnings ratio of 20?
- (b) What will happen to the expected return of firm A in part (a) if the firm size grows by 10 (million dollars)?
- (c) Explain why the factor premium can be interpreted as the marginal effect of the factor exposure on the expected return.

6.10. The following is the result of the estimation of a fundamental factor model:

$$r_i = 1.5 + 0.2\beta_i + \epsilon_i \quad \epsilon_i \sim N(0, 400)$$

- (a) If the sample variance of the factor exposure is 100, what would be the standard error of the coefficient estimate?
 - (b) What would be the standard error if the factor exposure were almost constant in the sample?
 - (c) Can you come up with a criterion to choose a factor based on your calculation in parts (a) and (b)?
- 6.11.** Some portfolio managers remove the outliers from the sample before estimating a model. For example, one may decide to drop observations whose values are more than three standard deviations away from the mean. Considering the properties of ordinary least squares, what would be the consequence? Does it matter whether the outliers are determined based on the values of the stock returns or based on the values of the factor exposure?
- 6.12.** Some portfolio managers may “winsorize” the outliers before using them in the estimation. For example, if the mean of a variable is μ and the standard deviation of the variables is σ , any value that is greater than $\mu + 3\sigma$ or less than $\mu - 3\sigma$ may be replaced with $\mu + 3\sigma$ or $\mu - 3\sigma$. Would the estimate of ordinary least squares still be unbiased (true on average)? Does it matter whether the variable we use for winsorization is the return or a factor exposure?
- 6.13.** One solution to the problem of the outliers is to use the minimum absolute deviation (MAD) estimation instead of the ordinary least squares (OLS) estimation. Under what circumstances do these two estimations produce identical estimates?

- 6.14.** Some researchers believe that small stocks experience higher variability in stock returns than large stocks do. That is, the variance of the error is believed to be greater for small stocks than for large stocks. Discuss the advantages and disadvantages of HAC standard errors in this situation.
- 6.15.** Define diversifiable risk and nondiversifiable risk. In what sense can the risk be diversified?

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- ¹ This notation requires explanation. It is the convention in statistics to use Greek letters for parameters that must be estimated and to use the more familiar Latin letters for variables whose values can be observed. We do not follow the convention here because in other models of stock returns the factor exposure is the unknown variable, whereas the factor premium is directly observable. To remain consistent across all models, we will always call the factor exposure β and the factor premium f .
- ² Even when we estimate the factor premiums using time series of stock returns and factor exposures, one can think of the regressions as pooled regressions, where the intercept term is a constant.
- ³ The *expected stock return*, not *the average stock return*, is the correct term to use. We will switch to the technically correct terminology in later chapters.
- ⁴ There is an extra constant term in the formula, but it will be zero if the factor exposure is properly normalized and the risk free rate has already been subtracted from the return. This will be discussed in more detail later.
- ⁵ Risk can be measured either with the variance or the standard deviation. In the formula shown here, risk is defined as the variance.
- ⁶ Please refer to [Chapter 4](#) for a comprehensive catalogue of factors.
- ⁷ For major indices or portfolios, some investors calculate the weighted average of the E/P ratio or related ratio and then invert that average so that it appears to investors in the more common form.
- ⁸ It is also important to use a total return risk-free rate index. Sometimes people use the yield of the risk-free rate. Although this might be approximately correct, it's much better to use the actual total return from the chosen risk-free instrument.
- ⁹ [Chapter 8](#) discusses this concept in greater detail.
- ¹⁰ Typically, when we estimate these factor models, we use information at time t to predict returns at time $t + 1$. For convenience, we sometimes write in this book the equation with time subscript t everywhere, as we do here. The factor exposure and the returns are at time t . The equation should still be interpreted as using information at t to predict returns at time $t + 1$. In other words, the factor exposure at time t is really the factor exposure value as of the end of the previous period, $t - 1$, which we may refer to, sometimes, as the beginning-of-month exposure for time t . There is also the issue of when a factor exposure was truly available for the portfolio manager. We discuss that issue later in this chapter.

- 11 Be careful not to confuse $\sim\alpha$ with the α of the model for total stock return in Eq. (6.2). The concept of $\sim\alpha$ is so important in the investment world that analysts sometimes refer to it simply as α . Refer to Chapter 2 for a discussion of the various α 's of investing and their corresponding models.
- 12 If we use the benchmark as a factor in the fundamental factor model, we need to estimate the factor exposure of each stock in some way. To estimate the factor exposure, we will need to employ the technique we use for the economic factor model.
- 13 See Chapter 9 for an explanation of this method.
- 14 As we explained in Chapter 5, there are many advantages to defining certain variables as their inverse. For example, one can analyze many more companies when using the E/P ratio rather than the P/E ratio. One might also notice that unlike in Chapters 4 and 5, we no longer have Tesla in our representative 10 stocks. This is because Tesla was not part of the S&P 500 in November of 2020.
- 15 We do this because in our estimations of factor premiums we estimate regressions over time and need to use the latest factor exposure (i.e., end of the month) against the subsequent monthly return of the stocks. Thus, we use end-of-November factor exposures and end-of-December returns to estimate factor premiums.
- 16 Though, as noted earlier in this chapter, the premium likely changes over the course of an entire decade.
- 17 We present the formulas for consistent estimators. For unbiased estimators, the degree-of-freedom correction is necessary.
- 18 Refer to Appendix C at www.ludwigbc.com under QEPM Exclusive Content for more on statistical significance, standard errors, and t -statistics.
- 19 See Appendix C at www.ludwigbc.com under QEPM Exclusive Content for more on specification error.
- 20 Eugene Fama would say that if you really believe a factor model, then you should not expect to see any sector effect on the estimates because the model already should have accounted for it. Nevertheless, it is industry practice to control for sectors with dummy variables.

CHAPTER 7

Economic Factor Models

Beta is the color of an angel's eye.

—Zvi Griliches

7.1 INTRODUCTION

Tenet 4 of quantitative equity portfolio management (QEPM) requires quantitative analysis to efficiently combine all the information relevant to an investment decision. In [Chapter 6](#), we looked at the fundamental factor model as one efficient way to combine relevant information about stocks into a system for determining returns and risks. The economic factor model is the counterpart to the fundamental factor model. It also combines relevant stock information in an efficient way, but with a different twist on the factor model framework.¹

The structure of the model remains the same as we move from the fundamental factor model to the economic factor model. The model still expresses the central idea that stock returns are the payoff for taking risk. As in the fundamental factor model, stock returns are determined by the product of factor exposures (i.e., exposures to risk) and factor premiums (i.e., payoffs for exposure to risk). In the economic factor model, however, the roles of the factor exposure and factor premium are, in a way, reversed. Recall that for the fundamental factor model, the factor exposure is observable in financial statements, whereas the factor premium must be estimated from a cross-sectional regression. In the economic factor model,

the factor premium is the known value (or, at least, the value that can be calculated from given data), whereas the factor exposure must be estimated by a regression of stock returns on factor premiums.

Take, for instance, an economic factor model with just one factor, inflation. In this model, the factor premium is the observed rate of inflation (or a rate that corresponds to the inflation rate). The factor exposure is the stock's sensitivity or reactivity to inflation, which is estimated as the relationship between the return on the stock and the rate of inflation. The economic factor model, therefore, takes as a given the premium that the market generally places on exposure to the risk but requires the estimation of a particular stock's exposure to the risk. This type of model especially makes sense for economic factors, which represent external risks in the marketplace that affect all stocks.² In this case, the particular type of risk that we are concerned with is inflation. The factor premium therefore shows the amount of reward investors demand when they invest in stocks affected by inflation. This amount of reward demanded most likely will be different from the actual inflation rate for two reasons. The first is that the units are different. The inflation rate is not expressed in risk terms, so we cannot say that the reward for taking 1% inflation rate should be 1% expected return. The second is that not all inflation corresponds to risk. Specifically, the expected component of inflation does not qualify as risk; only the unexpected component of inflation should be considered a risk. So why do we use the actual inflation rate as the factor premium?

The economic factor model does not assume that the amount of reward demanded by investors is the actual inflation rate. What it assumes is that the amount of reward is a linear function of the actual inflation rate. The reward for exposure to an $x\%$ inflation rate may not be $x\%$ expected return, but it may be stated as $a + bx$. This assumption removes the necessity to distinguish between the true factor premium and the observed inflation rate because the estimation of the model will not be affected whether we use the true factor premium or the observed inflation rate. This assumption also takes care of the problem caused by the expected component of inflation. The expected component will be adjusted by the constant a in the formula $a + bx$. For this reason, for the remainder of this chapter we do not distinguish between the true factor premium and the observed values of the economic variables.